

TROPICAL COLLEGE OF MEDICINE DEBRE MARKOS CAMPUS



COMMUNICABLE DISEASE CONTROL FOR NURSING STUDENTS

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CHAPTER I

INTRODUCTION TO COMMUNICABLE DISEASE CONTROL

Introduction

Definition

- **Communicable disease (infectious disease)** is an illness
 - due to a **specific infectious agent or toxic** product
 - that **arises** through transmission from a **reservoir** to a **susceptible host**
 - either **directly or indirectly**
- Many advances made in **the prevention and treatment** of infectious diseases.
- But they remain significant causes of morbidity and mortality in developed as well as developing countries
- Communicable diseases remain major health problems in Ethiopia
 - It accounts for about **60-80% of the health problems**

Specific features of communicable diseases

- ➡ Each infectious agent has a risk factor
- ➡ People may be immune
- ➡ An individual may be a source **without being recognized as a case**
- ➡ There is some times a need for urgency
- ➡ Preventive measures (usually) have a good scientific ground
- ➡ Intervention on infectious disease can have several effects
 - Direct effect the target of the intervention
 - Indirect effect on other

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- **What is the importance of studying infectious disease?**
 - To identify and treat **cases or carriers** as early as possible
 - Identification of causes of new/emerging infections
 - **Surveillance** of infectious diseases
 - Identification of source of **outbreaks**
 - To study route of transmission and natural history of infections
 - Identification of new **interventions**

Terminologies

- **Infection:** the **entry** and **development** or **multiplication** of an infectious agent in the body of human beings or animals
- **Disease:** change from a state of health
- **Carrier:** a person that carries specific infectious agent and can transmit to others but has no clinical sign or symptom
- **Reservoir:** any person, arthropod, animal, plant...in which the infectious **agent normally lives and reproduce itself**

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- **Case:** a person identified as having a given health problem or disease of interest
- **Chemoprophylaxis:** administration of antimicrobials **to prevent** the development or progression of an infection to the actual disease or to stop transmission to others
- **Chemotherapy:** use of drugs **to treat** a clinically manifesting disease or limit its further progression

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- **Contagious disease:** a disease that is transmitted easily **through close contact**
- **Contamination:** presence of infectious agent (IA) on body surface, clothes, beddings
- **Disinfection:** the **killing of IAs outside the body** by direct exposure to physical or chemical agents

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- **Disinfestations:** the procedure of **destroying or removing** undesired small forms of animals particularly **arthropods**
- **Sterilization:** **destruction of all forms of life** by heat, irradiation, gas or chemical treatment
- **Drug resistance:** the ability of infectious agent to survive despite the administration of anti microbial in a dose equal to or higher than the recommended dose

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- **Etiology:** study of the cause of a disease
- **Pathogenesis:** manner in which a disease develops
- **Susceptible host:** a person or animal **not possessing sufficient resistance** against a specific IA to prevent infectious disease acquisition
- **Inoculum size:** the minimum size of infectious agent or its products that can cause disease
- **Infestation:** the lodgment, development and reproduction of arthropods **on the surface of the body or in clothes**

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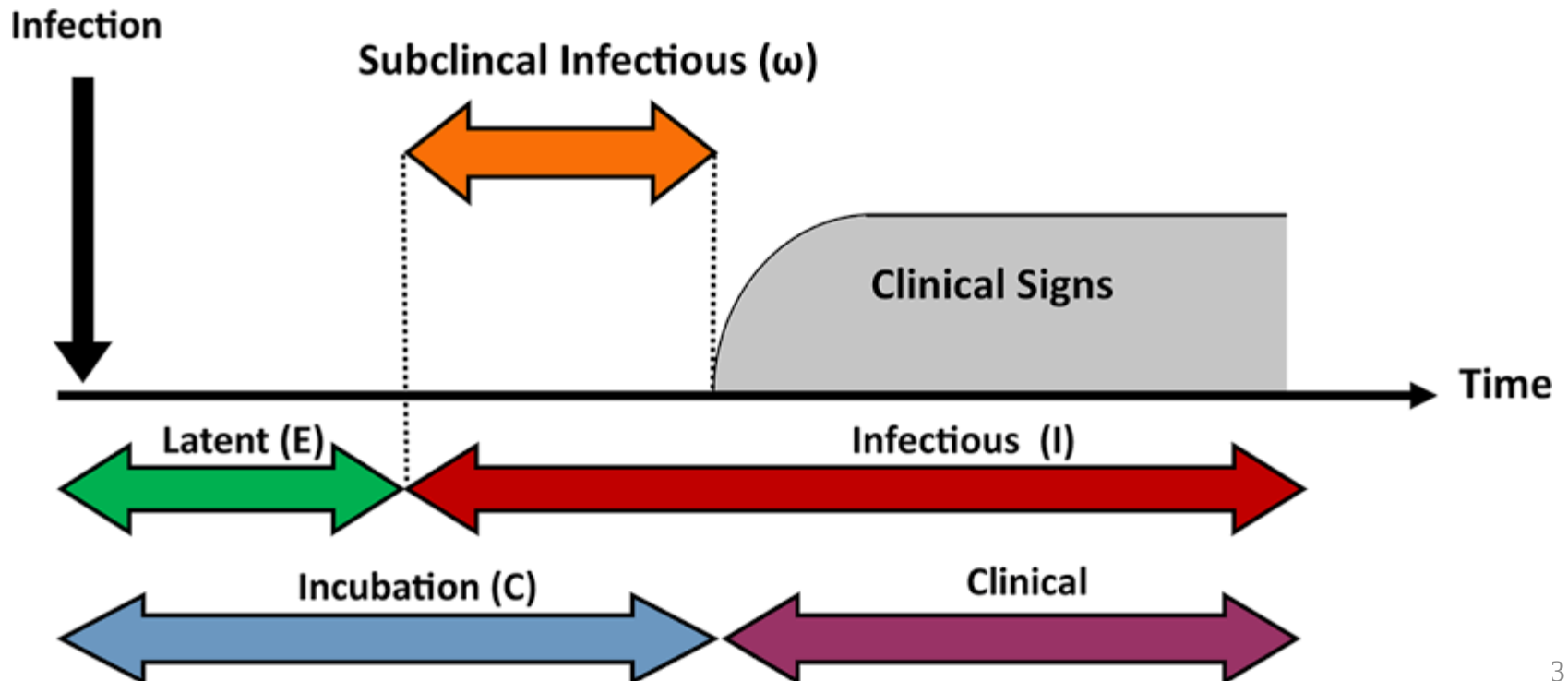
- **Communicable Disease:** an illness due to **a specific infectious agent or toxic product** that arises through transmission from a reservoir to a susceptible host either directly or indirectly
- **Zoonosis:** an infection transmissible under natural conditions from **vertebrate animals to man**
- **Nosocomial infection:** infection originating in a patient while in hospital or other health care facility in which the infection was **not present or incubating at the time of admission**
- **Community acquired infection:** infection that **occurs in the community (general population).**
- **Opportunistic infection:** an infection by organisms that take the opportunity of defect in the host defense to infect the host and cause disease

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- **Surveillance:** the continuous scrutiny/searching of the factors that determine the occurrence and distribution of disease and other conditions of ill health
- **Eradication:** **termination of infection transmission** by extermination of the infectious agent through surveillance and containment/control
- **Elimination:** eradication of disease **from large geographic** region or political jurisdiction
eg measles, polio, diphtheria
- **Control:** ongoing operation aimed at **reducing the prevalence of a disease** to a **level where it is not a public health problem**

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- **Incubation period:** the time interval between invasion by infectious agent and **appearance of first sign or symptom of the disease**
- **Latent period:** period from disease **initiation** to disease **detection**



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- **Quarantine:** restriction of the activity of well person or animal who were exposed to a case of communicable disease **for the duration of incubation period to see** if infection will occur
- **Isolation:** keeping infected persons or animals in separate place for the duration of communicability
- **Respiratory isolation:** use of private room to prevent transmission by air but similar patients may share the same room

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- **Communicable period:** time for an infectious agent to be transferred from infected to others
- **Primary(index) case:** a person who acquires a disease through exposure and **brings it in to the population**
- **Secondary case:** a person infected by primary case
- **Standard precaution:** simple standard procedure to be used during care of patients to minimize risk of transmission. Eg hand washing, gloving, safe medical waste disposal
- **Virulence:** the ability of the IA to invade and damage the host tissue and **cause death**

Classifying Infectious Diseases

1. Patient Appearance

A. Symptoms: subjective changes in body function (e.g. pain)

B. Signs: objective changes in body function that can be measured(e.g. fever)

C. Syndrome: specific group of symptoms & signs that may accompany a particular disease

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2. Spreadability

- A. **Communicable disease:** spreads from one **host to another** (e.g. herpes)
- B. **Contagious disease:** spreads easily from one **person to another** **by touching** (e.g. chicken pox)
- C. **Non-communicable disease:** **not spread** from one host to another
 - ✓ either resident flora that becomes an OI(e.g. **UTI**)
 - ✓ accidental inoculation from env't (**tetanus**)

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3. Spectrum of Disease Occurrence

- A. **Sporadic disease:** occurs only **occasionally** in population (e.g. RF)
- B. **Endemic disease:** **consistent and expected** level of a disease in a geographic area (e.g. malaria)
- C. **Outbreak:** **unexpected occurrence** of an infectious disease in a **limited geographical area during a limited period of time.**
- D. **Epidemic disease:** **unexpected increase** of an infectious disease in a geographical area **over an extended period of time** (e.g. influenza)
- E. **Pandemic disease:** **steady occurrence** of disease over a **very large geographical area** (e.g., continent of Africa) or worldwide (e.g. AIDS)

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4. Based on time course

- A. Acute disease:** develops rapidly, but lasts only a short time (e.g. cold)
- B. Chronic disease:** develops slowly, may have mild symptoms or signs, but is continual or recurrent for a long time (e.g. tuberculosis)
- C. Subacute disease:** intermediate between acute and chronic (e.g. endocarditis)
- D. Latent disease:** **agent remains inactive** for a period of time, but then activates to cause disease (e.g. shingles)

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5. Based on taxonomy of infectious agents

- Metazoan: e.g., helmenthiasis
- Protozoan: e.g., malaria, amoeba
- Bacterial: e.g., tuberculosis
- Fungal: e.g., candidiasis
- Viral: e.g., measles, polio

6. Based on clinical manifestation/system involved

- **Diarrheal diseases:** secretory/watery or invasive/dysentery
- **Febrile diseases:** malaria, typhus, relapsing fever
- **Respiratory diseases:** upper or lower respiratory
- **Central nervous system infections:** meningitis, encephalitis, brain abscess
- **Cardiovascular infection:** Endocarditis, Myocarditis
- **Urinary tract infection:** upper and lower urinary tract infections

7. Based on **mode of transmission**

- A. Common diseases, which use the respiratory tract as portal of entry and/or exit. E.g. pertussis, meningococcal meningitis, tuberculosis, measles
- B. Common diseases, which use gastrointestinal tract as portal of entry and/or exit. e.g. bacillary dysentery, salmonellosis, brucellosis, typhoid fever, cholera
- C. Common diseases, which use the mucous membrane and/or skin as their portal of exit/entry. E.g. STIs, scabies, trachoma, ringworm, candidiasis, hepatitis B and C
- D. Diseases transmitted by direct inoculation: e.g. schistosomiasis, hookworm infection
- E. Vector borne diseases: e.g. malaria, leishmaniasis, trypanosomiasis, relapsing fever, typhus
- F. Inoculation by bites of animals/contacts with animal product: e.g. rabies, tetanus, anthrax, brucellosis

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CHAPTER TWO

CHAIN OF DISEASE TRANSMISSION (DISEASES TRANSMISSION DYNAMICS)

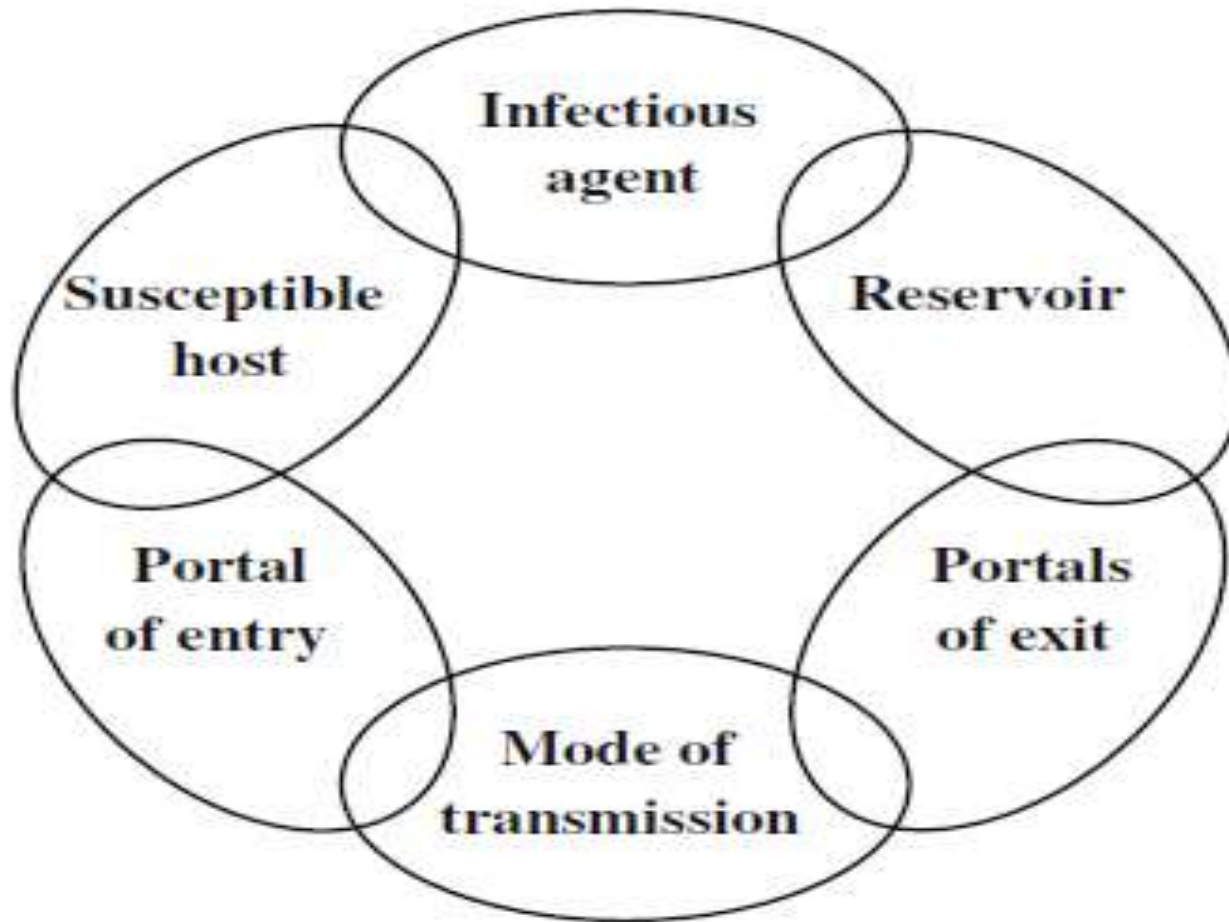
Infectious disease cycle

- It is also called **transmission cycle** or **chain of infection**
- It refers to the process by which infectious diseases are transmitted **from infectious host to susceptible host**
- The spread of an infectious disease through populations is determined by characteristics of the infectious agent, the host, and the environment

Components of the infectious process

- ✓ The Agent
- ✓ Its reservoirs
- ✓ Its portal of exits
- ✓ Its mode of transmission
- ✓ Its portals of entry
- ✓ Susceptible human host

Chain of infection



1. Infectious agent

- The **infectious agent** is virus, bacteria, parasite or other microbes
- The agent causes **infection and disease** depending on its basic biological characteristics like;
 - a. Infectiousness
 - b. Pathogenicity
 - c. Virulence
 - d. Immunogenicity

Infectivity:

- Is the characteristic of infectious agent **to enter, survive and multiply(produce infection) in the host**
- Measured by Proportion of susceptible close contacts who become infected(secondary attack rate)
- **Infection rate = $\frac{\text{total no. of infected people}}{\text{total no. of susceptible people exposed}} \times 100$**

Pathogenicity

- ❖ The ability of an infectious agent to **cause clinical disease among infected human hosts**
- ❖ It is measured by **clinical to sub-clinical ratio** or proportion of clinical cases among infected human hosts

$$\text{Pathogenesis} = \frac{\text{total number of clinical cases}}{\text{total number of sub-clinical cases}} \times 100$$

Examples:

- High pathogenecity: HIV, Rabies, Measles... etc.
- Moderate pathogenecity: Mumps virus and Rhino virus
- Highly infectious but less pathogenic: Poliovirus

Virulence

- It is the ability of an **infectious agent to cause Severe clinical disease** or **death** among clinical cases
- Enhanced by **virulence factors** that assist the bacterium colonize the host at cellular level.
- May be **secretary or membrane associated**
- Virulence of the infectious agent can be measured by:
 - ✓ Case fatality Rate (CFR)
 - ✓ Hospitalization Rate (HR)
 - ✓ Proportion of cases disabled or who have developed sequel
 - ✓ Proportion of cases that require different kinds of treatment

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$$CFR = \frac{\textit{Number of Fatal Cases}}{\textit{Total Number of Cases}} \times 100$$

$$HR = \frac{\textit{Number of Hospitalized Cases}}{\textit{Total Number of Cases}} \times 100$$

Immunogenicity

- ✓ The **infection's ability to produce specific immunity.**
- ✓ It is defined as the ability of a pathogen or a vaccine to inducing an immune response after an infection or a vaccination respectively
- ✓ It may lead to protection against re-infection or re-activation with the same or similar pathogen in the future

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- ❖ Immunity after infection may or may not be protective against re-infection or may last for variable periods of time.
- ❖ Some infectious diseases confer **lifelong immunity**
- ❖ Others confer partial immunity against severe symptomatic infection, but much less against sub-clinical infection
- ❖ Some confer **no or negligible immunity.**

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- Immunity can be acquired either after natural infection or indirectly
- Maternal antibodies protect the newborn child against many infections in the first few months of life
- Vaccine-induced immunity can be lifelong or temporary
- For temporary immunity, repeated booster vaccinations are necessary to ensure protection against the infection

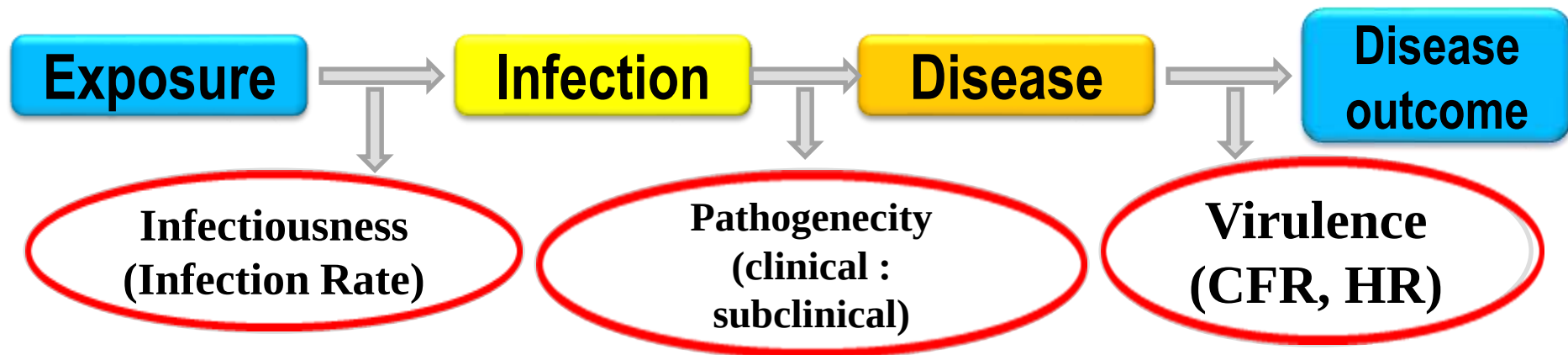
Example: TT vaccine

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□ Individuals gain protective antibodies in two ways:

- 1) They develop antibodies in response to infection, vaccine, or toxoid; immunity developed in these ways is called **active immunity**
- 2) They acquire their mothers' antibodies before birth through the placenta or they receive injections of antitoxins or immune globulin; immunity that is acquired in these ways is called **passive immunity**

Disease progression



2. Reservoirs

- ❖ **Reservoir** is an **organism or a habitat** in which an infectious agent lives, grows, transforms and/or multiplies itself
- ❖ **Reservoir** is **the starting point for the occurrence of CD**

Examples:

- **Human beings:** Most bacterial and viral respiratory diseases like Measles, typhoid, HIV/AIDS, STIs
- **Animals/zoonoses:** Examples: Rabies, anthrax, etc
- **Environment** (plant, soil, water): tetanus, hook worm etc...

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- All infected humans, whether showing signs and symptoms of the disease or not, are potential sources of infection to others
- A person who does not have apparent clinical sign/symptom but is a potential source of infection for others is **called carrier**.
- **Carriers could be:**
 - **Incubatory:** during incubation period e.g. **measles, mumps**
 - **Asymptomatic carrier:** with out ever showing s/s. e.g **amebiasis, polio**
 - **Convalescent carrier:** during recovery e.g **thyphoid**
 - **Chronic carrier:** transmitting for long period/indefinitely. E.g **thyphoid, viral hepatitis, HIV**

3. Portal of Exit

- ❖ The portal of exit is the route by which the infectious agent leaves the **infectious hosts or reservoirs**
- ❖ Many diseases have multiple reservoirs

☐ Portal of exit includes:

- **Body secretions and discharges** (mucus, saliva and tears, breast milk, urethral secretion, semen, vaginal secretion, pus, cervical secretion, exudates etc...)
- **Excretions** (feces and urine)
- **Blood and tissues** including placenta , etc...

4. Modes of Transmission

- ❑ Mode of transmission is the mechanism by which the infectious agent are **conveyed** to other **susceptible human host**

- ❑ There are two major mechanisms of transmission
 - Direct transmission
 - Indirect Transmission

I. Direct transmission

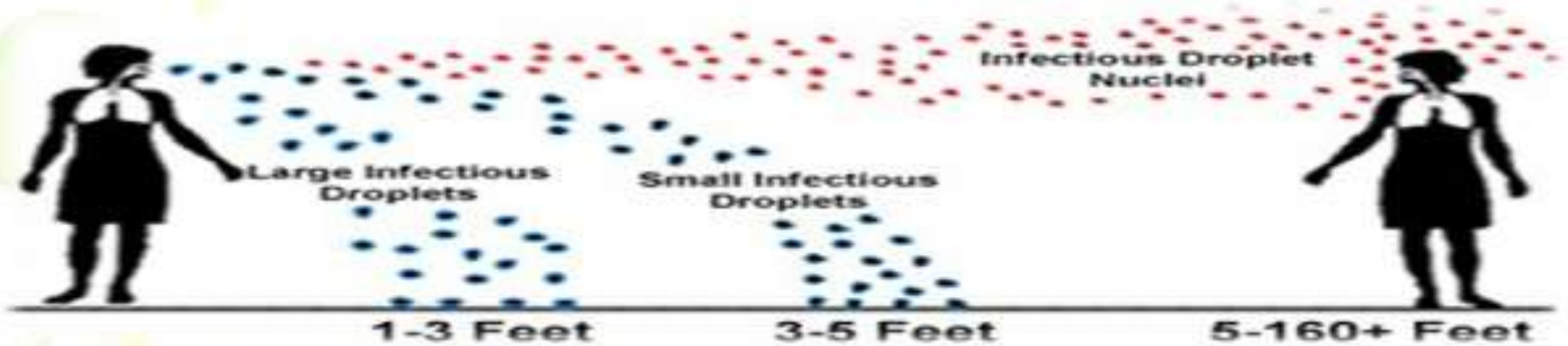
- a) Transmission by direct contact
- b) Transmission by direct projection
- c) Trans-placental transmission
- d) Blood transfusion
- e) Organ transplantation

a) Direct contact

- ❖ Transmission by contact of *skin-skin, mucosa-skin, or mucosa to mucosa of same or other persons*
- ❖ Direct and immediate transfer of IA from reservoir to host
- ❖ **It introduces larger dose of IA**
- ❖ No time interval of survival in the env't
- **Touching:**
 - Trachoma (eye-hand-eye)
 - **Common cold** (nose-hand-eye)
 - Shigellosis (feces-hand-mouth)
 - Viral hepatitis and HIV (through breaks in skin)
- **Sexual intercourse:** HIV/AIDS
- **kissing:** mononucleosis
- **Passing through birth canal:** Gonorrhea

b) Direct projection

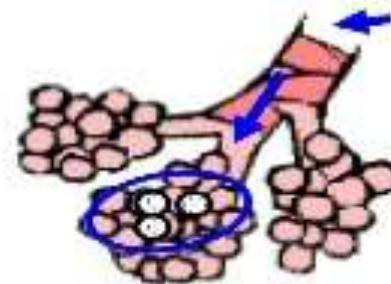
- ❖ Transmission by direct projection of saliva droplets/ nasopharyngeal secretion
- ❖ Created by expiratory activities such as breathing, coughing, sneezing, spitting, talking, singing etc...
- ❖ Droplets directly impinge on **conjunctiva, nasal mucosa or skin**
- ❖ Droplets size
 - Largest droplets fall to ground in seconds, may persist in dust
 - Medium size droplets
 - Smallest droplets- evaporate leaving droplet nuclei



Smallest droplets (<25 mm) evaporate leaving "**droplet nuclei**" of bacilli that can reach **alveoli** (e.g., TB).

Medium Droplets

Medium-sized droplets: trapped & cleared in upper airway.



- **H1N1**
- **Tubercle bacilli**
- **Measles**
- **Chickenpox**

Largest Droplets

Largest droplets fall to ground in seconds; may persist in dust, but not an important cause of infection.

c) Trans-placental transmission

➤ Transmission from **mother to fetus** via the placental membrane

Examples:

- ✓ HIV/AIDS, HBV
- ✓ TORCH Infections:
 - Toxoplasmosis,
 - Other (syphilis, varicella-zoster, parvovirus B19),
 - Rubella,
 - Cytomegalovirus (CMV), and
 - Herpes infections.
- ✓ Malaria etc...

d) Blood transfusion

- It is the transmission of disease through blood and blood products
- E. g. **HIV** and **Hepatitis-B viruses, malaria**

e) Organ transplantation

- It may include kidney, liver and cardiac transplantation which transmit some communicable diseases from the donors to the receivers
- E.g. **rabies virus, HIV, HCV, HBV**

II. Indirect transmission

- ❖ Uses Five Fs: **Fluid, fomites, finger, flies and food**
 - Airborne transmission
 - Vehicle borne transmission
 - Vector borne transmission
 - Non vector intermediate host

a) Airborne transmission

- It is the transmission of infectious agents by either suspended dust or droplet nuclei

Examples:

- Tuberculosis: droplet nuclei during sneezing, speaking...
- Brucellosis: blood aerosols created during slaughter

b) Vehicle borne transmission

- Transmission occurs through indirect contact with inanimate objects (**vehicle**)

Vehicle: is any non-living substance by which an infectious agent can be transported and introduced into a susceptible human host through a suitable portal of entry

Examples:

- **Food:** staphylococci, *C. botulinum*
- **Water:** typhoid, cholera
- **Soil and biological products**
- ***fomites*** (cooking utensils, towels, bed sheets, clothing, syringe, beddings, etc...)

c) Vector borne transmission

- The IA is conveyed by an arthropod(vector)

Vector: is an organism usually an arthropod, such as insect, tick, mite, which transports an infectious agent to a susceptible human host or to a suitable vehicle

- A vector is responsible for introducing the agent
 - **Biological vector: the agent needs multiplication in the vector before transmission** eg, **salivarian** transmission (malaria by mosquito)
 - **Mechanical vector: the agent is directly infective** to other hosts without having to go through a period of multiplication or development in the vector.
E.g flies for the transmission of trachoma.

d) A non-vector intermediate host

- ❖ These are hosts which are important for **development of the infectious agent**
- ❖ But, don't play an **active role in transporting** the agent to the susceptible human host
- ❖ **Example: Aquatic snail for schistosomiasis**

5. Portal of Entry

- Portal of entry is the route/site through which a microorganism enters into the susceptible human host
- Whether an agent will establish infection depends on the suitable **portal of entry**

Example: No harm in ingesting the *Clostridium tetani* (it is in fact the normal flora of GIT system)

6. Susceptible Human Host

- Host factors influence individual's exposure, susceptibility or response to infectious agent
- The susceptible human host is the **final link in the infectious** disease transmission process
- Host susceptibility or resistance could be:
 - **Individual level**: immunity acquired due to past infection
 - **Community level**: **herd immunity**
 - ❖ The resistance of the population to the introduction and spread of IA
 - ❖ Based on the immunity of the high proportion of individual members of the population
 - ❖ thereby decreasing of diseased person contact with susceptibles

Time course of a disease

- Pre-patent period
- Incubation period
- Communicable period
- Latent period

Pre-patent period

- It is the time interval between **infection and the time of first shedding of the agent by the human host**
- For some diseases the time when the agent can first be detected and the time of the first shedding do have difference
- i.e. the detection might not be possible at the beginning of the shedding of infectious agents
- **E.g. Window period of HIV/AIDS**

Incubation period

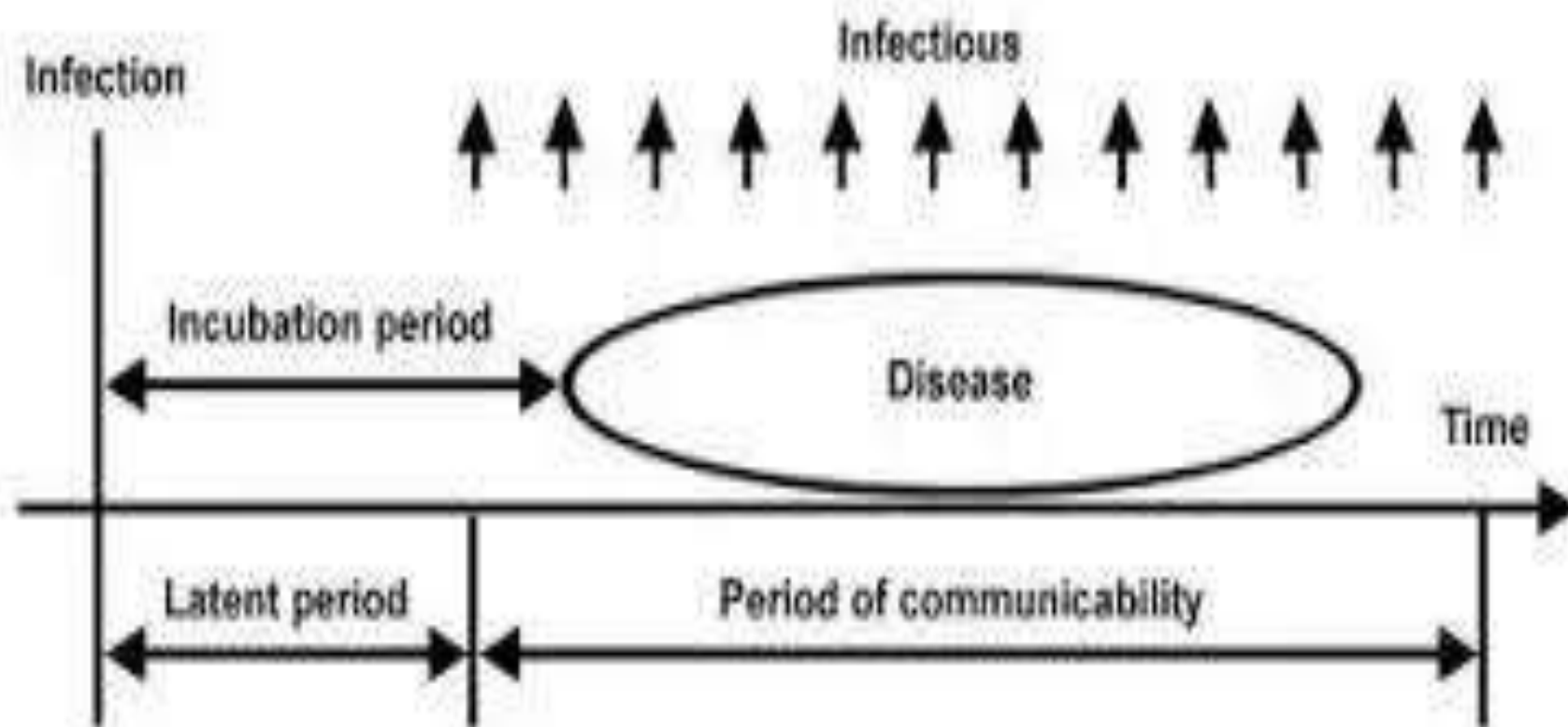
- It is the time interval between infection and the first clinical manifestation of disease

Communicable period

- It extends from the point of the **first shedding up to the last shedding of the infectious agent** by the infectious host
- It is the period during **which an infected host can transmit the infection to other host**
- It can be measured by the time interval during which the agent is shed by the host

Latent period

- **The period b/n exposure and the onset of the period of communicability which may be shorter or longer than IP**
- It is the time interval between **recovery and the occurrence of a relapse in clinical disease**
- Example: Typhus and malaria have latent period because they relapse after some time



Principles of communicable disease control

➤ The actions may be effected through:

- **Attacking the source**
- **Interrupting the mode of transmission and**
- **Protecting the host (decrease susceptibility)**

1. Attacking the source (reservoir) of infection

- **Domestic animals as reservoirs**
 - Immunization e.g. Brucellosis
 - Destruction of infected animals e.g. Rabies
- **Wild animals as reservoirs**
 - Post exposure prophylaxis e.g. Rabies
- **Humans as reservoirs**
 - **Isolation** of infected persons & separation of infected persons from other for the period of communicability
 - **Treatment**
 - Of cases (clinical) and carriers
 - Mass treatment: where large proportion are known to have a disease

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- **Quarantine:** the limitation of freedom of movement of apparently healthy persons or animals who have been exposed to a case or infectious disease.
- **Cholera, plague, and yellow fever** are the 3 internationally quarantinable diseases by international agreement.
- Since these diseases are very infectious, **cases shouldn't be referred** but seniors must be requested to visit the health center

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2. Interrupting the chain of transmission

- For Transmission by ingestion
 - Purification of water
 - Pasteurization of milk
 - Inspection procedures designed to ensure safe food supply
 - Improve housing conditions
- For Transmission by inhalation
 - Chemical disinfection of air
 - Use of UV-light
 - Improving ventilation
- Transmission by vector or intermediate hosts
 - Vector – control measures
 - Environmental manipulation

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3. Reducing host susceptibility

- **Immunization**

- **Active immunization:** when either the altered organism or its products is given to a person to induce production of antibodies. E.g. BCG
- **Passive immunization:** is provision of ready – made antibodies. E.g. TAT

- **Chemoprophylaxis:** use of antibiotics for known contacts to a case. E.g. Ciprofloxacin for contacts to a case of N. meningitis

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- **Better nutrition:** Malnourished children get infections more easily & suffer more severe complications.
- **Vector control:**
 - ❖ Repellants or detergents
 - ❖ Personal protection
 - ❖ Larvicides
 - ❖ Biological control
 - ❖ Environmental modification
 - ❖ Insecticides



Chapter Three

Natural history of disease and Level of Prevention

Learning objectives

By the end of this session, students will be expected to:

- Define natural history of a disease
- Discuss the stages of natural history of a disease
- Discuss levels of prevention
- Describe the levels of prevention to diseases of public health importance

Natural History of Disease

- It refers to the **progression of a disease process** in an individual over time, in the **absence of intervention**.
- Knowledge of the natural history of a disease helps
 - to **understand the effects and mechanism of actions**,
 - to **provide potential interventions**
- There are four stages in the natural history of a disease.
 1. Stage of susceptibility
 2. Stage of pre-symptomatic (sub-clinical) disease
 3. Stage of clinical disease
 4. Stage of disability or death

1.Stage of susceptibility (stage of exposure)

- The disease has not developed but the groundwork has been laid by the presence of factors that favor its occurrence

Examples:

- Unsafe sexual exercise to HIV/AIDS
- Alcohol consumption for liver cirrhosis
- High cholesterol to stroke,
- Smoking to hypertension
- Sedentary life style to chronic diseases
- Unvaccinated child to vaccine preventable disease
- Factors whose presence is associated with an increased probability that disease will develop later are called **risk factors**
 - Modifiable (smoking, chewing chat) or non
 - Non Modifiable(age, sex, race....)

2. Stage of asymptomatic disease

- In this stage, **pathologic changes of disease have started** but there is **no manifestation of a disease**
- Is a period of **subclinical or in—apparent pathologic change** which follows exposure
- This stage **may or may not be followed by the clinical disease**
- The disease can only be **detected through special tests** since the signs and symptoms of the disease are not present.

Examples:

- Detection of antibodies against HIV in an apparently healthy person.
- Ova of intestinal parasite in the stool of apparently healthy children.

3. Stage of clinical disease

- There is **signs and symptoms of the disease**
- The clinical stage of different diseases differs in duration, severity and outcome
- Examples
 - ❖ Common cold has a **short and mild clinical stage** and almost everyone **recovers** quickly.
 - ❖ Polio has a **severe clinical stage** and many patients develop paralysis becoming **disabled** for the rest of their lives.
 - ❖ Rabies has a **relatively short but severe clinical stage** and almost always results in **death**.
 - ❖ Diabetes Mellitus has a **relatively longer clinical stage** and eventually results in **death** if the patient is not properly treated

4. Stages of outcome of the disease

- **Outcome of a disease may be**
 - **recovery,**
 - **chronic carriers,**
 - **disability or death**
- Some diseases run their course and then resolve completely either spontaneously or by treatment.
- In others, the disease may result in a residual defect, leaving the person disabled for a short or longer duration.
- Still, other diseases will end in chronic carrier or death

Disability

- It is defined as **any temporary or permanent reduction** of an individual's activity as a result of an acute or chronic condition
- It encompasses not only physical loss of function but also **psychological, social and financial losses**
- *Examples:*
 - Trachoma may cause blindness
 - Meningitis may result in blindness or deafness.
 - Meningitis may also result in death.

Levels of disease prevention

- Definition: it means **interrupting or slowing the progression of disease** through appropriate intervention

□ There are four levels of prevention

- ✓ Primordial prevention
- ✓ Primary prevention
- ✓ Secondary prevention
- ✓ Tertiary prevention

1. Primordial prevention

- Primordial prevention consists of actions and measures that **inhibit/avoid the emergence of risk factors** in the form of
 - Environmental
 - Economic
 - Social
 - Behavioral conditions
 - Cultural patterns of living
- In primordial prevention, efforts are directed towards **discouraging from adopting harmful lifestyles**
- The main intervention in primordial prevention is through **individual and mass education**

2.Primary prevention

- Primary prevention keeps the disease process from becoming established by
 - **eliminating causes of disease** or
 - **increasing resistance to disease.**
- Helps to **limit the incidence of disease** by controlling specific causes and risk factors
- Primary prevention is mainly done through:
 - ❖ Health promotion
 - ❖ Prevention of exposure
 - ❖ Prevention of disease

2.1. Health promotion

- It is a general **non-specific** intervention that **enhances health and body's ability to resist disease**

Examples:

- Balanced diet
- Health education
- Physical exercise
- Emotional and social supports etc ...

2.2. Prevention of exposure

- ❖ It is the **avoidance of factors which may cause disease if an individual is exposed to them**
- ❖ It acts **before exposure to the risk factors**

Examples:

- Use of bed nets: prevention of malaria
- Consistent and correct use of condom: prevention of HIV/AIDS
- Proper ventilation: control of acute respiratory tract infection
- Proper excreta disposal

2.3. Prevention of disease

- ❖ It is **prevention of disease development** after the individual **has become exposed to the disease causing factors**
- ❖ This intervention aims to **prevent initiation of disease**

Examples:

- ✓ INH prophylaxis for tuberculosis for HIV positive people
- ✓ Active Immunization acts after exposure has taken place.
 - Being exposed to the vaccine
- ✓ Passive Immunization
 - Passive: TAT and rabies vaccines

Duration of protection in women following 1–5 doses of TT vaccine.

Dose (0.5ml)	When given	Duration of protection
TT1	At first contact with women of childbearing age, or as early as possible in the pregnancy	No protection
TT2	At least 4 weeks after TT1	3 years
TT3	At least 6 months after TT2	5 years
TT4	At least 1 year after TT3	10 years
TT5	At least 1 year after TT4	All child bearing year

3. Secondary prevention

- **Stop or slow the progression of disease** so as to prevent or limit permanent damage
- Prevents complications
- Achieved through detecting people who already have the disease as early as possible and treat them
- **The specific interventions are:**
 - **Early diagnosis** (e.g. screening tests, and case finding programs....) and
 - **Adequate treatment**

- It attempts to
 - **Arrest** the disease process
 - **Restore health** by seeking out unrecognized disease
 - Treating it before irreversible pathological changes take place /prevent disability
 - **Reverse communicability** of infectious diseases to others
- It protects others in the community from acquiring the infection
- Thus, provide at once
 - ❖ Secondary prevention for the infected ones and
 - ❖ Primary prevention for their potential contacts

Early diagnosis and treatment strategies

The diagram consists of three overlapping shapes: a yellow rectangle at the top, a blue oval on the left, and a green oval on the right. A red pentagon is positioned at the bottom, overlapping the bottom of the blue and green ovals. The yellow rectangle is connected to the blue oval by a yellow triangular shape. The text is in bold black font.

**Better prognosis of
the case**

**Prevention of the
occurrence of other
secondary cases**

**Treatment of active
TB disease**

3. Tertiary prevention

- The goal of tertiary prevention is to **slow or block the progression of a disease**:
 - ➔ Reducing impairments and disabilities
 - ➔ Improving the quality of life and
 - ➔ Improving survival among diseased individuals

Disability limitation

- ❖ It refers to limiting the continuation of loss of function or limiting the impact of the damage

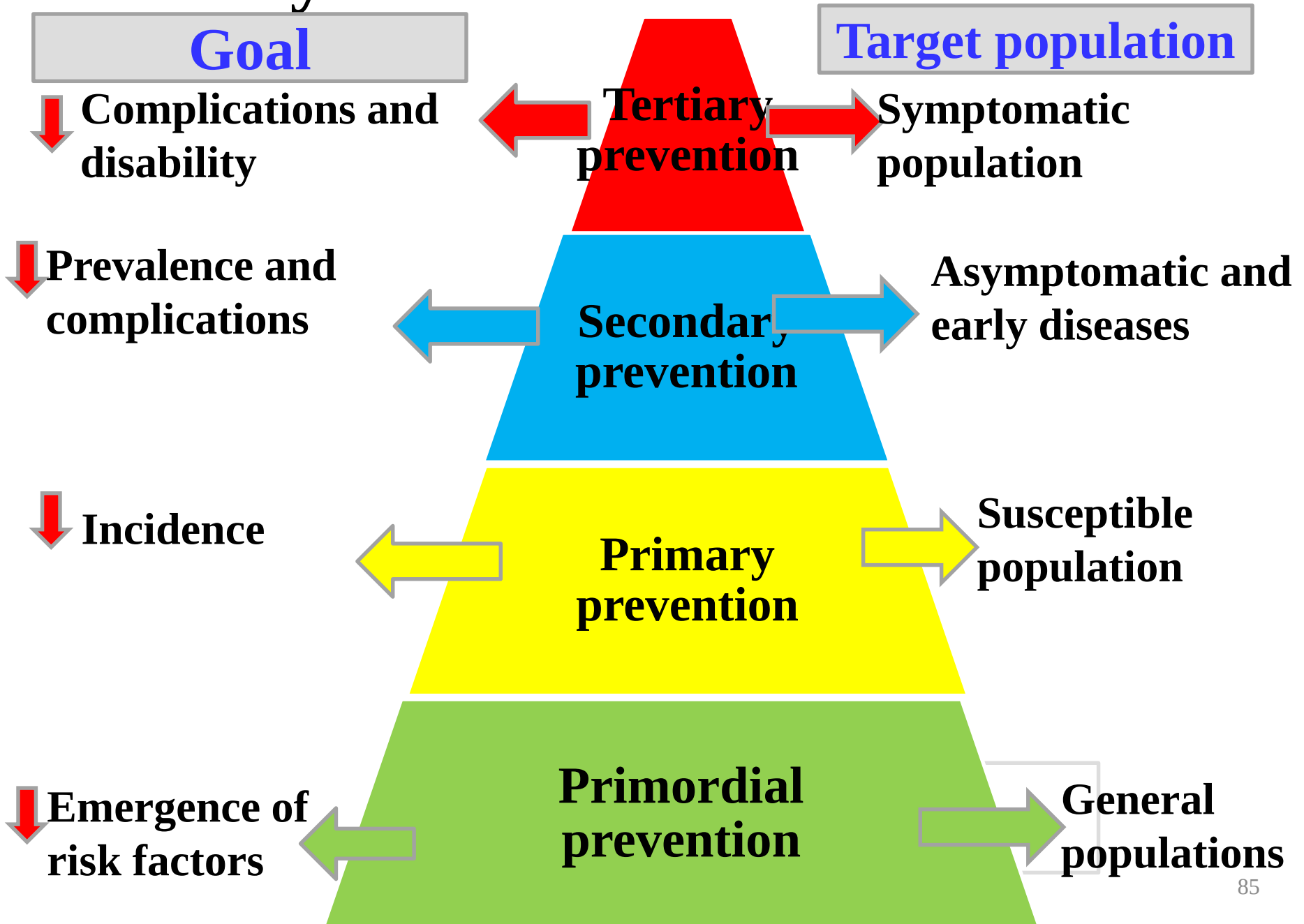
Examples:

- Foot care for diabetic patients to prevent injuries
- Passive joint movement for polio to prevent deformity

Rehabilitation activities

- ❖ It is “ the combined and **coordinated use of medical, social, educational, and vocational measures**
- ❖ It refers to the **retraining of remaining functions for maximal effectiveness**
- ❖ used for training and retraining the individual to the highest possible level of functional ability”
- ❖ Rehabilitative activities include:
 - plastic surgery for burn patients
 - wheel chairs
 - Prosthesis (artificial body part)

Summary





Chapter Four

Faeco - Oral Transmitted Diseases

Faeco-oral transmitted Diseases

- The transmission of causative organisms **from the faces of infected person to the GIT of a susceptible host**

Common features:

- The causative organisms are **excreted in the stools of infected person**
- The portal of entry for these diseases is **mainly mouth**
- Transmission mostly occurs through **unapparent fecal contamination of food, H₂O, hands**

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- In feco-oral transmission **food takes the central** role b/c it can be contaminated via
 - Polluted water
 - Dirty hands
 - Contaminated soil
 - Flies
- The five “**5Fs**” play an important role
 - ✓ Fluid
 - ✓ Finger
 - ✓ Fomites
 - ✓ Flies
 - ✓ Food

Classifications of feco-oral transmitted diseases

1. Diseases as a result of Faecal contaminated Water and food

- The diseases in this group are mainly transmitted through faecally contaminated **water and food**
- Infective agents enter the body through
 - **Ingestion** of contaminated food or water
 - **Using contaminated eating utensils**

2. Feces Mainly in Soil

1. Typhoid fever/ enteric fever

- Typhoid fever is a **systemic infection** characterized by **fever and abdominal pain** caused by dissemination of *Salmonella typhi* and occasionally by *S. paratyphi A* and *B* and *S. typhimarium*, all of which are non capsulated, gram negative motile bacteria.
- **Cause or Infectious Agent**
 - *Salmonella Typhi*
 - *Salmonella Paratyphi A* and *B*
 - *Salmonella typhimarium* (rare)

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- **Epidemiology**
 - Human are the only hosts for *S. typhi* and *S. paratyphi*.
 - Chronic carriers are the source of infection harboring the organisms in their **gall bladder (especially in the presence of gall stones)**
 - It affects people of **all ages and both sexes**.
 - Enteric fever is endemic in most developing countries including Ethiopia
 - Currently the disease is observed at a great frequency in AIDS patients than the general population.
- **Reservoir:** Humans
- **Mode of transmission**
 - **Feco-oral transmission:** by water and food contaminated by feces and urine of patients and carriers.
 - **Flies** may infect foods in which the organisms then multiply to achieve an infective dose.
- **Incubation period** –1-3 weeks

Pathogenesis

- Following ingestion of the organism in contaminated food or drink, *Salmonella typhi* passes the gastric barrier and **reach the upper small intestine** where the bacilli invade the intestinal epithelium and they are engulfed by phagosomes which reside in the **Peyer's patches**.
- The bacilli multiply and enter the blood stream and cause transient bacteremia.
- At this stage the *Salmonellae* disseminate throughout the body in **macrophages via lymphatic and colonize reticuloendothelial tissue** (liver, spleen, lymph nodes, and bone marrow).

...

- Patients have relatively fewer or no signs and symptoms during this initial incubation period.
- Signs and symptoms, including **fever and abdominal pain** result when a **critical number of bacteria have replicated.**
- During weeks after initial colonization, further inflammation of the Peyer's patches may result enlargement and necrosis which may result intestinal hemorrhage and perforation.
- Infection may become persistent and **invade the gall bladder.**
The clinical phase of the disease depends on host defense and bacterial multiplication.

Period of communicability

- As long as the bacilli appear in excreta, **usually from the first week throughout convalescence.**
- About 10% of untreated patients will discharge bacilli for 3 months after onset of symptoms, and
- 2%-5% become chronic carriers.

Susceptibility and resistance

- Susceptibility is **general** and increased in individuals with **gastric achlorhydria** or those **who are HIV positive.**
- Relative specific immunity follows recovery from clinical disease, inapparent infection and active immunization increase resistance

Clinical features

First week

- ➡ Fever is high grade, with a daily increase in a **step-ladder pattern** for the one week and **then becomes persistent.**
- ➡ Headache, **malaise**, **joint pain** Abdominal pain
- ➡ Initially diarrhea or loss stool followed by constipation in adults, diarrhea is dominant feature in children
- ➡ Relative bradycardia
- ➡ Splenomegaly and Hepatomegaly
- ➡ “**Rose spots**” not commonly seen in black patients. In whites it appears as small, pale red, blanching macules commonly over chest & abdomen, lasting for 2-3 days.

Second week

...

- ➡ **Fever becomes continuous**

- ➡ The patient becomes very ill and withdrawn confused, delirious and sometimes may be even **comatose**

Third Week

- ➡ The patient goes to a pattern of “typhoidal state” characterized by extreme **toxemia, disorientation**, and “**pea-soup**” **diarrhea** and sometimes may be complicated by **intestinal perforation** and **hemorrhage**.

Fourth Week

- ➡ **Fever** starts to **decrease** and the patient may deferveresce (experience an abatement of a fever) with resolution of symptoms.
- ➡ At this point patient may lose weight.
- ➡ Relapse may occur in 10% of cases.

Diagnosis

C/M Can be suggested by the presence of

- Persistent fever
- Relative bradycardia, which was found to occur in 86% of Ethiopians.
- Rose spots, which occurs in 70% of whites and 20% of Ethiopians.
- Leucopenia

...

Definitive diagnosis of the disease requires **laboratory tests**.

- Isolation of the organism by **blood, stool** or **urine** culture is diagnostic.
- **Blood culture:** mostly (up to 90%) patients have positive culture in the 1st week, and only 50% by the 3rd week. The yield is much lower if patient has taken antibiotics prior to the test.
- **Stool culture** is negative in the first week and becomes positive in 75% of patients in the 3rd week.
- **Urine culture parallels stool culture.**
- **Widal test**

Management

- Ciprofloxacin(first line) 500 mg bid (PO) or 400 mg q12h (IV) 5–7 days
- Azithromycin (1 g orally once followed by 500 mg once daily for 5 days
- Chloramphenicol:1g, IV bolus QID until 48 hrs after fever has settled, followed by 500mg P.O., QID for a total of 14 days
- Ceftriaxone 1–2 g/d (IV) for 7 days
- ❖ **Severe typhoid fever** (fever; delirium; stupor; coma; or septic shock; and a positive culture for *S.typhi* or *S. paratyphi*) →
Dexamethasone: a single dose of 3mg \kg followed by eight doses of 1mg\kg; given every 6 hrs).
- ⇒ **Ceftriaxone** 2-4gm once a day for 3 days and then 1- 2gm IV/IM for a total of 10- 14 days.
- ⇒ **Intravenous Chloramphenical** 1gm IV QID for 2-3 days and then start PO medication as soon as the patient can take oral medication.

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Alternative

- ⇒ Azithromycin 1 gm PO daily for 5 days
- ⇒ Chloramphenicol 500 mg Po QID for 14 days
- ⇒ Norfloxacin 400mg twice daily for 10 days
- Intravenous drugs are recommended for critically sick patients who are admitted or for patients who are unable to take oral drugs

Complication

- Intestinal perforation /bleeding
- Hepatic /splenic abscesses
- Meningoencephalitis
- Nephritis, Arthritis /osteomyelitis
 - ❖ Relapse rate despite Rx =10%
 - ❖ Chronic carriers: High incidence in women, individuals with biliary tract abnormalities (stone /tumor) and GI malignancy

Prevention and control

- Treatment of patients and carriers
- Education on hand washing, particularly food handlers, patients and child care givers
- Sanitary disposal of feces and control of flies.
- Provision of safe and adequate water
- Safe handling of food.
- Exclusion of typhoid carriers and patients from handling of food and patients
- Immunization for people at special risk
 - Eg. Travelers to endemic areas
- Regular check up of food handlers in food and drinking establishments

2. Bacillary Dysentery (Shigellosis)

- It is an acute bacterial disease involving the **large and distal small intestine**, caused by the bacteria of the genus shigella.
- **IA: *Shigella dysenteriae*** (most common cause)
- **Epidemiology**
 - It occurs worldwide
 - Endemic in both tropical and temperate climates.
 - Outbreaks commonly occur under conditions of crowding; and where personal hygiene is poor such as in **jails, institutions for children, day care centers, mental hospitals and refugee camps.**
 - It is estimated that the disease causes 600,000 deaths per year in the world.
 - Two third of the cases, and most of the deaths are in children under 10 years of age.

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- **Reservoir:** Humans

Mode of transmission

- Mainly by **direct or indirect fecal-oral transmission** from a patient or carrier.
- Transmission through **water and milk** may occur as a result of direct fecal contamination.
- **Flies can transfer organisms** from latrines to a non refrigerated food item in which organisms can survive and multiply.

Incubation period: 12 hours - 4 days (usually 1-3 days)

Period of communicability

- During acute infection and until the infectious agent is no longer present in feces, usually **within four weeks after illness.**

Susceptibility and resistance

- Susceptibility is general. The disease is more severe in **young children, the elderly and the malnourished.**
- Breast-feeding is protective for infants and young children.

Pathogenesis

- Resistance to low-pH conditions allows shigellae to survive passage through the gastric barrier
- The pathogenesis of *Shigella* involves

Invasion of colonic mucosal cells

Induction of an intense inflammatory response

Formation of colonic mucosal ulcerations and abscesses

death of epithelial and immune cells

Clinical Manifestation

- ➡ Fever
- ➡ Rapid pulse
- ➡ Vomiting and abdominal cramp are prominent
- ➡ **Diarrhea** usually appears after 48 hours with dysentery supervening two days later
- ➡ **Dehydration** is **common and dangerous** & it may cause **muscular cramp, oliguria and shock**
- ➡ Generalized abdominal tenderness
- ➡ Tenesmus is present
- ➡ Feces are bloody, mucoid and of small quantity.

Diagnosis

- Based on clinical grounds(**small, oderless, dark stool**)
- Stool microscopy (presence of **pus cells**)
- Stool culture confirms the diagnosis

Treatment

- **Fluid and electrolyte replacement**
- Ciprofloxacin 500 mg, PO bid for 3 days
- Antispasmodic; hyosin 10mg po prn
- In severe cases nalidixic acid 1gm po qid for 5-7days

Prevention and control

- Detection of carriers and treatment of the sick will interrupt an epidemic
- Hand washing after toilet and before handling or eating food
- Proper excreta disposal especially from patients, convalescent and carriers.
- Adequate and safe water supply
- Control of flies
- Cleanliness in food handling and preparation

3. Cholera

- It is an acute illness caused by an enterotoxin
- **Infectious agent:** *Vibrio cholerae*
- **Epidemiology**
 - had made periodic outbreaks in different parts of the world and given rise to pandemics. Endemic predominantly in children
- **Reservoir:** humans (**The natural habitat of *V. cholerae* is coastal salt water**)
- **Incubation period:** from a **few hours** to 5 days, usually 2-3 days.
- **Period of communicability:** for the duration of the stool positive stage, usually only a few days after recovery. Antibiotics shorten the period of communicability
- **Susceptibility and resistance:** Variable. Gastric achlorhydria increases risk of illness. Breast fed infants are protected.

Mode of transmission

- Transmitted by ingestion of food or water directly or indirectly contaminated with feces or vomitus of infected person.
- Ingestion of water contaminated by human feces is **the most common means of acquisition**
- Consumption of raw or improperly cooked seafood

Pathogenesis

- Cholera is **a toxin-mediated disease**
- The watery diarrhea is due to the action of cholera enterotoxin elaborated by the organism in the small intestine
- *V. cholerae* survive and multiply in (colonize) the **small intestine**
- In intestinal epithelial cells
 - **absorptive sodium transport system** in villus cells are inhibited and
 - the **secretory chloride transport system** in crypt cells are activated
 - these events lead to the accumulation of sodium chloride in the intestinal lumen
 - Since **water moves passively to maintain osmolality**, isotonic fluid accumulates in the lumen.
 - When the volume of that fluid exceeds the capacity of the rest of the gut to reabsorb it, watery diarrhea starts
- ❖ Cholera is **severely dehydrating secretory diarrheal illness.**

Clinical manifestations

- ❖ Incubation less than 1 day to 5 days.
- ❖ **sudden onset, profuse, painless and watery life-threatening diarrhoea (rice water stools)** may develop in < 10% cases
- ❖ **Profuse Vomiting (common)**
- ❖ **Rapid dehydration**
- ❖ Postural hypotension, weakness, tachycardia & ↓ skin turgor, shock and death.
- ❖ **No fever**
- ❖ CFR exceeds 50% in severe untreated cases. If proper treatment with rehydration therapy it will reduced to 1%.

Diagnosis

- Clinically
- Confirmed by culturing *V. cholerae* sero group

Specific treatment

- Death from cholera is due to **hypovolemic shock**
- The cornerstone of treatment is **timely and adequate rehydration**
- Patients with suspected cholera need to be treated prior to laboratory confirmation
- **Fluid + electrolyte replacement, IV fluid**
- **Administration of ORS takes advantage of the hexose- Na^+ co-transport mechanism to move Na^+ across the gut mucosa**
- Cl^- and water follow
- This transport mechanism remains intact even when cholera toxin **is active**

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- Doxycycline (a single dose of 300 mg) or
- **Tetracycline**, 500mg P.O., QID for 3-5 days.
- Sulfamethoxazole + trimethoprim,
 - Children 6 weeks – 5 months: 100/20mg
 - Children 6 months – 5 yrs: 200/40mg
 - Children 6 – 12 yrs: 400/80mg BID for 5 days
 - Adults: 800mg/160mg P.O., BID for 5 days. *OR*
- **Ciprofloxacin**, 500mg P.O., BID, for 3 days

Prevention and control

1. Case treatment
2. Safe disposal of human excreta and control of flies
3. Safe public water supply
4. Handwashing and sanitary handling of food
5. Control and management of contact cases

4. Amoebiasis (Amebic dysentery)

1) Identification

- A protozoan parasite infection
- exists in 2 forms
 - ❑ the hardy **infective cyst** and
 - ❑ more fragile potentially **pathogenic trophozoite**

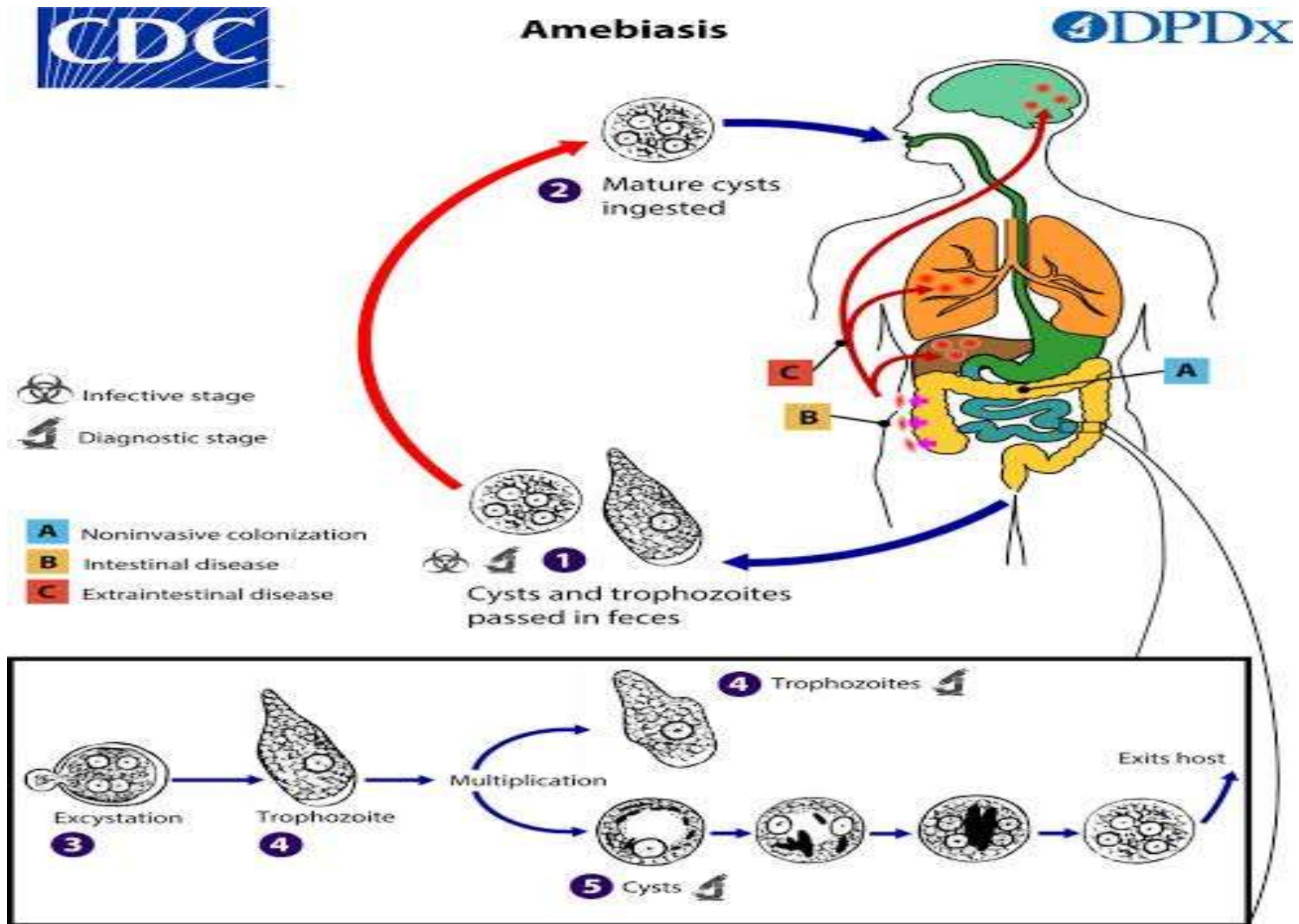
2. Infectious agent: Entamoeba histolytica, a parasitic organism

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- 3) **Reservoir:** Humans, usually a chronically ill or asymptomatic cyst passer
- 4) **Mode of transmission:** ingestion of faecally contaminated food or water containing **chlorine resistant amoebic cysts**
- Patients with **acute amoebic dysentery pose limited danger to others**
 - because **no cysts in dysenteric stools and the fragility of trophozoites**

Pathophysiology

- ❖ *E. histolytica* acquired by the ingestion of infectious viable **cysts** from feces contaminated water, food or hands → release of motile Trophozoite from cysts in *small intestine* → **colonic invasion by trophozoites** → **colitis**
- ❖ Large bowel mucosa invasion with sub mucosal extension → inoculation of amoebas in to the portal system → **distant metastatic abscess in liver /lung/brain**
- ❖ Encystation of trophozoites(Cysts) and/or motile trophozoites passage in stool.



Spectrums of amoebiasis

- 1) asymptomatic carrier state
 - 2) acute amoebic dysentery
 - 3) amoebic liver abscess
 - 4) amoeboma
- Intestinal
amoebiasis

Amoebic dysentery

Intestinal amoebiasis

- **Mostly asymptomatic cyst passage (90%)**
- Symptomatic cases (10%):
- Presentation
 - **Bloody, mucoid diarrhea**
 - **Fever < 40 %, chills**
 - **lower Abdominal pain**
 - **Dysentery**, with 10–12 small-volume, blood- and mucus-containing stools daily

Diagnosis

- Amoebic (**hematophagous trophozoites**) in stool (repeated stool exam increases the yield)
- Mixed WBCs in stool
- Patchy inflammation seen on colonoscopy

Amoebic liver abscess

Presentation

1. **Persisting fever**
2. **RUQ** or epigastric pain and/or **shoulder pain**
3. **Rarely diarrhea**

Diagnosis

1. ultrasound(u/s)
2. raised WBC
3. serology
4. aspirate microscopy
5. response to metronidazole

Diagnosis

- Demonstration of *Entamoeba histolytica* **cyst or trophozoite in stool**

Differential diagnosis

- Other causes of acute diarrhea or bloody stools, particularly bacterial pathogens including
- *Shigella*, *Salmonella*, *Campylobacter*, *Escherichia coli*, *Clostridioides difficile*, and some *Vibrio* species

Treatment

Tissue amebicides

- Reaches high concentration in the blood and tissue after oral or parenteral administration
- **Does not eradicate cyst**
- Metronidazole 500—750 mg PO tid 5–7days
- Tinidazole 2gm PO/day for 3 days with food

Luminal amebicides

- **Poorly absorbed and reach high concentration in the bowel**
- **Activity limited to cysts and trophozoites close to mucosa**
- **Used to treat asymptomatic carriers**
- **Diloxanide** 500 mg po TID for 10 days
- Paramomycin 500mg po TID for 10days
- Lodoqoinol 650mg po TID for 20days

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Extra intestinal amebiasis

- Metronidazole 750mg po TID **PLUS**
- Diloxanide furoate 500mg tid for 10days

Preventive measures

- 1) Educate on personal hygiene, particularly in
 - ☐ sanitary disposal of feces
 - ☐ Hand washing after defecation and before preparing or eating food
- 2) Disseminate information regarding the risks involved in **eating unclean or uncooked fruits and vegetables and in drinking water of questionable purity**

...

3) Protect public water supplies from fecal contamination

- ❖ **Sand filtration of water removes nearly all cysts**
- ❖ Water of undetermined quality can be made safe by boiling for 1 minute (at least 10 minutes at high altitudes)
- ❖ **Chlorination of water as generally practiced in municipal water treatment does not always kill cysts**

4. Treat known carriers-thorough hand washing after defecation to avoid re-infection

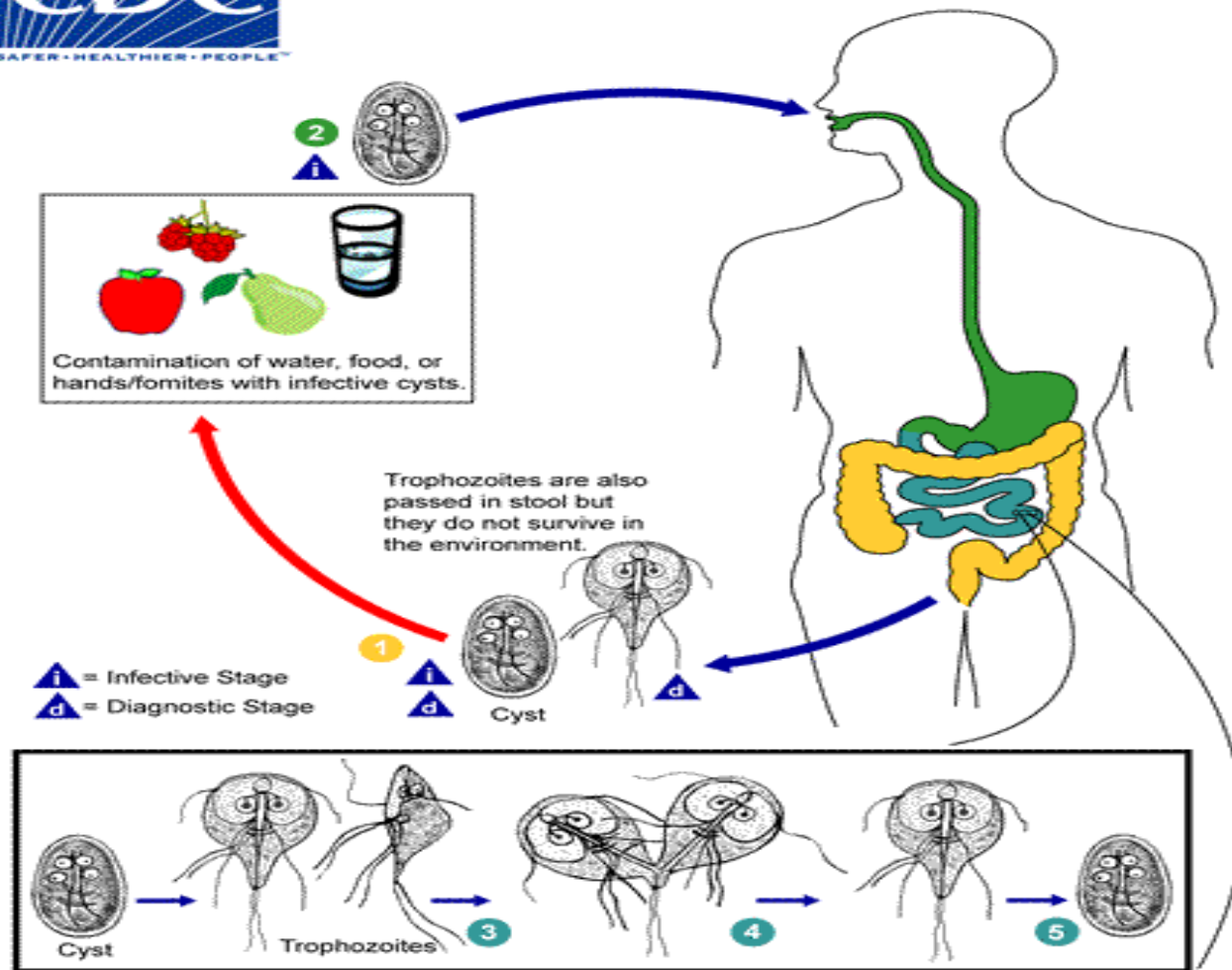
5. Giardiasis (Beaver Fever)

- A protozoan infection principally affects the **upper small intestine**
- **Infectious agent:** *Giardia lamblia*, also known as *G. duodenalis* or *G. intestinalis*
- **Reservoirs:** zoonosis **human** and **found in most mammals** esp. beaver (“**beaver fever**”), cattle, cats, dogs...
- **IP:** usually 3 - 25 days
- **Period of communicability:** Entire period of infection often months
- **Susceptibility and resistance-** Variable. Gastric achlorhydria increases risk of illness. Breast fed infants are protected

Mode of transmission

- ❖ Fecal oral spread: **Ingestion of 10 to 25 cysts** can lead to giardiasis;
- ❖ **Chlorine used in routine water treatment do not kill Giardia cysts**
- ❖ Following cyst ingestion, excystation occurs in the proximal small bowel with release of trophozoites
- ❖ Prevalence **30-40% carrier rate**, more in children, and seasonal (**increase during rainy seasons**).
- ❖ Most common in HIV patients

Life cycle



Clinical manifestations

- **Prominent early symptoms include**

- Abdominal pain,

- **Bloating,**

- **belching,**

- **Flatus,**

- Nausea, vomiting;

- **Frequent, loose, pale, fatty Malodorous stools,**

Symptoms of chronic giardiasis

- Loose stools but usually **not diarrhea**
- **Steatorrhea**
- Profound weight loss (10 to 20 percent of body weight)
- **Malabsorption**
- Malaise
- Fatigue
- Depression
- Abdominal cramping
- **Flatulence**

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Diagnosis:

- ✓ Microscopic examination of fecal specimens
- ✓ Established by identifying *Giardia lamblia* trophozoite or cyst from fecal or duodenal samples.
- ✓ **No leukocyte**
- ✓ **No mucus in stool**

Treatment

Supportive treatment:

- Correction of fluid and electrolyte losses due to the diarrhea

Medical Treatment

- **Asymptomatic cases** usually resolve without treatment without any complication.
- **Symptomatic patients:**
 - **Tinidazole (2 g orally, single dose)** is the first line option for all, 3 years and above individuals.
 - Alternatives
 - **Nitazoxanide, 500 mg PO two times per day for three days**
 - Metronidazole, 250-500mg P.O., TID for five days
 - Albendazole, 400 mg orally once daily for 5 days

Prevention:

- Avoid drinking water or eating foods likely contaminated (E.g. purify drinking water, do not swallow water while swimming, and eat cooked foods at least in risk conditions).
- Strict hand washing with water and soap, **alcohol based hand rubs are effective for trophozoites but poor for cysts**
- Careful diaper disposal and treatment of symptomatic children will prevent spread.
- Breastfeeding is protective against giardiasis in nursing infants in endemic areas

6. Infectious Hepatitis

- Hepatitis is an **inflammation of the liver**
- Hepatitis viruses are the most common cause of hepatitis
- **Hepatitis A and E** are typically cause the acute form of hepatitis.
- **HEV** is a common cause of hepatitis **outbreaks** in developing parts of the world.
- **IP:** 1wks - 64 days
- **Reservoir:** human and animals
- **MOT:** ingestion of contaminated food or water



Basic Features of Hepatitis Viruses

<u>Virus</u>	<u>Transmission</u>	<u>Incubation Period(week)</u>	<u>Chronic Infection</u>
A	fecal-oral	4 (2-6)	No
B	parenteral	8-12 (6-24)	Yes
C	parenteral	6-9 (2-24)	Yes
D	parenteral	(2-10)	Yes
E	fecal-oral	4-5 (2-9)	No

Viral Hepatitis - Overview

	A	B	C	D	E
Source of virus	Feces	Blood*	Blood*	Blood*	Feces
Route of transmission	fecal-oral	Percutaneous	Percutaneous	Percutaneous	fecal-oral
Chronic infection	no	yes	yes	yes	no
Vaccine	yes	yes	no	yes**	no
Pre/Post prophylaxis	pre/post (IG)	post (HBIG)	no	no	no

* Blood and blood-derived body fluids

**Prevention of Hep B with vaccine

...

Clinical manifestation

- Asymptomatic or
- Symptomatic
 - ✓ **Jaundice,**
 - ✓ **Dark urine,**
 - ✓ Extreme fatigue,
 - ✓ Nausea,
 - ✓ Vomiting and
 - ✓ Abdominal pain

DX

- Epidemiologically
- **Serology test for IGM anti-A or E**

Treatment

- Supportive management
- **Rest, high carbohydrate diet with low fat and protein**
- Vaccine

Prevention

- Public education about good sanitation and personal hygiene, with special emphasis on careful hand washing and sanitary disposal of feces.
- Proper water treatment and distribution systems and sewage disposal.
- Proper management of day care centers i-e to minimize possibility of fecal-oral transmission.
- HA vaccine for all travelers to intermediate or highly endemic areas.

2. Feces Mainly in Soil

- The diseases in this category are mainly transmitted through **fecal contamination of soil**
- These infections are acquired **through man's exposure to faecally contaminated soil**

1. Ascariasis

- A helminthic infection of the **small intestine** generally associated with few or no symptoms
- **Infectious agent:** *Ascaris lumbricoides*
- **Reservoir:** Humans; ascarid eggs in soil
- **Incubation period:** 4 - 8 weeks
- **Period of communicability:** As long as mature fertilized female worms live in the intestine
- The prevalence of ascariasis is highest in tropical countries where **warm, wet climates** favor year-round transmission of infection.
- In dry areas, transmission occurs predominantly **during the rainy months**

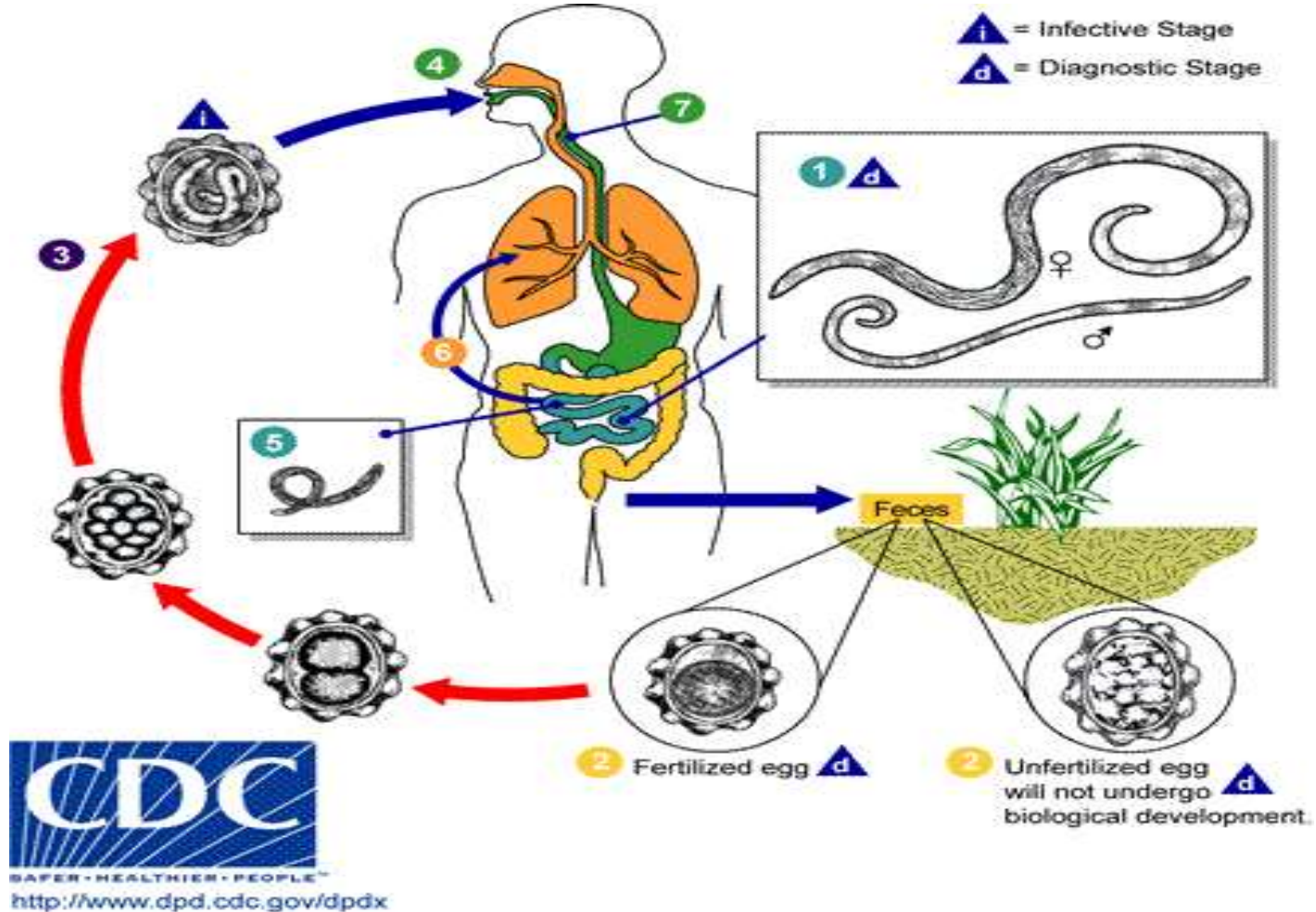
the longitudinal incision of the Reum;



Mode of transmission

- ❖ Ingestion of infective eggs **from soil contaminated with human feces** or
- ❖ From uncooked vegetables contaminated with soil containing infective eggs
- ❖ **But not directly from person to person or from fresh feces**
- ❖ Ingested embryonated eggs
- ❖ heavy infections in children- **ingesting soil**

Life cycle



Transmission

1. Infective eggs ingested in food or from contaminated hands

Human Host

2. Larvae hatch. Migrate through liver and lungs.
3. Pass up trachea and are swallowed
4. Become mature worms in small intestine
5. Eggs produced and passed in feces.

Environment

6. Eggs become infective (**embryonated**) in **soil in 30-40 days.**
7. Infective eggs contaminate the environment.

Clinical Manifestation

- Live worms, passed in stools or from the mouth, anus, or nose
first sign
- Migrant larvae may cause itching, wheezing, dyspnea, fever, cough productive of bloody sputum
- Heavy parasite burdens may aggravate nutritional deficiency
- Abdominal pain may arise from bowel obstruction by a bolus of worms

Diagnosis and Rx

- Microscopic identification of eggs in feces
- Adult worms passed from anus, mouth or nose

Treatment

- First line-options
- Albendazole, 400mg P.O. as a single dose, for children: 1 – 2 years, 200mg as a single dose.
- Mebendazole, 100mg P.O.BID for 3 days or 500mg, once

Alternative (pregnant women)

- Pyrantel pamoate, 700mg P.O. as a single dose

Prevention and control

1. Treatment of cases
2. Educate on latrine construction for sanitary disposal of feces
3. Prevent soil contamination in areas where children play
4. Promote good personal hygiene (hand washing)

2. Strongyloidiasis

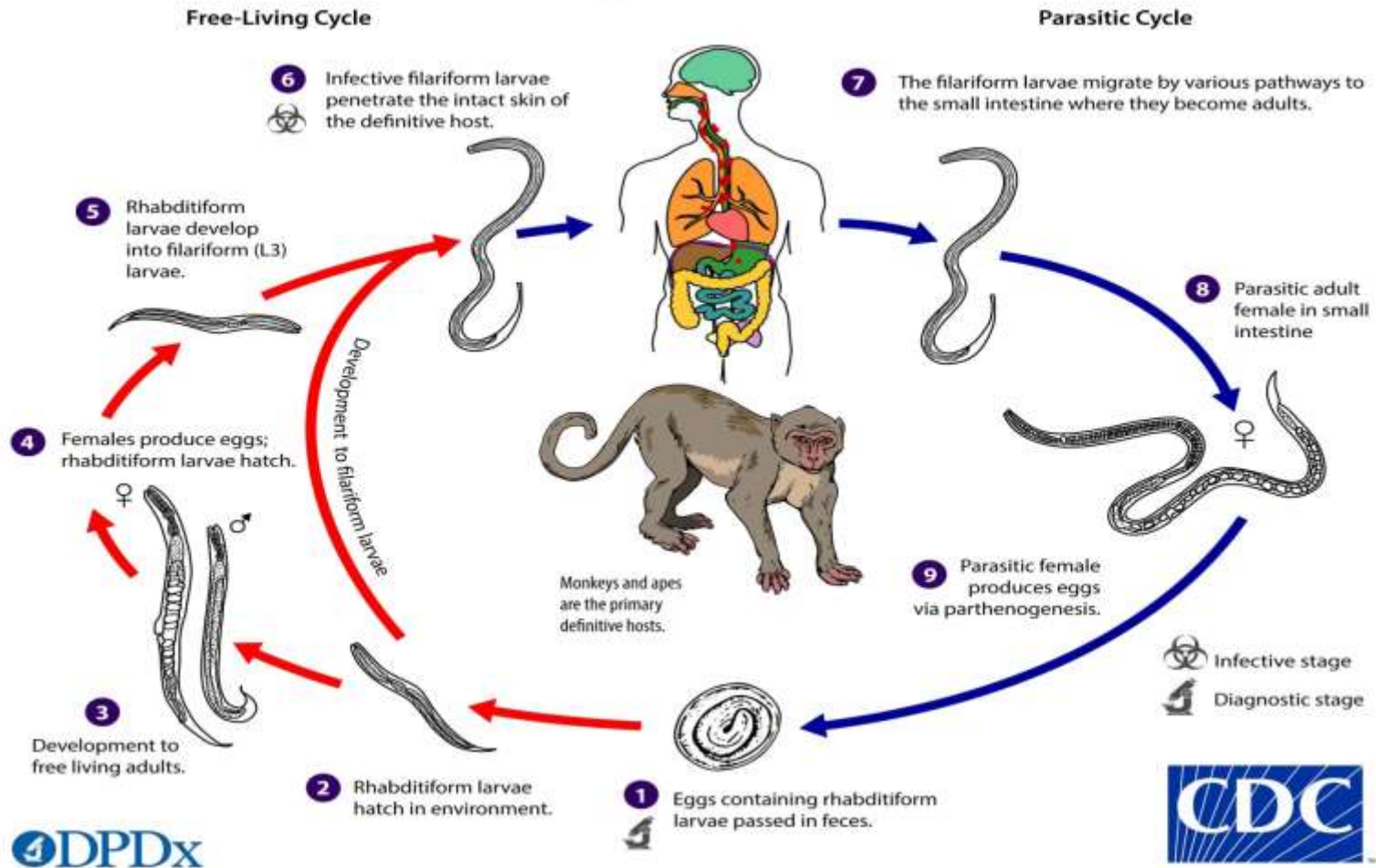
- ✓ An often asymptomatic helminthic infection of the **duodenum** and **upper jejunum**.
- **Infectious agent:** **Strongyloides stercoralis**
- **Period of communicability**-As long as living worms remain in the intestine; **up to 35 years** in cases of **auto-infection**
- **Reservoir**-Human and soil
- **Incubation period:** 2-4 weeks (from skin penetration up to when **rhabditi form** larvae appear in the feces)

Mode of transmission

- Infective (filariform) larvae develop in feces or moist soil contaminated with feces → **Penetrate the skin and enter the venous circulation**
- **Autoinfection-** (rhabiditi form → filariform in the intestine) → penetrate intestinal mucosa or perianal skin

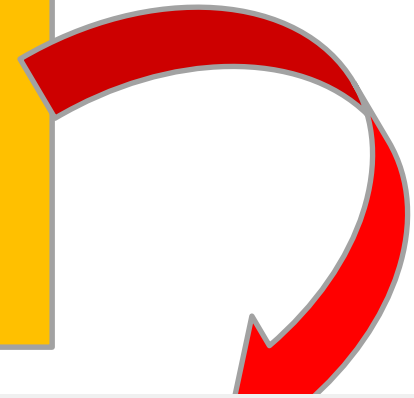
Life Cycle

Strongyloides fuelleborni



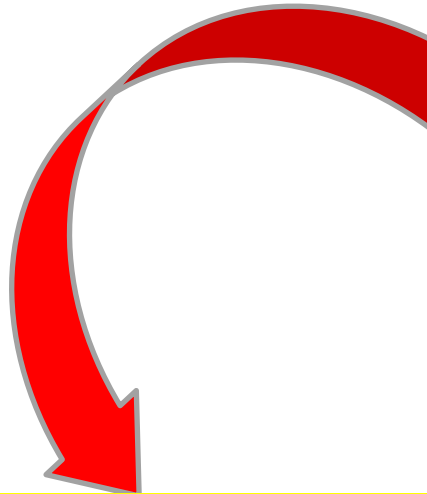
Transmission

1. Infective filariform larvae penetrate skin, e.g. feet. Autoinfection also occurs.



Human Host

2. Larvae migrate, pass up trachea and are swallowed.
3. Become mature worms in small intestine
4. Eggs laid. Hatch rhabditiform larvae in intestine.
5. Rhabditiform larvae:
 - Passed in feces, or
 - **Become filari form larvae in intestine, causing autoinfection.**



Environment

6. In soil larvae become free living worms produce more rhabditi form larvae*

* Free-living cycle can be repeated several times

7. Become infective filari form larvae in the soil

Classic symptoms

- **Upper abdominal pain** (due to duodenitis)
- **Epigastric pain may mimic a duodenal ulcer**, except that food ingestion may aggravate the pain of strongyloidiasis
- **Chronic enterocolitis** and **malabsorption** can result from a high intestinal worm burden
- sometimes nausea, weight loss, vomiting, weakness and constipation
- Intensely **pruritic dermatitis radiating from the anus**

Diagnosis

- Identification of **larvae** in stool specimen

Treatment

First line

- Ivermectin, 200mg/kg daily for 2 days. For disseminated strongyloidiasis, treatment with ivermectin should be extended for at least 5–7 days or until the parasites are eradicated.

Alternatives-options

- Albendazole 400mg P.O.BID for three consecutive days (less effective than ivermectin). OR
- Thiabendazole, 1500mg, P.O. BID, for children: 25mg/kg p.o. for three consecutive days (comparable efficacy to ivermectin).

Prevention and control

1. Proper disposal of human excreta (feces)
2. Personal hygiene including **use of footwear**
3. Examine and treat infected dogs, cats
4. Case treatment

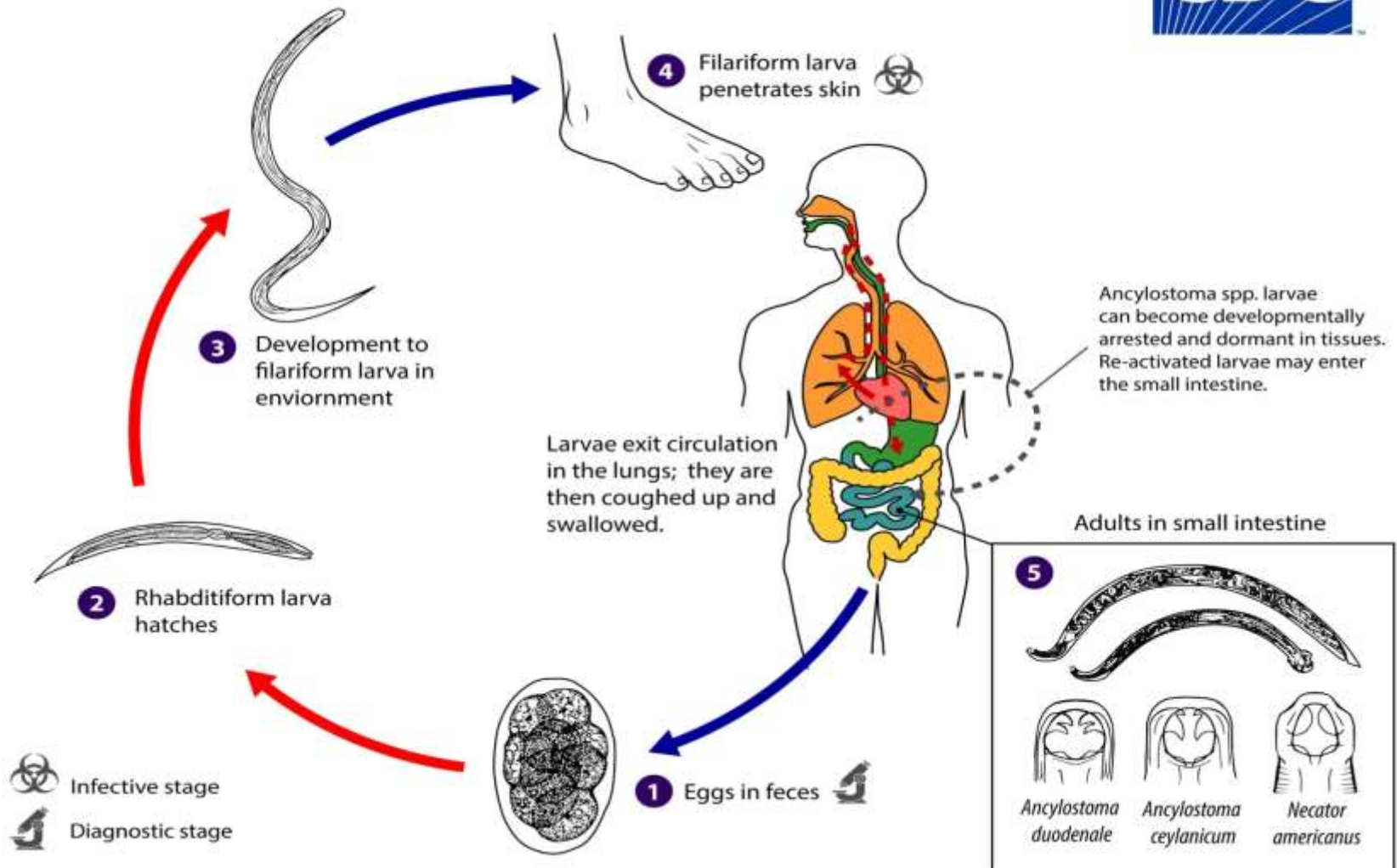
3. Hookworm disease

- **Infectious agent**
 - *Ancylostoma duodenale* and
 - *Necator americanus*
- **Reservoir:** Humans
- **Mode of transmission-**Through skin penetration by the infective larvae
- **Incubation period-** a few weeks to many months depending on intensity of infection and iron intake of the host
- **Period of communicability-** contaminate the soil for several years

Life cycle

4DPDx

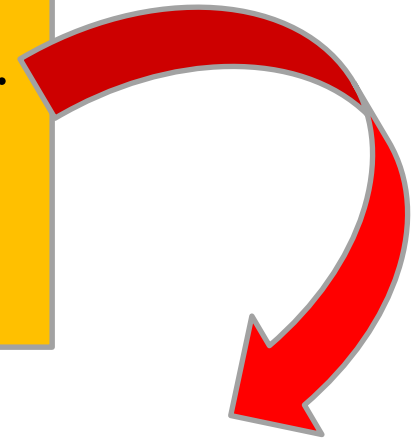
Intestinal Hookworm



Transmission

1. Infective filariform larvae penetrate the skin, e.g. feet.

A. duodenale also transmitted by ingestion of larvae.

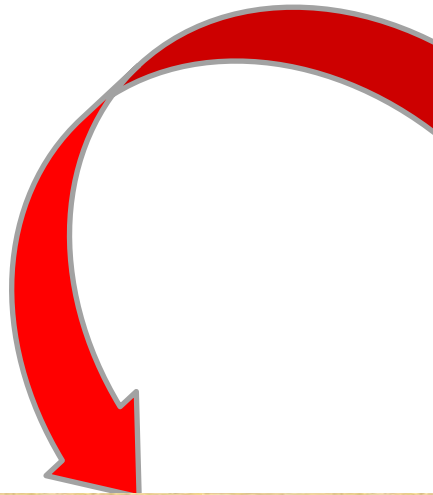


Human Host

2. Larvae migrate. Pass up trachea and are swallowed.

3. Become mature worms in small intestine (attach to wall and suck blood).

4. Eggs produced and passed in feces.



Environment

5. Eggs develop; Rhabditi form larvae hatch. Feed in soil.

6. Develop into infective filariform larvae in about 1 week.

7. Filariform larvae contaminate soil

Clinical Manifestation

The clinical manifestation is related to:

1. Larval migration of the skin

- Produces transient, **localized maculopapular rash**
- associated with itching called **ground itch**

2. Migration of larva to the lungs.

- Produces cough, wheezing and transient pneumonitis

3. Blood sucking

- Light infection-no symptoms
- Heavy infection-result in symptoms of **peptic ulcer disease** like **epigastric pain and tenderness**
- major impact of hookworm infection is on nutritional status
- **The daily losses of blood, iron, and albumin can lead to anemia**

Diagnosis and Rx

- Demonstration of eggs in stool specimen.

Treatment

First line-options

- Albendazole, **400mg P.O. as a single dose (preferred)** OR
- Mebendazole, 100mg P.O. BID for 3 days or 500mg stat

Alternatives:

- Pyrantel pamoate, 700mg P.O. as a single dose

Prevention and control

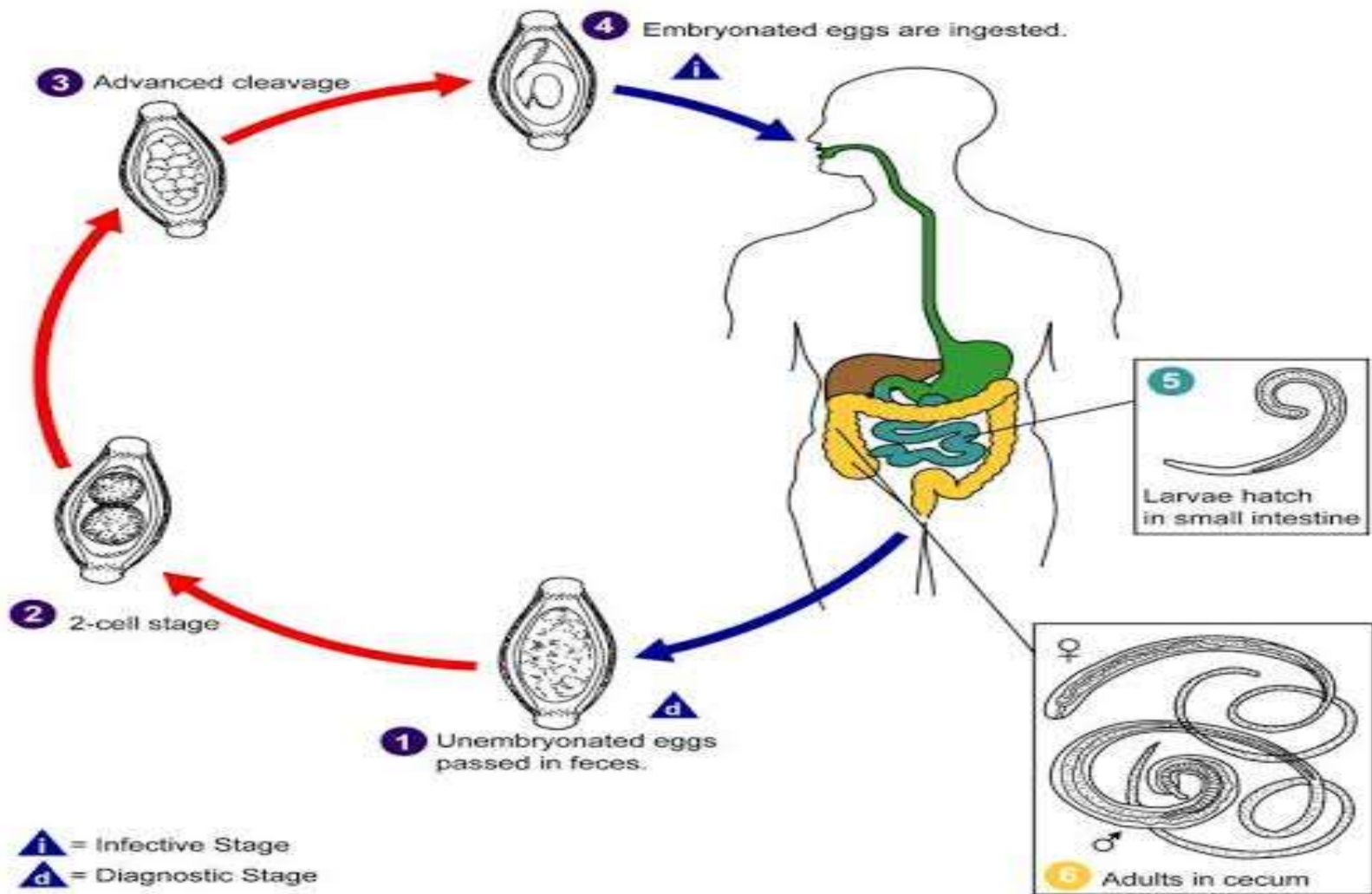
1. Sanitary disposal of feces
2. Properly cleaning and cooking food, hand washing
- 2. Wearing of shoes**
3. Case treatment

Trichuriasis

- A nematode infection of the **large intestine, usually asymptomatic in nature.**
- **Infectious agent:** Trichuris trichuria (**whip worm**)
- **Occurrence:** Worldwide, **especially in warm moist regions. Common in children 3-11 years of age.**
- **Reservoir:** Humans
- **Mode of transmission:** Indirect, **particularly through pica or ingestion of contaminated vegetables. Not immediately transmissible from person to person.**
- **Incubation period:** Indefinite
- **Period of communicability-** Several years in untreated carriers.
- **Susceptibility and resistance-** Susceptibility is universal.

Trichuriasis

(*Trichuris trichiura*)



Transmission

1. Infective eggs ingested in food or from contaminated hands

Human Host

2. Larvae hatch. Develop in small intestine. Migrate to caecum.
3. Become mature worms.
4. Eggs produced and passed in feces

Environment

6. Eggs become infective(embryonated) **in soil after 3 weeks.**
7. Infective eggs contaminate the environment

Clinical manifestation

- Severity is directly related to the number of infecting worms.
- **Most infected people are asymptomatic.**
- Abdominal pain, tiredness, nausea and vomiting, diarrhea or constipation are complaints by patients.
- **Rectal prolapse** may occur in heavily infected very young children.

Diagnosis

- Demonstration of eggs in feces.

Treatment

First line-options

- **Mebendazole, 500mg P.O., single dose** (preferred over Albendazole) OR
- Albendazole, 400mg,P.O. for three days
- Oxantel pamoate 15 to 30 mg/kg (if available) **plus**
- Albendazole 400 mg on consecutive days is superior than other therapies

Prevention and control

1. Sanitary disposal of feces
2. Maintaining good personal hygiene (**i.e. washing hands and vegetables and other soil contaminated foods**)
3. **Cutting nails especially in children**
4. Treatment of cases.

Guinea Worm Infection



Guinea Worm Infection

- An infection of the **subcutaneous and deeper tissues** by large nematode.
- **Infections agent:** *Dracunculus medinensis*, a nematode
- **Epidemiology Occurrence:** In Africa (16 countries south of the Sahara) and in Asia (India and Yemen) especially in regions with dry climates. Local prevalence varies greatly. In some locales, nearly all inhabitants are infected, in others, few, mainly young adults.
- **Reservoir-** Humans

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➤ **Mode of transmission:**

- **Larvae discharged by the female worm into stagnant fresh water** are ingested by minute crustacean copepods (Cyclops species).
- In about 2 weeks, the larvae develop into the infective stage.
- People swallow the infected copepods in drinking water **from infested stepwells and ponds.**
- The larvae are liberated in the stomach, cross the duodenal wall, migrate through the viscera and become adults.
- The female, after mating, grows and develops to full maturity, then migrates to the subcutaneous tissues (**most frequently of the legs**).

➤ **Incubation period- About 12 months**

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- **Period of communicability:** From rupture of vesicle until larvae have been completely evacuated from the uterus of the gravid worm, **usually 2-3 weeks.**
- In water, the larvae are infective for the copepods for about 5 days.
- After ingestion by copepods, the larvae become infective for people after 12-14 days at temperatures $>25^{\circ}\text{C}$ and remain infective in the copepods for about 3 weeks.
- **Susceptibility and resistance-** Susceptibility is universal. No acquired immunity; multiple and repeated infections may occur in the same person.

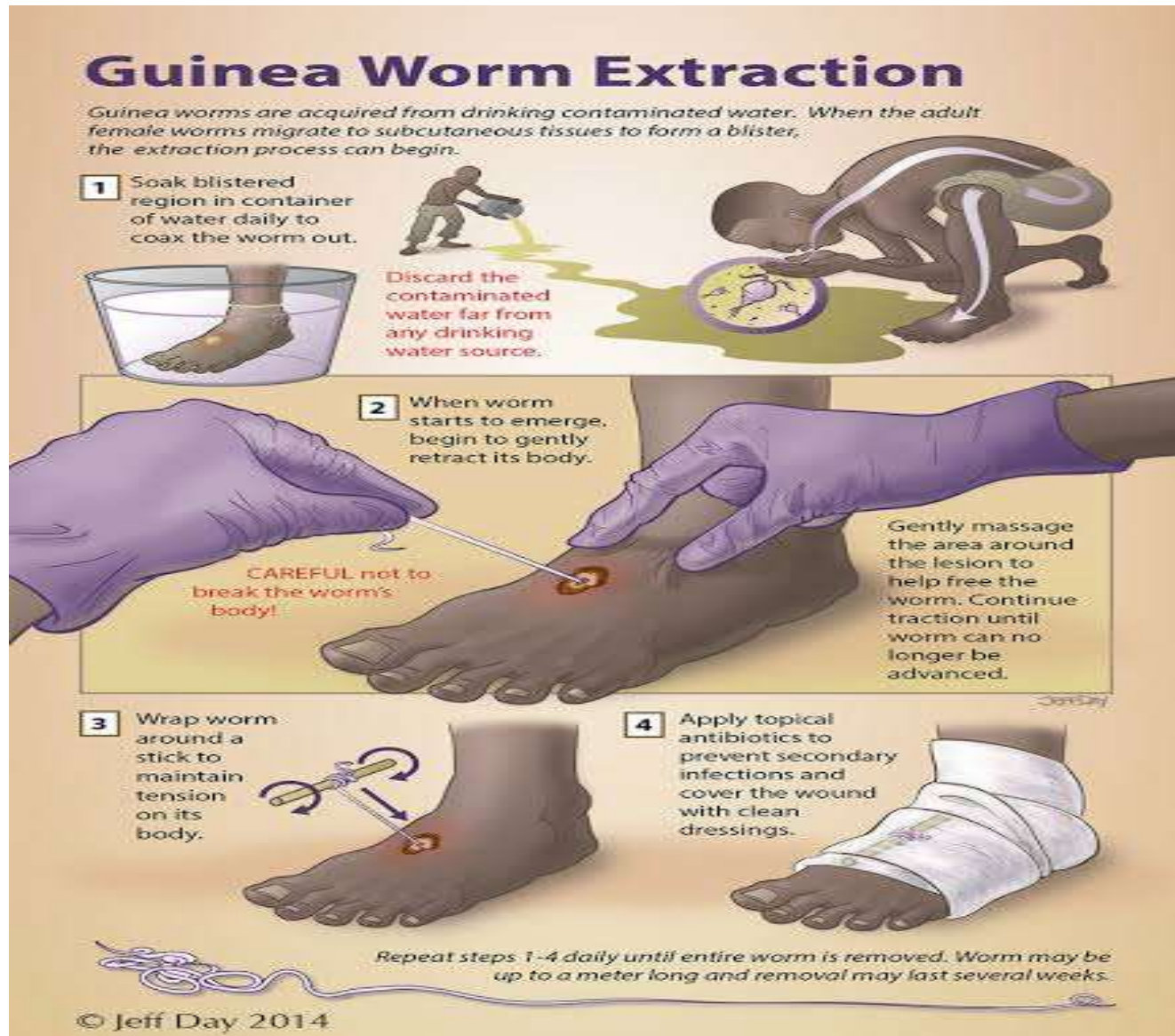
Clinical Manifestation

- Few or no clinical manifestations are evident **until just before the blister forms.**
- Fever and generalized allergic symptoms, including periorbital edema, wheezing, and urticaria.
- The emergence of the worm is associated with local pain and swelling.
- When the blister ruptures, the adult worm releases larvarich fluid and this is associated with a relief of symptoms.
- The shallow ulcer surrounding the emerging adult worm heals over weeks to months.

Diagnosis

- Based on clinical and epidemiological grounds

Treatment



Treatment

- Gradual extraction of the worm by winding of a few centimeters on a stick each day remains the common and effective practice.
- Worms may be excised surgically.
- Administration of thiabendazole or metronidazol may relieve symptoms but has no proven activity against the worm.

Prevention and control

- Provide health education programs in endemic communities to convey three messages:
- The guinea-worm infection **comes from their drinking water**
- Villagers with blisters or ulcers **should not enter any source of drinking water and**
- That drinking water should be **filtered through fine mesh cloth** to remove copepods
- Provision of safe drinking water

Entrobiasis

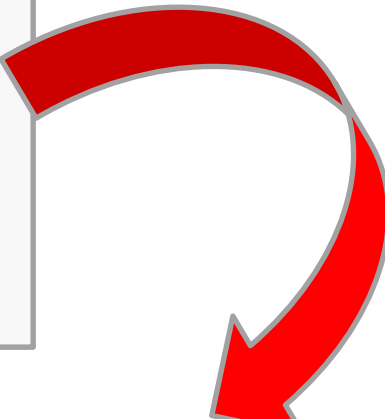
(Oxyuriasis, pinworm infection)

- A **common intestinal helminthic infection** that is often asymptomatic.
- **Infectious agent:** *Entrobis vermicularis*
- **Epidemiology Occurrence:** Worldwide, affecting all socio-economic classes with high rates in some areas.
- Prevalence is **highest in school-aged children**, followed by preschools and is lowest in adults except for mothers of infected children.
- Prevalence is often high in domiciliary institutions.
- Infection usually occurs in more than one family member

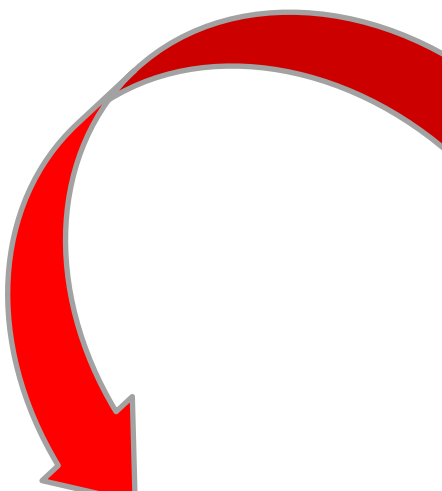
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- **Reservoir:** Human
- **Mode of transmission-** **Direct transfer of infective eggs** by hand from anus to mouth of the same or another person or indirectly through clothing, bedding, food or other articles contaminated with eggs of the parasite.
- **Incubation period-** 2-6 weeks
- **Period of communicability:** As long as gravid females are discharging eggs on perianal skin. **Eggs remain infective in an indoor environment for about 2 weeks.**
- **Susceptibility and resistance-** Susceptibility is universal

Larvae hatch in duodenum → migrate to caecum → within 2-3 weeks Adult worms in Caecum



Gravid females migrate through the anus to the perianal skin and **deposit eggs (usually during the night)**

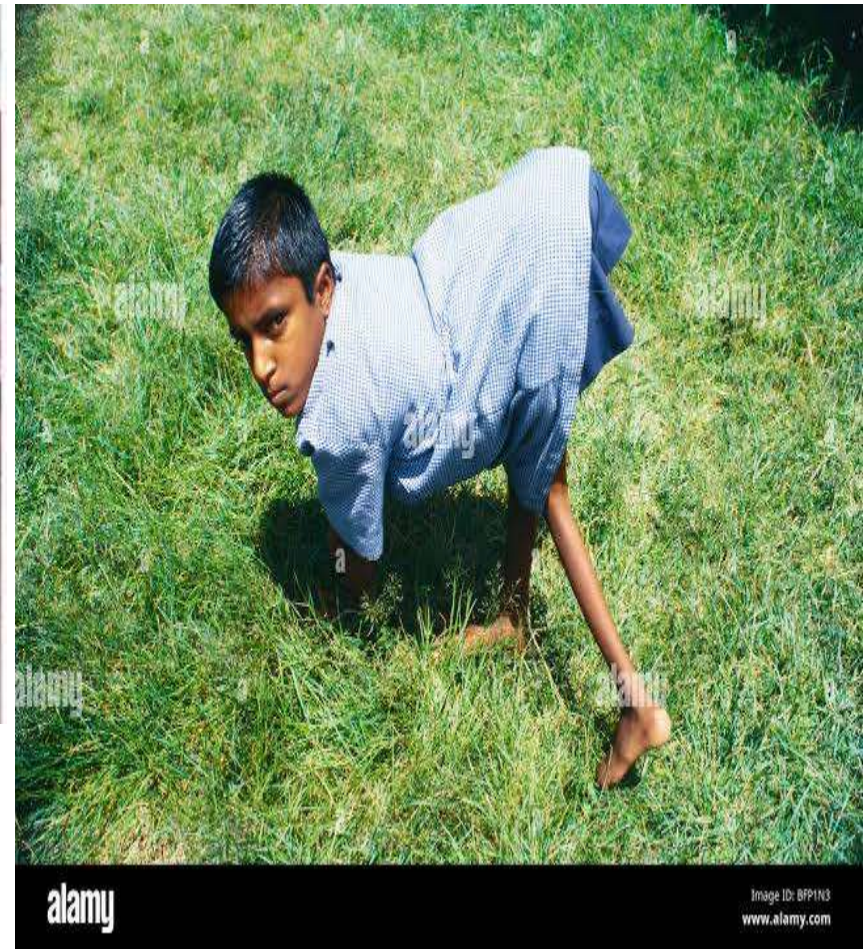


Eggs become infective in a few hours in perianal area → Ingestion of eggs by Man

Clinical manifestation

- **Perianal itching, disturbed sleep, irritability** and some times secondary infection of the scratched skin.
- **Diagnosis:** Stool microscopy for eggs or female worms.
- **Treatment:** Mebendazole.
- **Prevention and control**
 - ✓ Educate the public about hygiene (i.e. hand washing before eating or preparing food,
 - ✓ **Keeping nails short and discourage nail biting).**
 - ✓ Treatment of cases
 - ✓ **Reduce overcrowding in living accommodations.**
 - ✓ Provide adequate toilets.

Poliomyelitis



Poliomyelitis

- A viral infection most often recognized by the **acute onset of flaccid paralysis**.
- **Infectious agent:** Polio viruses (type I, II and III)
- **Epidemiology Occurrence:** Worldwide prior to the advent of immunization. Cases of polio occur both sporadically and in epidemics. Primarily a disease of infants and young children. 70-80% of cases are **less than three years of age**. More than 90% of infections are unapparent. Flaccid paralysis occurs in less than 1% of infections.
- **Reservoir** – humans, especially children
- **Mode of transmission-** **Primarily person-to-person, spread principally through the fecal-oral route**. In rare instances, milk, food stuffs and other materials contaminated with feces have been incriminated as vehicles.
- **Incubation period-** commonly 7-14 days
- **Period of communicability** – not precisely known, but transmission is possible as long as the virus is excreted.

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- **Susceptibility and resistance:** Susceptibility is common in children but paralysis rarely occurs. **Infection confers permanent immunity**

Clinical manifestation

- Usually asymptomatic or **non-specific fever** is manifested in 90% of cases.
- If it progresses to major illness, severe muscle pain, stiff neck and back with or without flaccid paralysis may occur.
- **Paralysis** is asymptomatic and occurs within three to four days of illness.
- The legs are more affected than other part of the body.
- **Paralysis of respiratory and swallowing muscles is life-threatening.**

Diagnosis

- Based on clinical and epidemiological grounds

Treatment

- Symptomatic

Prevention and control

- Educate public about the advantage of immunization in early childhood.
- Trivalent live attenuated vaccine (OPV) at birth.
- Safe disposal of human excreta (feces)



Chapter Five

Air born diseases

Air born diseases

- Diseases transmitted by dissemination of the **microbial agent by air** to a suitable portal of entry; usually respiratory tract
 - When a patient or carrier **talks, coughs, or laughs... fluid droplets are discharged**
1. **Smallest of these droplets remain in the air** for some time and inhaled by new host
 - These 1-5microns are quite easily drawn in to the lungs and retained there
 2. **Droplets with larger size fall on ground**
 - Dry and mix with dust
 - When the pathogens able to survive drying, they become airborne by wind
 - Spread more easily when overcrowding

Measles

- An acute **highly communicable viral d/se**
 - ❖ Especially **contagious during catarrhal stage before rash appears**
- It is a severe disease caused by the rubeola virus, which damages **epithelial surfaces and the immune system**
- It **increases susceptibility** to other infectious agents such as **pneumococcus, H.influenzae and S.aureus**
- Maximum incidence **6mo—5yrs old children**
- Attacks are **commonest during winter and spring**

Mode of Transmission

- **Primarily** by respiratory **droplets over** short distances
- less commonly, by **small-particle aerosols** suspended in the air for long periods
- **by direct contact with nasal secretions**
- >94% herd immunity to interrupt community transmission
- Spreads easily in overcrowded areas
- The IP is **10 days to fever** onset and **14 days to rash** onset
- **PC: From 4 days before rash onset to 4 days after rash onset**
- But minimal after the **2nd day of rash**

Pathogenesis

- Infection is initiated when the virus is deposited on epithelial cells in the **respiratory tract, oropharynx, or conjunctivae**
- During, **the first 2–4 days after infection, virus proliferates** locally in the respiratory mucosa and spreads to draining lymph nodes
- **Virus then enters the bloodstream** in infected leukocytes (primarily monocytes)
- Produces the primary viremia that disseminates infection throughout the **reticuloendothelial system**
- Further replication results in secondary viremia that begins 5–7 days after infection
- This **disseminates measles** virus throughout the body
- Replication of the virus together with the host's immune response, is responsible for the signs and symptoms

Clinical features

Case definition:

The disease has **prodromal** period during which:

- The **lymphoid tissue become enlarged**
- **High fever(39.5—40°C)**

By 2nd day of prodromal period

- Coryza (rhinitis and sore throat)
- Cough
- Conjunctivitis with photophobia
- Runny nose
- Koplik spots (**small, irregular, bright—red spots** with blue—white center points on **the buccal mucosa**)

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By the 4th day of fever

- **Deep-red** rash beginning on the face; **hair borderline of forehead, behind the ear, and back of the neck appears**
- **This red rash- fades on pressure**
- **Spreads cephalocaudally and centrifugally**
- involving the neck, upper trunk, lower trunk, and extremities within 3- 4 days
- It is maculopapular, but not itchy
- After several days, the rash typically **turns from red to brown**(do not fade on pressure)
- After 5-6 days, rash completely fades



CONSIDER MEASLES

In patients presenting with febrile rash illness and clinically compatible measles symptoms (cough, coryza, and conjunctivitis)

Ask patients about recent travel internationally or to domestic venues frequented by international travelers, as well as a history of measles in the community.

www.cdc.gov/media/releases/2013/s0101-measles.html



Complications

- Unimmunized <5 children and infants are at highest risk of measles and its complications
 - ❖ **Pneumonia (most common cause of death)**
 - ❖ **Diarrhea** (dysentery and persistent diarrhea)
 - ❖ **Corneal ulceration**
 - ❖ Malnutrition
 - ❖ **Otitis media, mastoiditis**
 - ❖ Febrile convulsion
 - ❖ AGE in children <2yrs
 - ❖ **Blindness**

Treatment

A. Nondrug treatment

- **Tepid sponging and administration of paracetamol**
- Keep the patient well hydrated
- treat diarrhoea with ORS
- Observe the patient closely for complications

B. Drug treatment

- Vitamin A
 - 50,000 IU for <6mo
 - 100,000 IU 6-12mo
 - 200,000 IU >1yr po single dose

Complicated Cases

- Admit cases
- Treat pneumonia, malnutrition, diarrhea
- Treat eye infection with TTC eye ointment
- Treat mouth ulcer with gentian violet(GV)

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- Maintain an adequate protein–calorie intake by giving frequent and small meals
- Continue breastfeeding
- Provide supplementary feeding if available
- Do not admit patients to general feeding centers until after the infectious period

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Use antimicrobials only when indicated

- A. Documented complications such as pneumonia, otitis media and **dysentery**

- B. Children at significant risk of secondary bacterial infection(e.g. severe malnutrition, HIV or xerophthalmia)
 - A broad-spectrum antimicrobial such as ampicillin or co-trimoxazole

Preventive measures

- 1) Public education to encourage immunization for all infants at 9 month of age
- 2) Early response to measles outbreaks
- 3) Immunization of all children (less than 5 years of age) who had contact with infected children
- 4) Initiate measles **vaccination at 6 months of age during epidemic and repeat at 9 months of age**

Acute Bacterial Meningitis

- Acute Bacterial meningitis is an inflammation of the meninges in response to bacterial infection
- It is mainly caused by **N. meningitides, S. pneumoniae, and H. influenzae**
- The only bacterial pathogen with a potential to cause **epidemics is N. meningitidis** and therefore is a reportable disease.
- Ethiopia is one of the countries in the so called "meningitis belt" of the Sub-Saharan Africa which spans from Gambia in the West to Ethiopia in the East of Africa
- One striking feature of the epidemic has been its seasonality by which it tends to occur during the **dry and windy season** of the years between January and May.

Mode of transmission

- The bacteria that cause meningitis are transmitted from person-to-person **through droplets of respiratory or throat secretions from carriers.**
- Close and prolonged contact – such as **kissing, sneezing or coughing** on someone, or living in close quarters with an infected person, facilitates the spread of the disease.
- The average incubation period is 4 days but can range between **2 and 10 days.**

Clinical Manifestation

- ✓ Sudden onset of fever, **intense headache**, nausea and **often vomiting**
- ✓ Neck stiffness
- ✓ Photophobia (frequently)
- ✓ Change in mental status(defined as GCS <14)
- ✓ **Kernig's sign may be positive**
- ✓ **Brudinski's sign may be positive**
- ✓ Delirium and coma often appear
- ✓ Skin rash: maculopapular, petechial, or purpuric

Diagnosis

Based on clinical and epidemiological grounds

- **Blood cultures before antibiotics are administered**
- **WBC count: $>10,000$ in 83% of bacterial meningitis**
- Lumbar puncture- CSF analysis
- **CSF Gram stain has 60–90% Sensitivity**
- **Culture has 70–85% Sensitivity if LP done prior to antibiotic administration**
- **Rule of 2s:**
 - **CSF WBC $> 2,000$,**
 - **Glucose < 20 &**
 - **Total protein > 200 has $>98\%$ Specificity for bacterial meningitis**

Treatment

- Acute bacterial meningitis is a medical emergency
- Institute empiric antimicrobial therapy promptly
- **Objectives**
 - **Rapidly clear the organisms from the CSF**
 - **Prevent acute complications and long term sequelae**

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A. Community acquired, bacterial etiology unknown

- First line: **Ceftriaxone**, 4 g/day , I.V., divided in 2 doses for 10-14 days PLUS
- Where Penicillin resistance is common (particularly *S.pneumoniae*), Empiric treatment should include:
Vancomycin, 1gm IV BID for 10-14days

B. Empiric treatment, hospital acquired meningitis

- **First line**
 - **Vancomycin, 15mg/kg IV Q8h PLUS**
 - **Ceftazidime, 2 g IV q8h**

Prophylaxis for contacts-N. meningitides

- Rifampin (oral) Adults 600 mg every 12 hours Two days
- Ciprofloxacin (oral) Adults 500 mg Single oral dose
- Ceftriaxone (intramuscular) Children <15 years 125 mg Single IM dose
 - Older children and adults **250 mg** Single IM dose
- Azithromycin 500 mg Single oral dose

Prevention and control measures

- 1) Educate the public on the need to reduce **direct contact and exposure to droplet infection**
- 2) Reduce overcrowding in living quarters and workplaces
- 3) Vaccines containing groups A, C, Y and W-135 meningococcal polysaccharides
- 4) Implementing mass vaccination campaign**
5. Chemotherapy of cases
6. Chemo prophylaxis
7. Report for concerned health authorities

Tuberculosis(Tb)

- caused by an organism called *Mycobacterium tuberculosis*, a rod-shaped bacillus with “acid-fast”
- Occasionally caused by *Mycobacterium bovis* and *Mycobacterium africanum*
- usually affecting the lungs—*pulmonary TB*
- **Pulmonary TB is the most common type**
- TB is a major cause of morbidity & mortality

Transmission and pathogenesis of TB

- main source of TB infection is a person with **active pulmonary TB disease**
- Usually transmitted through the **air by droplet nuclei** that are produced when a person with pulmonary especially smear positive or laryngeal TB coughs, sneezes, talks, sings and laughs
- Can transmitted by **ingestion of raw milk from cattle affected by M. Bovis**
- A sneeze liberates **4,500 to 1 million organisms at once**

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- On inhalation, the droplet nucleus ($< 5 \mu\text{ms}$ in size) reaches an alveolus –TB infection
- **Bacilli may spread to other parts via the blood stream, the lymphatic system or directly**
- Tuberculosis that affects organs other than the lungs such as lymph nodes, bones & joints is called *extra pulmonary TB*
- **Patients with extra pulmonary TB are usually not infectious**

Symptoms of TB disease

- General symptoms of TB disease (*PTB or EPTB*)
 - Weight loss
 - Fatigue
 - Malaise
 - Fever
 - Night sweats
 - Loss of appetite

Symptoms of PTB disease

- Coughing, Coughing up sputum or blood
- Pain in the chest when breathing or coughing
- Dyspnoea (shortness of breaths)
- ***Symptoms of extra-pulmonary TB disease:***
 - Depend on the body part affected by the disease

Diagnosis

1. Sputum smear examination

- Smear microscopy has variable sensitivity (40 - 60%), the sample examined **must contain at least 5000-10,000 bacilli** per ml in order to be positive on microscopy
- The specificity of smear microscopy is high (99%)
- Three sputum specimens (**SPOT-EARLY MORNING-SPOT**) need to be collected and examined in two consecutive days

2. Sputum culture

- a highly sensitive diagnostic method which permits detection of a minimum of **10 to 100 viable bacilli** in the volume of cultured materials (**usually 0.1 ml**)
- Allows diagnosis of less infectious cases
- The specificity of culture $> 99\%$
- **gold standard in mycobacterial detection** because of its higher sensitivity and specificity

3. GeneXpert MTB/RIF

- Is a **new rapid DNA test for TB** which is fully automated
- Simultaneously detects TB and Rifampicin drug resistance (a proxy indicator for MDR-TB) from sputum sample
- Provides accurate results in 100 minutes— immediate treatment

4. Radiology (Chest x-ray)

- For the diagnosis of pulmonary TB, X-ray is sensitive but less specific
- *Chest X-ray is supportive evidence to microscopy!*
- ***It is a major error to omit sputum examination and diagnose TB based on Chest X-ray only***

5. Histopathology

- Is also supportive diagnostic methods where applicable
- Can be needle aspirate or biopsy
- For extra pulmonary tuberculosis and commonly for **tuberculous lymphadenitis**

Classification of cases

- TB cases are classified according to the site
 - **pulmonary (80-85%)**
 - **extra-pulmonary (15% -20%)**
- **Pulmonary (PTB)** cases are further classified depending on the sputum AFB microscopy
 - **sputum-smear-positive PTB** (75-80% of PTB cases)
 - **sputum smear negative PTB** (20-25% of PTB cases)

Definition of types of patient

A. New patient

- New patients **have never had** treatment for TB, or have taken anti-TB drugs for **less than 01 month**

B. Previously treated

- Previously treated patients have **received 1 month or more** of anti-TB drugs in the past

C. MDR

- TB patient with mycobacterial strain resistant to the first line anti-TB drugs **at least to Isoniazid & Rifampicin**

Drugs used for treatment of TB

- The drugs used for the TB treatment are safe and effective if properly used
- The following drugs are used as **first line treatment of TB** in Ethiopia:
 - Rifampicin(R)
 - Ethambutol (E)
 - Isoniazid (H)
 - Pyrazinamide (Z) and
 - Streptomycin (S)
- Streptomycin is administered by injection while the other anti TB drugs are to be taken orally

Treatment Regimens

- Has 2 phases— intensive and continuation phases
- New –**2(RHZE)/ 4(RH)**
- Previously treated adults cases: **2(RHZE) / 4(RH)**
- The following extrapulmonary forms of Tb require prolonged continuation phase
 - CNS (TB meningitis or Tuberculoma)
 - Bone or joint TB (Vertebral(TB spondylitis), joint and osteomyelitis),
 - o Regimen (a total of 12 months : 2months intensive phase and 10 months continuation phase); **2RHZE/10RH**

Preventive measures

- 1) Promptly identify, diagnose and treat potentially infectious patients with TB disease
- 2) Provision of drugs and facilities for early and complete treatment of cases
- 3) Educate the public in mode of spread and methods of control and regarding the importance of early diagnosis and continued adherence to treatment
4. Reduce or eliminate those social conditions that increase the risk of infection
5. Set up TB prevention and control programs in institutional settings where health care is provided
6. Preventive chemotherapy with isoniazid for 6 mo to prevent progression of latent TB infection to TB disease
- 7. BCG immunization**

Leprosy (Hansen's disease)



Leprosy (Hansen's disease)

- A chronic bacterial disease of the **skin, peripheral nerves** and, in lepromatous patients, the **upper airway**
- **Infectious agent:** *Mycobacterium leprae*
- **Epidemiology:** Although common in rural tropics and subtropics, **socio-economic conditions** may be more important than climate itself. Endemic in south and southeast Asia, tropical Africa and Latin America.
- **Reservoir:** Humans

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- **Mode of transmission:** Not clearly established. Household and **prolonged close contact** appear to be important.
- Millions of bacilli are liberated daily in the nasal discharges of untreated lepromatous patients. Cutaneous ulcers in lepromatous patients may shed large number of bacilli.
- **Organisms probably gain access (entrance) through the URT** and possibly through broken skin.
- In children less than one year of age, transmission is presumed to be transplacental.

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- **Incubation period-** **9 months to 20 years.**
- **Period of communicability-** Infectiousness is lost in most instances within 3months of continuous and regular treatment with dapsone or clofazamin or **within 3 days of rifampicin treatment.**
- **Susceptibility and resistance:** The presence and form of leprosy depend on the ability to develop effective cellmediated immunity.

Clinical Manifestation

- Clinical manifestations vary between two polar forms: **lepromatous** and **tuberculoid** leprosy.

Lepromatous (Multibacillary form):

- Nodules, papules, macules and diffused infiltration are bilaterally symmetrical and usually numerous and extensive.
- Involvement of the nasal mucosa may lead to crusting, obstructed breathing and epistaxis.
- Ocular involvement leads to iritis and keratitis.

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Tuberculoid (Paucibacillary form):

- **Skin lesions are single or few, sharply demarcated**, anesthetic or hyperesthetic and bilaterally symmetrical.
- Peripheral nerve involvement tends to be severe.
- **Borderline** Has features of both polar forms and is more liable to shift toward the lepromatous form in untreated patients and toward the tuberculoid form in treated patients.

Diagnosis

- Complete skin examination (hyperesthesia, anesthesia, paralysis, muscle wasting or trophic ulcer which are signs of peripheral nerve involvement) with bilateral palpation of peripheral nerves (ulnar nerve at the elbow, peroneal nerve at head of fibula and the great auricular nerve) for enlargement and tenderness.
f
- **Skin lesion are tested for sensation** (light touch, pink prick, temperature discrimination).
f
- **Demonstration of AFB** in skin smears made by scraped incision method.
f
- **Skin biopsy** confined to the affected area should be sent to the experienced pathologists in leprosy diagnosis.

Classification of cases

Multibacillary (MB) Leprosy:

- Usually presents with **multiple (>5) poorly defined**, hypopigmented or erythematous lesions associated with hypoesthesia.
- **six or more skin lesions**. Or
- Slit skin smear result **positive for AFB**, irrespective of the number of skin lesion. Or
- If there is **involvement (enlargement) of more than one nerve**.

...

Paucibacillary (PB) leprosy:

- Presents with one or few (**usually <5**) **hypopigmented** and hypoesthetic lesions.
- **Only one nerve trunk** enlarged and
- Slit skin **smear negative for AFB.**

Pure neural leprosy

- These are patients who **do not have any skin lesion**, but who have **clearly thickened nerves with or without signs of nerve damage.**
- Patients with pure neural leprosy should be reported and treated as a MB case

Treatment

Treatment of Multibacillary Leprosy (MB-MDT regimen):

- Three drug regimen for 12 months: Rifampicin, Clofazimine and Dapsone (12 MDT blister packs).
- **Rifampicin**, 10 mg/kg (600mg P.O., for adults), once-monthly, supervised for 12 months.

PLUS

- **Dapsone**, 2 mg/kg (100 mg P.O., for adults), daily self-administered for 12 months

PLUS

- **Clofazimine**, 6mg/kg (300mg P.O., for adults) monthly, supervised and 1mg/kg (50mg for adults) daily, self-administered for 12 months.

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Treatment of Paucibacillary leprosy

- **Rifampicin**, 10mg/kg/day body weight (600mg/day for adults).
Supervised once-monthly x 6 months
PLUS
- **Dapsone** 2 mg/kg (100mg /day for adults), daily self-administered X 6 months

Pertusis (whooping cough)

- An acute bacterial disease of tracheo-bronchial **tree**
- Chinese name for pertussis is "**the 100-day cough**"
- Infectious agent: **Bordetella pertusis**
- Most dangerous in children <1year old
- **Youngest non-immunized children most affected**
- >90% attacks in <6yrs olds
- **Most have coughing w/c lasts up to 8wks**

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Reservoir: Humans

Mode of transmission

- **Primarily** by **direct contact with discharges** , probably by **droplets**
- Indirectly by handling objects freshly solid with nasopharyngeal secretions
- Incubation period: 1-3 weeks

Clinical feature

- The Chinese name for pertussis is "the 100-day cough," **which accurately describes** the clinical course of the illness
- Pertussis is a **prolonged coughing illness** with clinical manifestations that vary by age
- **Insidious** onset with 3 phases

...

A. The initial **catarrhal** stage

- ✓ an irritating cough
- ✓ Sneezing, lacrimation
- ✓ Rhinorrhea/coryza
- ✓ low-grade fever
- ✓ Cough becomes gradually becomes paroxysmal
- ✓ Lasts 1-2 weeks

B. Paroxysmal phase

- C:\Users\Addisu Computer\Downloads\Video\Whooping Cough in an Adult - NEJM.mp4
- Cough becomes more frequent and spasmodic
- **Repetitive bursts of 5–10 coughs without intervening inhalation**
- **Posttussive vomiting is frequent**
- A mucous plug occasionally expelled at the end of an episode
- The episode may be terminated by **an audible whoop**, which occurs upon rapid inspiration against a closed glottis at the end of a paroxysm
- **Cough is prolonged, paroxysmal and sleep-disturbing**, looks healthy b/n paroxysms but increasing fatigue
- During a spasm, **there may be impressive neck-vein distension, bulging eyes, tongue protrusion, and cyanosis**
- **More severe attack at night, no fever**

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C. Convalescent phase

- After 2–4 weeks, the coughing episodes become less frequent and less severe
- Within 1—2 weeks cough diminish and is characterized by gradual resolution of coughing episodes
- ~~f~~After improvement the disease may **recur**

Complications

- Most complications result from increased intrathoracic pressure generated by severe fits of coughing
- **Subconjunctival hemorrhages**
- **Abdominal and inguinal hernias**
- Pneumothoraces, and
- Pneumonia
- **weight loss and rib fracture**

Diagnosis

- History and physical examination at paroxysmal phase
- Marked lymphocytosis (as high as 20,000 to 30,000)

Treatment

A. Non—drug treatment

- Bed rest until the paroxysm subsided
- Avoid cigarette, dust, strenuous activity that initiate coughing episodes
- Infants may need admission

B. Drug treatment

- Helps to treat infection in phase 1 but to decrease transmission in phase 2
- Antibiotics for super—infection
 1. Erythromycin 500mg(8mg/kg) po tid for 7 days
 2. Azithromycin 500mg po/d for 5 days
 3. Antibiotics for super infections like pneumonia because of bacterial invasion due to damage to cilia

Prevention and control

1. Educate the public about the dangers of whooping cough and the advantages of initiating immunization at 6 weeks of age
2. Consider protection of health workers at high risk of exposure by using erythromycin for 14 days

Influenza

- An acute viral disease of the respiratory tract
- **Infectious agent**
Three types of influenza virus (A,B and C)
- **Epidemiology:** In pandemics, epidemics and localized outbreaks.
- **Reservoir-** Humans are the primary reservoirs for human infection.
- **Mode of transmission-** Airborne spread predominates among crowded populations in closed places such as school buses.
- **Incubation period-** short, usually 1-3 days
- **Period of communicability-** 3-5 days from clinical onset in adults; up to 7 days in young children.
- **Susceptibility and resistance-** when a new sub-type appears, all children and adults are equally susceptible. Infection produces immunity to the specific infecting agent.

Clinical Manifestation

- Fever, head ache, myalgia, prostration, sore throat and cough
- Cough is often severe and protracted, but other manifestations are self-limited with recovery in 2-7days

Diagnosis

Based on clinical ground

Treatment

1. Same as common cold, namely: Anti-pain and antipyretic
- High fluid intake
 - Bed rest
 - Balanced diet intake

Prevention and control

- Educate the public in basic personal hygiene, especially the danger of unprotected coughs and sneezes and hand to mucus membrane transmission.
- Immunization with available killed virus vaccines may provide 70-80% protection.
- Amantadine hydrochloride is effective in the chemoprophylaxis of type A virus but not others



Chapter 6

VECTOR-BORNE DISEASES

Introduction

- **Vectors** are **living organisms** that can transmit infectious diseases between humans or from animals to humans
- Vectors **includes insects** (mosquitoes, ticks, fleas, black flies and sand flies)
- Many of these vectors are bloodsucking insects
- They ingest MOs during a blood meal
- Later inject it them into a new host

1. Malaria

- Protozoan parasites-genus **Plasmodium**
- Five plasmodium species develop in humans
 - ☐ P. falciparum (**life-threatening malaria**)
 - ☐ P. vivax
 - ☐ P. ovale
 - ☐ P. malariae
 - ☐ P. knowlesi
- Persistent liver forms; relapses after cured-hypnozoites
- The **commonest causes of Malaria in Ethiopia** are **P. Falciparum and P. vivax**
- P.falciparum is the commonest species throughout the tropics
- Up to 80–90% cases in sub-Saharan African are due to P. falciparum

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- **Reservoir:** Humans; except *P.malariae*, the African apes and some South American monkeys
- **Mode of transmission:** bite of an infective **female**
Anopheles mosquito
- Most species feed at ***night***; some at ***dusk*** or in the early ***morning***
- **Transfusion of blood** and occasionally from **mother to fetus**

Pathogenesis

- The mosquito **ingests** blood containing gametocytes
- Male and female gametes unite in the mosquito stomach to form an ookinete
- This penetrates the stomach wall to form a **oocyst** on the outer surface
- Thousands of sporozoites develop inside
- **Sporozoites** penetrate the wall of the oocyst, reach the salivary glands
- During mosquito bite sporozoites travel via bloodstream of the host to the **liver, invade hepatocytes**
- Divide many 1000-fold until mature tissue schizonts are formed

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- Each schizont contains thousands of daughter merozoites
- This exoerythrocytic stage is **asymptomatic**
- hepatocytes rupture and asexual parasites(merozoites) invade the erythrocytes
- Within RBCs the parasites digest red cell proteins, primarily hemoglobin
- new daughter merozoites are released from the infected erythrocytes
- **Most** released merozoites continue in the **asexual cycle- causes symptoms**
- **a few** differentiate into male or female gametocytes (sexual forms) which cause no symptoms
- They circulate in the bloodstream

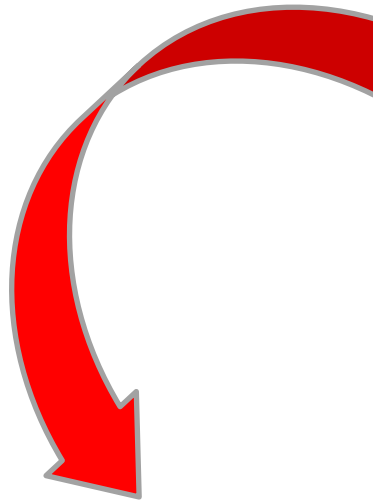
A. Mosquito

1. Mosquito suck blood from patient (gamatocyte)
2. Male and female gamatocytes unite → ookinete → oocyte
3. Oocytes → sporozoites



Liver and tissue

4. Mosquito injects sporozoites
5. Sporozoites reach liver and invade hepatocytes
6. Inside hepatocytes, sporozoites → schizonts
7. schizonts → merozoites (**Asexual phase**)
8. Merozoites goes to RBCs



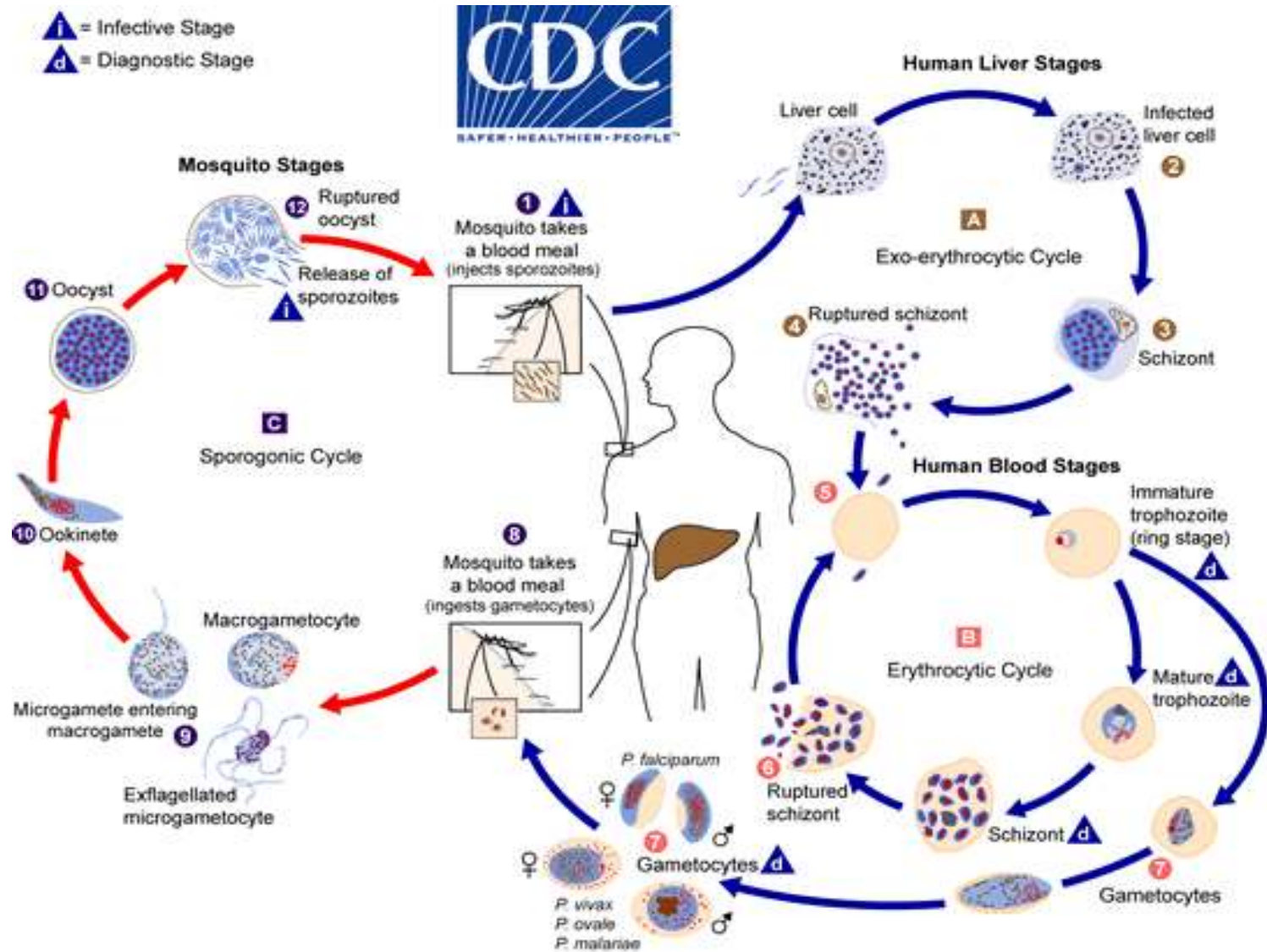
Erythrocyte

9. New merozoites digest Hgb → **symptoms**
10. Few merozoites → gamatocytes (**sexual phase**)
- 11. Gametocytes sucked by mosquitos**

Incubation periods

- 9 – **14 days** for *P. falciparum*
- 12–18 days for *P. vivax* and *P. ovale*
- 18 – 40 days for *P. malariae*

Life cycle



Period of communicability

- Humans infective-**gametocytes** in the blood
 - varies with parasite species and with response to **therapy**
- **Untreated or insufficiently treated patients**
 - several years in **malariae**
 - up to 5 years in vivax
 - generally not more than 1 year in falciparum malaria
- The mosquito **remains infective for life**

Clinical Manifestation-**uncomplicated malaria**

- **Fever, Chills, Rigors, Sweating**
- Severe Headache, Generalized body and joint pain
- Nausea and or vomiting, Loss of appetite, Abdominal pain (especially in children)
- Irritability and refusal to feed (in infants), flu-like symptoms, fever above 38°C,
- **Splenomegaly,**
- **Pallor**

Clinical features of **complicated malaria**

- Inability to take in fluids (or breast milk in children)
- Repeated vomiting
- **Circulatory collapse or shock**
- **Dark or 'cola-coloured' urine**
- **Spontaneous unexplained heavy bleeding**
- Difficulty in breathing, Tachypnoea, cyanosis
- Generalized weakness, inability to walk or sit without assistance
- Sleepiness, change of behavior
- Repeated generalized convulsions
- Altered consciousness: Confusion, delirium, convulsions, coma
- **Oliguria, renal failure**
- Hypoglycemia
- **Severe anaemia (Hb < 6 g/dL)**
- **Hyperpyrexia (axillary temperature >38.5°C)**

Diagnosis

- Blood film (thick or thin smear)
- Rapid diagnostic tests (**RDT**)
- In highly endemic areas of Africa, people gradually develop immunity in adults
 - the clinical picture - treatment decisions
- In non-immune populations- all parasitaemias treated

Treatment

- First line- *P.falciparum* or mixed– uncomplicated
 - **Artemether + Lumefantrine**, 20mg + 120mg in a fixed dose combination
- Treatment of uncomplicated *P. Vivax* malaria
- First line
 - **Chloroquine** phosphate, 1 g, then 500 mg in 6 hours followed by 500 mg p.o QD for 2 days or
 - 1 g at 0 and 24 hrs followed by 0.5 g at 48 hrs P.O
 - Followed by **Primaquine**, 15mg base P.O.QD for 14 days

Treatment of complicated malaria

- Administer drugs parenterally to ensure adequate blood-serum concentrations
- First line
 - **Artesunate**: 2.4mg/Kg IV or IM given on admission (time = 0), then repeat at 12 hours, and 24 hours, then once a day for up to 5 days

Prevention and control measures

- Rapid diagnosis and effective case management – important in reducing malaria transmission.
- Insecticide-treated mosquito nets (ITN)
- Permethrin-sprayed blankets, sheets
- Permethrin-treated outer clothing worn in the evening or in bed
- Indoor residual spraying of insecticide (“house spraying”)
- Environmental control

2. Lymphatic Filariasis

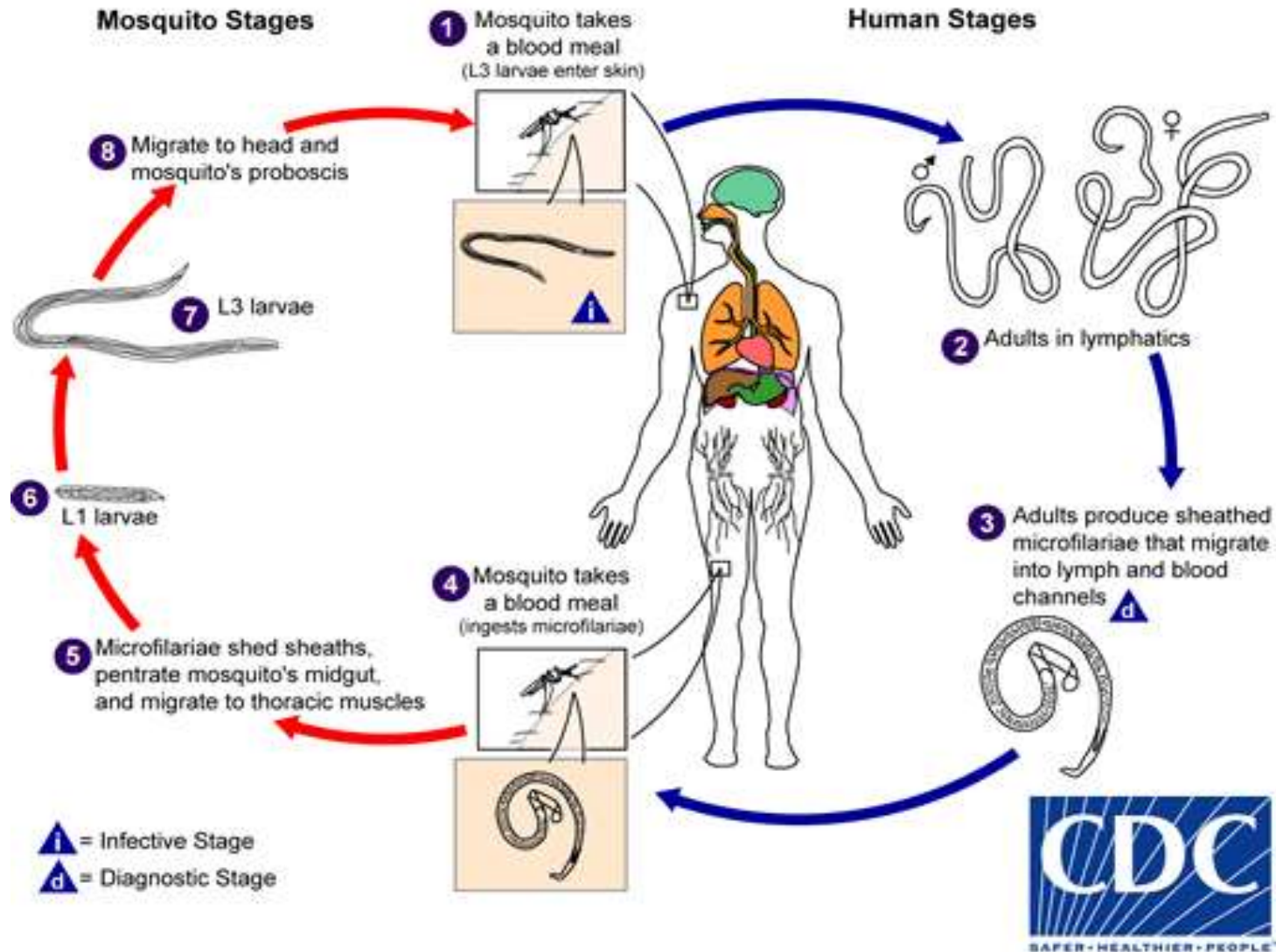
- A disease caused by **the reaction of the body to worms in the lymphatic system**
- **Agents**
 - **Wuchereria bancrofti**
 - Brugia malayi
 - Brugia timori

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- **Reservoir:** Humans are definitive hosts
- **Mode of transmission:** **by bite of mosquito** harboring infective larvae
- **Incubation period-** **one month**
- **Period of communicability:** Not directly transmitted from person to person
 - Humans may infect mosquitoes when microfilaria are present in the peripheral blood

Life cycle

Wuchereria bancrofti



Clinical Manifestation

Three phases may be distinguished

A. Acute phase

- Starts within a **few months** after infection
- **Toxic products of living or dead worm** → Inflammatory manifestations: **due to hypersensitivity reaction**
- Lymphadenopathy (neck and axilla)
- Fever, chills
- **Eosinophilia**

B. Sub--acute phase

- This occurs **after about one year**
- **Microfilaria: peripheral blood**
- Reactions to the adult worms cause attacks of fever with **lymphangitis or Epididymitis**
- Recurrent attacks lead to **hydrocele**
- Lesions associated with hypereosinophilia

C. Chronic phase

- Lymph vessels and glands: **obstructed →** as a result **lymph edema develops**
- Lymph edema mostly in **legs or scrotum (elephantiasis)**
- Hard, thick, rough and fissured skin
- Worms **died**, no microfilariae in the blood



Diagnosis

- Obstructive signs with history and residence in endemic areas
- microfilariae in the peripheral blood (blood film)

Treatment

- a) Diethyl carbamazoin Citrate (DEC) 8 to 10 mg/kg for two to three weeks for filarial infections
- b) Ivermectin (200 mcg/kg), albendazole
- c) Doxycycline
- d) Refer the patient for surgical treatment of hydrocele

Prevention

- Reducing the vector population
- Mass and selective treatment
- Personal protection against mosquito bite

Flea-Borne Diseases

Endemic Typhus (**Flea-borne typhus**)

- resembles louseborne typhus, but is **milder**

Infectious agent

- *Rickettsia typhi*

Reservoir: Rats and mice

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Mode of transmission:

- **Infective rat fleas** defecate **rickettsiae** while sucking blood
- this contaminates the bite site and other fresh skin wounds

Incubation period:

- From 1 to 2 weeks, commonly 12 days.

Period of communicability

- Not directly transmitted from person to person
- Once infected, **fleas remain for life (up to 1 year) → transfer it to their progeny**

Clinical Manifestation

- Prodromal symptoms of **headache**, myalgia, arthralgia, nausea, and **malaise** developing 1 to 3 days before the **abrupt onset of chills and fever**
- **Rash** is present in only 13% of patients
- Pulmonary involvement: **non-productive cough and pneumonia**

Diagnosis

- Epidemiological ground
- Weilfelix agglutination test (Serology)

Treatment

1. Doxycycline-adults 100 mg orally twice a day or
2. Chloramphenicol 50 mg/kg per day orally in four divided doses for **seven days**

Prevention and control

1. Destroy rats from burrows and harborages.
2. Use insecticides to abolish flea
3. Treatment of patients

Louse-Borne Diseases

A. Epidemic Typhus

- is a potentially lethal, louse-borne disease caused by **Rickettsia prowazekii**
- The principal vectors are
 - *Pediculus humanus corporis* (**the body louse**) and
 - *P. humanus capitis* (**the head louse**)

Mechanisms of transmission

- During a blood meal, the louse defecates highly **infective feces** at the site of its feeding → skin irritation at the site of a louse bite
- Rickettsiae in feces then introduced into **abraded skin**
- Lice feces may remain infectious for as long as 100 days
- Transmission occur from sharing clothes or transfer of **feces**

Symptoms

- The majority experience the **abrupt onset of fever** and **severe headache**
- Cough
- Tachypnea
- Muscle tenderness
- Abdominal tenderness
- Arthralgias

Diagnosis

- In a suspected epidemic laboratory testing to confirm the diagnosis for the first cases
- After confirmation clinical
- Serological techniques are used

Treatment

- The drug of choice is **doxycycline**
- tetracycline and chloramphenicol are effective agents for epidemic typhus

Prevention

- **Delousing**: eradication of human infestation with lice using permethrin powder 0.5%
- Antibiotic prophylaxis: chloramphenicol or tetracycline
- Vaccination
- Clothing and bedding that have not been used should also be treated

B. Relapsing Fever

- Relapsing fever occurs in two major forms: tick-borne (TBRF) and louse-borne (LBRF)
- Caused by spirochetes of the genus **Borrelia**
- characterized by **recurrent episodes of fever** that accompanies spirochetemia

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- **LBRF** (Louse-borne relapsing fever) is caused only by **B. recurrentis**
- **TBRF** caused by **B. duttoni** other causes possible
- The vector of *B. recurrentis* is the body louse, which only feeds on humans
- Humans are probably the only **reservoir** of *B. recurrentis*
- ***B. recurrentis* is currently endemic in Ethiopia and Sudan**
- The highlands region of Ethiopia may have hundreds to thousands of cases of LBRF annually
- The highest incidence during the rainy season when the poor gather together in shelters
- Lice move from one person to another,

Mode of transmission

- Acquired by crushing an infected louse → so that it contaminates the bite wound by hemo-lymph fluid
- **Incubation period:** 5-10 days usually 8 days

Clinical Manifestations

- Presents with the **sudden onset of fever**, punctuated by an intervening afebrile period
- The temperature may be as high as 43°C and is **usually above 39°C**
- First episode of fever is **unremitting for three to six days**
- Headache, neck stiffness, arthralgia, myalgia, and nausea may accompany the first and subsequent episodes
- **A non-productive cough**

Diagnosis

- Characteristic **fever pattern**, especially if the recurrent fevers are accompanied by the **crisis** phenomenon- Jirish Heximer reaction
- Giemsa or Wright stain (blood film)

Treatment

- Penicillins and tetracyclines are the antibiotics of choice

Prevention

1. Control of vectors (louse)
2. Personal hygiene
3. Health education about hygiene and modes of disease transmission
4. Delousing of patient's clothes and his/her family
5. Chemotherapy of cases and Chemoprophylaxis for contacts

Snail-Borne Diseases

Schistosomiasis

- referred to as “**bilharziasis**”
- Human schistosomiasis is caused by five species of the parasitic trematode genus *Schistosoma*:
- Infection may cause considerable morbidity in the **intestines, liver, and urinary tract**
- The **intestinal** species *S. mansoni*, *S. japonicum*, *S. mekongi*, and *S. intercalatum* and
- The **urinary** species: *S. haematobium*.

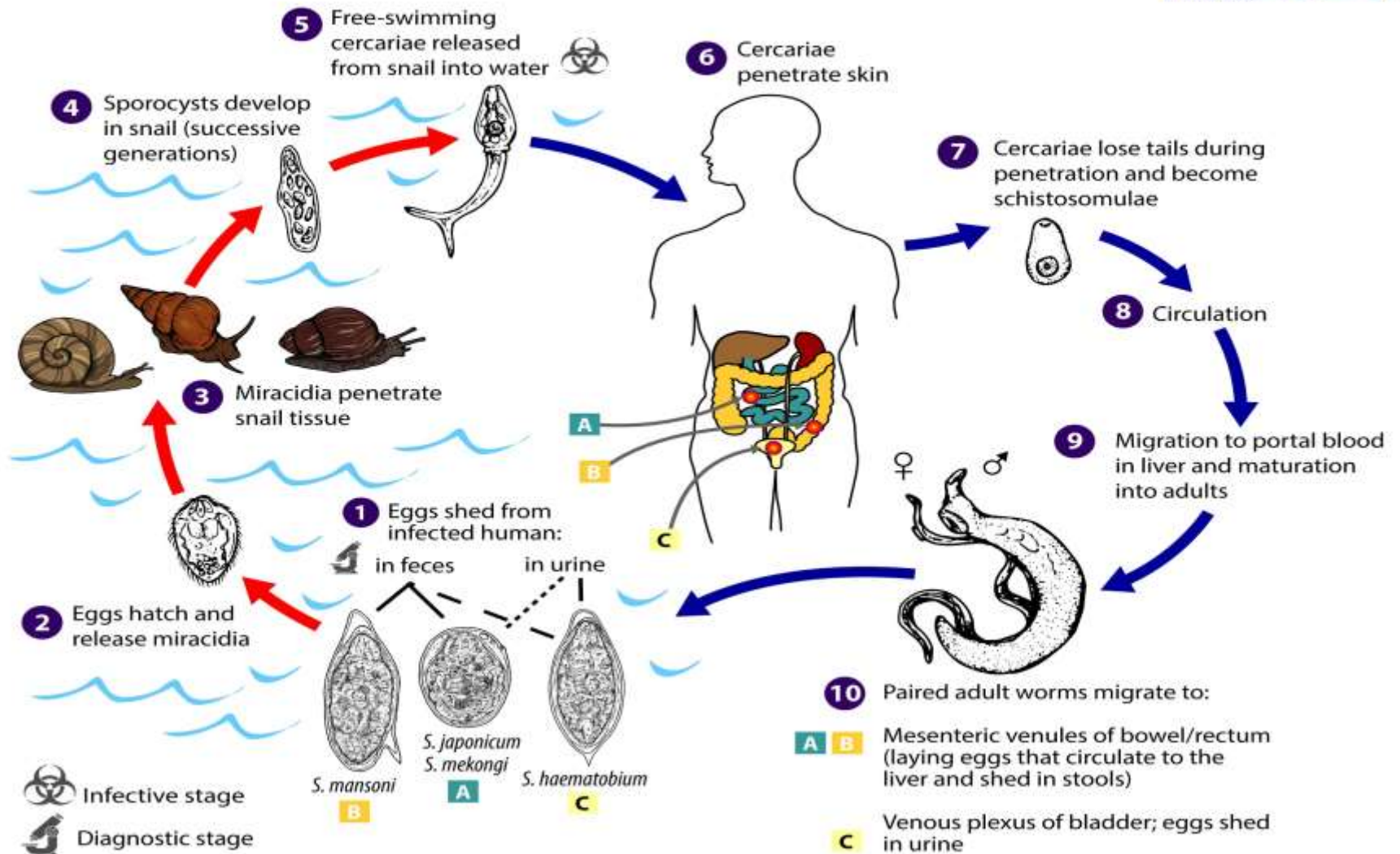
Mode of transmission

- Human infection is initiated by **penetration of intact skin with infective cercariae**
- These organisms are released from infected snails in **freshwater bodies**
- Cercariae → schistosomula → adult → eggs → faeces → miracidia → Cercariae
- Schistosomula begin their migration within 2–4 days via venous or lymphatic vessels, **reaching the liver parenchyma**.
- Sexually mature worms descend into the venous system at specific anatomic locations:
 - intestinal veins (*S. mansoni*, *S. japonicum*, *S. mekongi*, and *S. intercalatum*) and
 - vesical veins (*S. haematobium*).
- After mating, adult gravid females deposit **their ova intravascularly**.

Life cycle



Schistosoma spp.



Clinical Features

- Schistosomiasis occur in three stages, which vary not only by species but also by intensity of infection

1. During the phase of cercarial invasion:

- a **form of dermatitis** may be observed
- This so-called **swimmers' itch** occurs most often with *S. mansoni* and *S. japonicum* infections
- manifests **2 or 3 days after invasion**
- It has an itchy maculopapular rash on the affected areas of the skin
- Cercarial dermatitis is a self-limiting clinical entity.
- During worm maturation and at the beginning of oviposition **(i.e., 4–8 weeks after skin invasion) acute schistosomiasis starts**

2. Acute schistosomiasis or Katayama fever manifesting with

- **fever, generalized lymphadenopathy**, and hepatosplenomegaly may develop
- Individuals with acute schistosomiasis show a high degree of peripheral-blood **eosinophilia**
- Parasite-specific antibodies may be detected before schistosome eggs are identified in excreta
- The course of acute schistosomiasis is generally benign

3. Chronic schistosomiasis

- are species-dependent
- Intestinal species (*S. mansoni*, *S. japonicum*) cause **intestinal and hepatosplenic disease** as well as several manifestations associated with **portal hypertension**.
- During the intestinal phase symptomatic patients characteristically have **colicky abdominal pain, bloody diarrhea, and anemia**
- Patients may also report **fatigue and an inability to perform daily** routine functions
- may also show evidence of **growth retardation**
- The severity of intestinal schistosomiasis is often related to the intensity of the worm burden

- The hepatosplenic phase of disease manifests early (during the first year of infection, particularly in children) with
 - Liver enlargement due to parasite-induced granulomatous lesions
 - Hepatomegaly is seen in 15–20% of infected individuals
 - It correlates roughly with intensity of infection, occurs more often in children
 - In subsequent phases of infection, presinusoidal blockage of blood flow leads to portal hypertension and splenomegaly

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- Moreover, **portal hypertension** may lead to **varices** at the **lower end of the esophagus**
- Patients with schistosomal liver disease may have **right-upper-quadrant "dragging" pain during the hepatomegaly phase**
- this pain may move to the **left upper quadrant as splenomegaly progresses**
- Bleeding from esophageal varices may, however, be the first clinical manifestation of this phase

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- Patients may experience repeated bleeding but seem to tolerate its impact, since an adequate total hepatic blood flow permits normal liver function for a considerable duration
- In late-stage disease, typical fibrotic changes occur along with
 - **liver function deterioration**
 - the **onset of ascites**
 - **hypoalbuminemia**
 - **defects in coagulation**

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- The clinical manifestations of *S. haematobium* infection occur relatively early and involve a high percentage of infected individuals
- Up to 80% of children infected with *S. haematobium* have **dysuria, frequency, and hematuria**, which may be terminal
- Urine examination reveals **blood and albumin**
- **Unusually high frequency of bacterial urinary tract infection**

Diagnosis

- **Microscopy:** Demonstration of parasite eggs in **stool or urine**

Treatment :

- All patients with evidence of infection should be treated **regardless of symptoms**
- **Praziquantel** : The treatment of choice for **all** species
- 40 mg/kg in one or two doses
- But *S. japonicum* needs **60 mg/kg** in two or three doses at least three hours apart

Prevention

- Water sanitation programs, mass treatment and vaccine development
- Proper disposal of human feces and urine
- **Avoid swimming in water bodies known to have the infection**

Leshmaniasis

- Vector-borne disease caused by genus leishmania, intracellular protozoa
- Vectors – *Phlebotomus* and *Lutzomyia*
- Zoonotic form, **dogs** as main reservoir in the Mediterranean basin, China, the Middle East, and South-America; cause is ***L. infantum***
- The anthroponotic form, **humans** as reservoir, is caused by ***L. donovani*** ; prevalent in East Africa and the Indian subcontinent

Major forms:

- **Visceral:** *L. donovani*, *L. infantum*
- **Cutaneous:** *L. aetopica*, *L. major*, *L. tropica*
- **Mucocutaneous:** *L. aetopica*

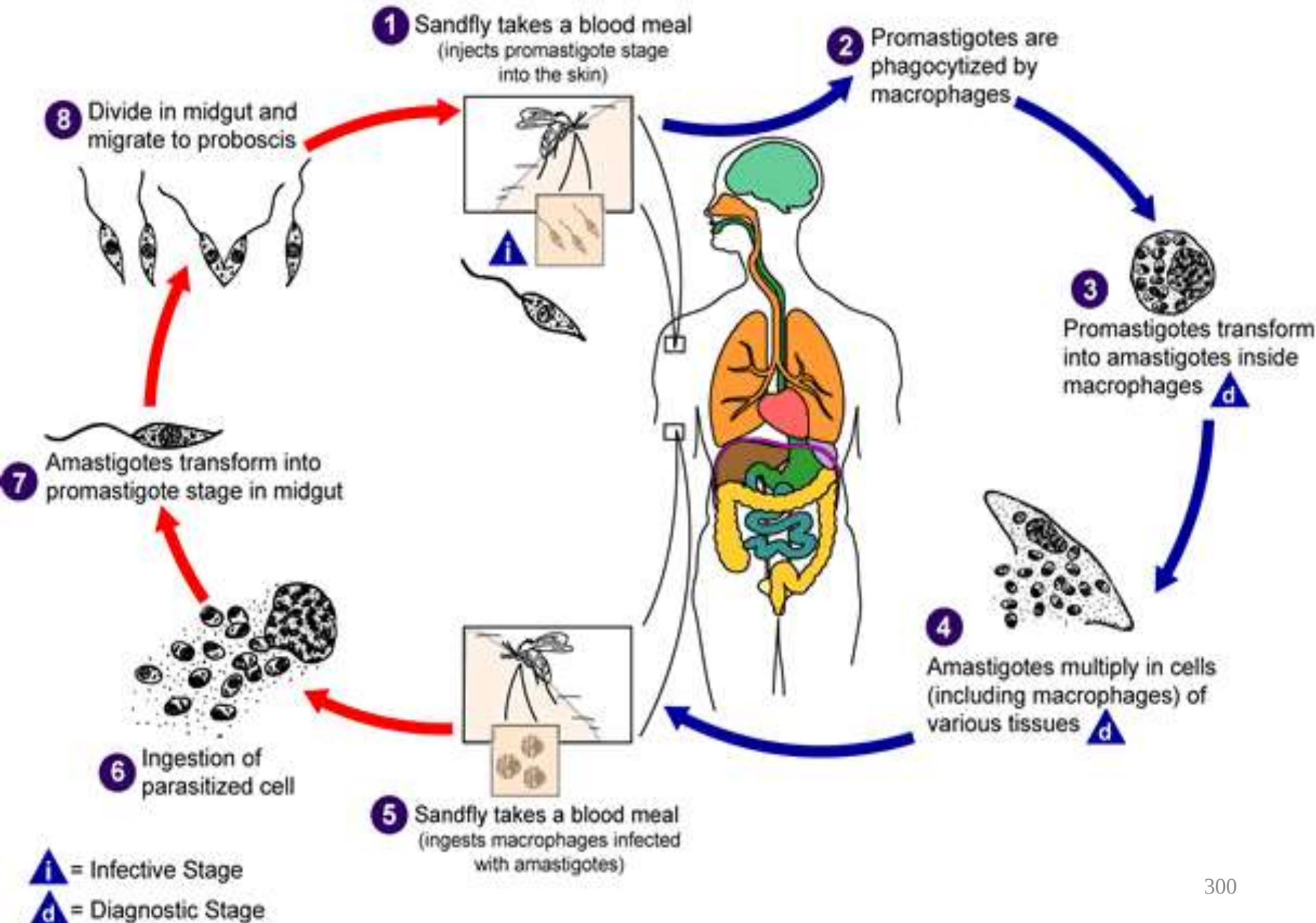


Transmission VL

- **Female sand flies** (Phlebotomus and Lutzomyia Spp.) get infected after sucking blood from infected reservoir eg. human or other mammals
- **Amastigotes** transform to **promastigotes** in sand fly gut
- Promastigotes are regurgitated and injected into the skin during subsequent blood meal

Sandfly Stages

Human Stages



Epidemiology in Ethiopia

- Lowlands of Ethiopia with varying endemicity
 - N.West: **Metema and Humera, Wolkayit, Libo/Fogera**
 - N.East: Ethio-Djibouti Awash Valley
 - S. West: Dawa, Genale, Woito, Konso, Omo, Gambella Sudan border
- Main risk factor: population migration
- VL cause:
 - ✓ *Leshmania donovani* complex
 - ✓ *L. donovani* and *L. Infantum*

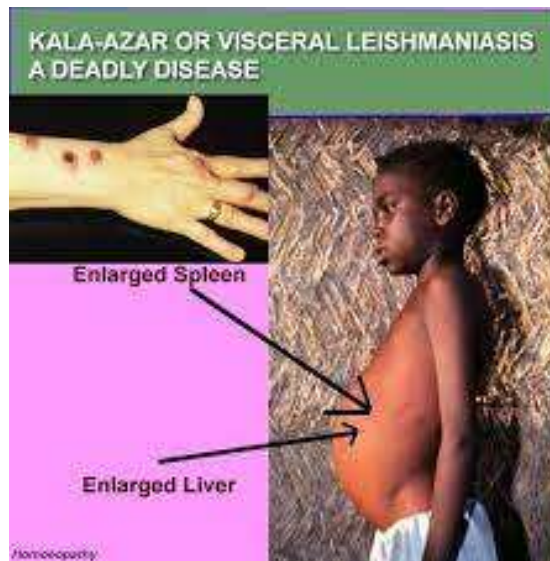
Visceral Leishmaniasis

Presentation

- Kala-azar (Black fever in Hindu)
- Fever, night sweats, weakness, weight loss
- Cachexia (Severe malnutrition, weakness, and muscle wasting resulting from a chronic disease), Pallor
- **Non tender, soft massive splenomegaly**
- **Hepatomegaly $\pm$ LAP**

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- **Darkening of face/ashen grey appearance**
- Bleeding 2^o to thrombocytopenia
- Susceptibility for 2^o infection
- **Pancytopenia (Anemia, Thrombocytopenia, Leukopenia, Neutropenia)**
- **Marked eosinopenia**
- Hypergammaglobulinemia
- Hypoalbuminemia



Diagnosis

- Gold-standard: **Demonstration of amastigotes in tissue aspirates**
- Diagnostic sensitivity – Spleen - > 95%; Bone marrow → 70%, Lymph nodes → 58%
- Detection of Rk 39- Sn 72%, Sp 82%
- Culture

Treatment

- Sodium stibogluconate (SSG) 20mg/kg iv/im for 30 days
- Alternative: Amphotericin B 1mg/kg every other day for 30 days/total 15 doses.
- AmBisome 4mg/kg for 5-7 doses: 1-5days, 10th, 14th

Prevention and control

1. The avoidance of outdoor activities
2. The use of mechanical barriers such as screens and bed nets
3. Wearing of protective clothing
4. Application of insect repellent
5. Treatment of cases
6. Vector control and elimination of reservoir host (e.g. domestic dogs)

African Trypanosomiasis

- A systemic disease caused by protozoa characterized by **fever → general weakness and cerebral involvement → death.**

Infectious agent

- The commonest agents are:
 - T. Brucei rhodesiense
 - T. Brucei gambiense
- Other species which are less important are: **T. Cruzi**, which causes American Trypanosomiasis
- Vectors for all species are **tsetse flies of Genus Glossina**

Epidemiology

- The trypanosomes that cause **sleeping sickness** are found only in Africa. 20,000 new cases are reported each year.
- In Ethiopia, the distribution of Trypanosomiasis is mostly found in **Jinca, Afar, Setitu Humera, Konso, Moyale, Woito, and Dilla.**
- **Reservoir-** for *T. brucei gambiense* it is only **humans**. For *T. brucei rhodesiense* the reservoir is **dog, cattle, fox, wolf and human beings.**

Mode of transmission:

- by the bite of infective *Glossina* **Tsetse fly** during blood meal.
- **Congenital transmission** can occur in humans.
- **Direct mechanical transmission** is possible by blood on the proboscis of *Glossina* and other man-biting insects, such as houseflies or in laboratory accidents

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Incubation period

- *T. brucei rhodensiense*: 3 days to few weeks.
- *T. brucei gambiense*: several months up to one year.

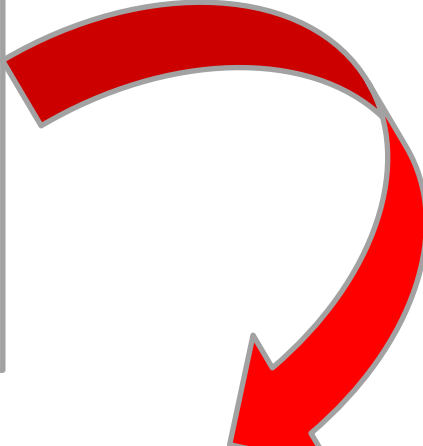
Period of communicability

- The disease is transmitted as long as the parasite is present in the blood of infected person or animal and infected Tsetse fly

Susceptibility and resistance

- All persons are equally susceptible for the disease

1. Trypomastigotes injected through skin when tsetse fly takes a blood meal



Human host

2. Trypomastigotes multiply in blood, lymph and **in later stages, in CNS**. T.b. rhodesiense infects a wide range of game animals and domestic animals



Tsetse fly

3. **Trypomastigotes** ingested by tsetse fly
4. parasites multiply in mid gut. Migrate to salivary glands.
5. Become **epimastigotes** and multiply. Develop into infective metacyclic **trypomastigotes**.

Clinical Manifestation

Stage I (Signs & symptoms)

- 1. Painful trypanosoma chancre**
2. Hematogenous and lymphatic dissemination
- 3. High body temperature**
4. Lymphadenopathy discrete
5. Winter bottom's sign (classic), painless enlargement of lymph node
6. Malaise
7. Headache
8. Weight loss
9. Edema
10. Hepatomegally and
11. Tachycardia

- **Stage II**
 1. Abnormality in CSF
 2. Day time somnolence
 3. Tremors
 4. Parkinson's disease may appear
 5. Hypertonia
 6. Congestive heart failure
 7. CNS disease develops
 8. Coma and death
- **Diagnosis**
 - ƒWet blood smear
 - ƒThick blood smear
 - ƒSerological test
 - ƒCSF analysis
 - ƒBlood film
 - ƒBone marrow biopsy

Treatment

- **For stage I** (Normal CSF): T.b.
fSuramin or Eflornithine or Pentamidine
- **For stage II:** Trypansamide
- **Prevention and control**
 1. Public education on personal measures to protect against insect bite.
 2. Eradication of vectors.
 3. Drug treatment of infected humans.
 4. Avoiding areas to be known by harboring infected insects.
 5. By wearing protective cloth and by using insect repellents.
 6. Reducing tsetse fly number by identifying and studying the breeding habits of local vector

Thank
Very
Much

Chapter 7

Sexually Transmitted Infections and HIV/AIDS

Learning Objectives

- Describe the association between STIs and HIV
- Identify the etiologies and mode of transmission of some of the most common STIs
- Understand the major symptoms of commonly encountered STIs
- Determine the most effective treatment approaches for commonly encountered STIs

STI versus STD

- **STI:** Infections acquired through sexual intercourse (may be **symptomatic or asymptomatic**)
- **STD :** **Symptomatic disease** acquired through sexual intercourse
- STI is most commonly used because it applies to both symptomatic and asymptomatic infections

Introduction

- **Most STIs are treatable:** however, resistance to many of the older antibiotics is a challenge.
- Other **have no cure, such as herpes, genital warts, HIV**
- Many STIs can lead to related conditions such as:
 - Pelvic inflammatory disease
 - Cervical cancer
 - Complications in pregnancy
- STIs can have socio-economic consequences
- Education about these diseases and prevention are important

Link Between STIs & HIV/AIDS

- **STIs facilitate HIV transmission and acquisition**
 - Chancroid, chlamydia, gonorrhea, syphilis, and trichomoniasis may increase the risk of HIV transmission by two to nine times
- **STI control reduces transmission of HIV**
 - The prevalence of HIV in heterosexual populations is higher in Africa than in Europe and the US where STIs are more often treated and cured
- STIs and HIV infection share similar epidemiologic determinants
 - Result from risky sexual behavior
 - Affect the same age group
- HIV affects the clinical presentation and management of STIs
 - In people with HIV infection, other STIs may be more resistant to treatment

Interventions to Reduce Transmission

- Decrease duration of infectivity: **early diagnosis & curative or suppressive therapy**
- STI/HIV risk assessment, STI screening, and client-centered risk reduction counseling should be provided routinely to persons with HIV infection
- Decrease efficiency of transmission: **Promote safer sexual behavior**
- Decrease susceptible persons' exposure rate to infected individuals: **Counsel to modify sexual behavior**
- **Prevention Methods**
 - ❖ **Abstinence and reduction of number of sex Partners**
 - ❖ Pre-exposure Vaccination: is one of the most effective methods for preventing transmission of human papillomavirus (HPV), Hepatitis A Hepatitis B virus
 - ❖ **Condoms use, Male Circumcision, Partner Services**

Commonly Encountered STIs

- Syphilis
- Gonorrhea
- Chlamydia
- Chancroid
- Genital Herpes
- Genital warts
- Trichomoniasis
- Lymphogranularoma Venereal

Management approach for STI

- There are two basic approaches in the management of STIs:
- ✓ **Etiologic Management:** etiologic diagnosis using laboratory tests to identify the causative agent
- Benefits: focused, specific therapy, avoiding the cost and toxicity of unnecessary medications.
- ✓ **Syndrome Management approach**
- Considers the likely causative agent(s) for a given clinical syndrome and treats accordingly, without regard for identifying the specific infection.

Components of syndromic approach of STI

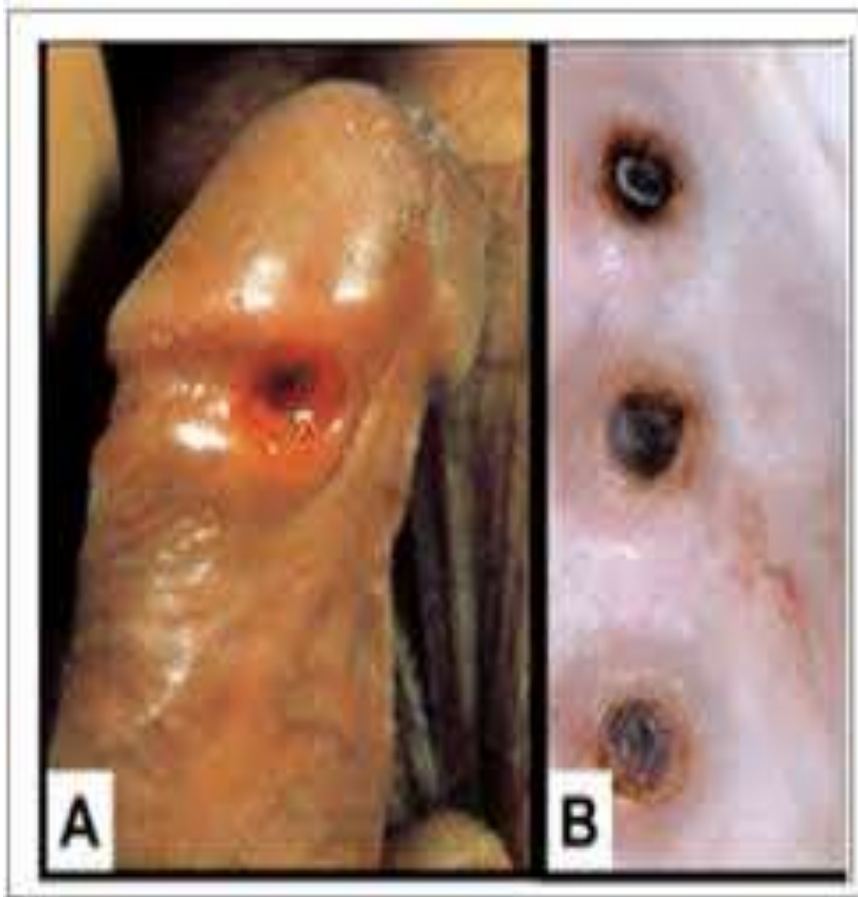
- Drug treatment
- Partner notification
- Health education and risk reduction
- Condom provision
- PITC
- Follow up
- Monitoring and evaluation
- Reporting

Syphilis (Hard chancre)

- A disease characterized by a **primary lesion, a later secondary eruption on the skin and mucus membranes**, then a long period of latency, and finally late lesions of skin, bones, viscera, CNS and cardiovascular systems.
- **Infectious agent:** **Treponema pallidum**, a spirochete.
- **Epidemiology:** Worldwide spread. Primarily involving **sexually active young people between 20 and 29 years**. More common in urban than rural areas.
- **Reservoir:** Humans
- **Mode of transmission:**
 - ✓ by direct contact with lesion mainly **during sexual intercourse**.
 - ✓ Accidentally **by touching** infective tissues. Or
 - ✓ Via **blood transfusion**. Or
 - ✓ **Congenitally**, which may occur before birth,

• • •

- **Incubation period:** 10 days to 3 months, usually 3 weeks.
- **Period of communicability:** variable and indefinite, during primary and secondary stages and also in mucocutaneous recurrences that may occur during the first 4 years of latency. Extent of communicability through sexual activity during this latent period is not established. Adequate penicillin treatment usually ends infectivity within 24 – 48 hours.
- **Susceptibility and resistance:** Susceptibility is universal, although only approximately 30% of exposures result in infection. Infection leads to developing immunity against *T. pallidum* gradually and to some extent, but immunity usually fails to develop because of early treatment in the primary and secondary stages.



Clinical Manifestation

- The clinical presentation is divided into three groups:

Primary syphilis:

- consists of hard chancre, the primary lesion of syphilis, together with **regional lymphadenitis**.
- The hard chancre is a **single, painless ulcer** on the genitalia or elsewhere (lips, tongue, breasts) and heals spontaneously in a few weeks without treatment.
- The lymph glands are bilaterally enlarged and not painful.
- There will not be suppuration.

...

Secondary syphilis

- After 4 – 6 weeks of the primary infection, **a generalized secondary eruption appears**, often accompanied by mild constitutional symptoms.
- These early rashes tend to be symmetrical, quickly passing, and do not itch. These early skin lesions **are highly infective**

Tertiary syphilis

- characterized by **destructive, non-infectious lesions of the skin, bones, viscera, and mucosal surfaces**. Other disabling manifestations occur in the cardiovascular system (aortic incompetence, aneurysms) or central nervous system (dementia paralytica, tabes dorsalis).

Syphilis in pregnancy

- According to the severity, congenital syphilis can result in congenital abnormalities, still birth, or repeated spontaneous abortions.

Diagnosis

- Serological test : will be positive 6 to 8 weeks after infection
- Dark field microscopy of smears from primary lesion (hard chancre) or from skin lesions in the early secondary stage will show the spirochaetes.

Treatment

1. Primary and secondary syphilis

- Benzathin penicillin 2.4 M IU Im stat or
- Tetracycline or Erythromycin 500mg PO Qid for 2 weeks for penicillin sensitive people

2. Tertiary syphilis

- Benzathin penicillin 2.4 M IU Im single dose every week for 3 consecutive weeks or
- Tetracycline or Erythromycin for one month for penicillin sensitive individuals.

3. Early congenital syphilis

- Crystalline penicillin 50,000 IU/ Kg per dose IV or Im bid in the first 7 days of life and Tid then after for 10-14 days.

Prevention and control

1. Treatment of cases
2. Treatment of contacts and source of infection
3. Health education on safe sex
4. Controlling STDs **among commercial sex workers**
 - Monthly check up and treatment of cases
 - Provision of condom
5. **Screening of pregnant women and early treatment** to prevent congenital syphilis
6. **Screening of blood before transfusion.**

Chancroid (**soft chancre**)

- An acute bacteria infection localized in the genital area and characterized clinically by **single or multiple painful narcotizing ulcers at the site of infection.**
- **Infectious agent.:** Haemophilus ducreyi, the Ducrey bacillus
- **Epidemiology:** endemic in many developing countries. The commonest cause of genital ulcer in many developing counties. Most frequently diagnosed in men, especially those who frequently **prostitutes.**
- **Reservoir :** Humans

• • •

- **Mode of transmission** : by direct sexual contact with discharges from open lesion and pus from buboes. Infected males don't pass the infection farther because of the painful ulcer.
- **Incubation period**: from 3 to 5 days, up to 14 days after sexual contact.
- **Period of communicability**: until healed and as long as the infectious agent persists in the original lesion or discharging regional lymph nodes, which lasts for several weeks or months without antibiotic treatment. Antibiotic therapy eradicates *H. ducreyi*, and lesions heal in 1 – 2 weeks.
- **Susceptibility and resistance**: Susceptibility is general. The **uncircumcised are at higher risk than the circumcised**. No evidence of natural resistance.

Clinical manifestation

- Classic Chancroid ulcer begins as a **tender papule that ulcerates within 24 hours.**
- The ulcer is **painful, irregular and sharply demarcated** from the nearby skin.
- **About 50% of men will have single ulcer.**

Diagnosis

- Clinical, but always rule out syphilis
- Gram stain of smear from ulcer shows typical rods in chain
- Culture

Treatment

1. Co- trimorazele or
2. Erythromycin or
3. Tetracycline can be used

N.B. Do not incise lymph nodes even with fluctuation because they will completely heal with treatment.

Prevention and control

1. Case treatment
2. Investigation of contacts, source of infection and treatment
- 3. Thorough washing of genitalia with soap and water promptly after intercourse is very effective.**
4. Controlling STDs among commercial sex workers
5. Sex education for high risk groups

Gonorrhea

- An acute or chronic **purulent infection of the urogenital tract.**
- **Infectious agent:** Neisseria gonorrhea, the gonococcus
- **Epidemiology:** worldwide, affecting both genders, especially **sexually active adolescents and young adults. Common in rural areas.** Prevalent in communities of lower socio-economic status. In most industrialized countries, the incidence has decreased during the past two decades.
- **Reservoir:** Strictly a human disease
- **Mode of transmission:** **almost always as a result of sexual activity**
- **Incubation period:** usually 2-7 days
- **Period of communicability:** may extend for months in untreated individuals. Effective therapy ends communicability within hours.
- **Susceptibility and resistance:** susceptibility is general.

Clinical manifestations

- **Males:** usually involves the urethra resulting in **purulent discharge, dysuria and frequency.**
- **Females:** females are usually asymptomatic. **Vaginal discharge is common.**
- Most common site of infection is cervix, followed by urethra, anal canal and pharynx.
- Bartholinitis occurs unilaterally.
- Salpingitis as a complication occurs in 20% of women.
- Neonates borne to infected mothers develop a purulent discharge which exudes from between eyelids which are edematous and erythematous 2 -3 days postpartum.

Diagnosis

- Gram stain of discharge (urethral, cervical, conjunctival discharge)
- Culture on selective media
- **Treatment**
 1. Co - trimoxazole or
 2. Erythromycin or
 3. Ceftriaxone can be used
- **Prevention and control**
 1. The same as syphilis
 2. Application of 1% tetracycline in both eyes of newborn as soon as delivered.

STI Syndromes

1. Urethral discharge or burning on urination in men
2. Vaginal discharge
3. Genital ulcer
4. Lower abdominal pain in women
5. Scrotal swelling
6. Inguinal bubo
7. Neonatal conjunctivitis

1. Urethral Discharge Syndrome

- abnormal secretions from the distal part of the urethra
- it is the characteristic manifestation of urethritis
- Possible etiologies:
 - Gonococcal infection (81%)
 - Chlamydia Trachomatis (36.8%)
 - T.vaginalis- usually asymptomatic in men

Clinical Presentation

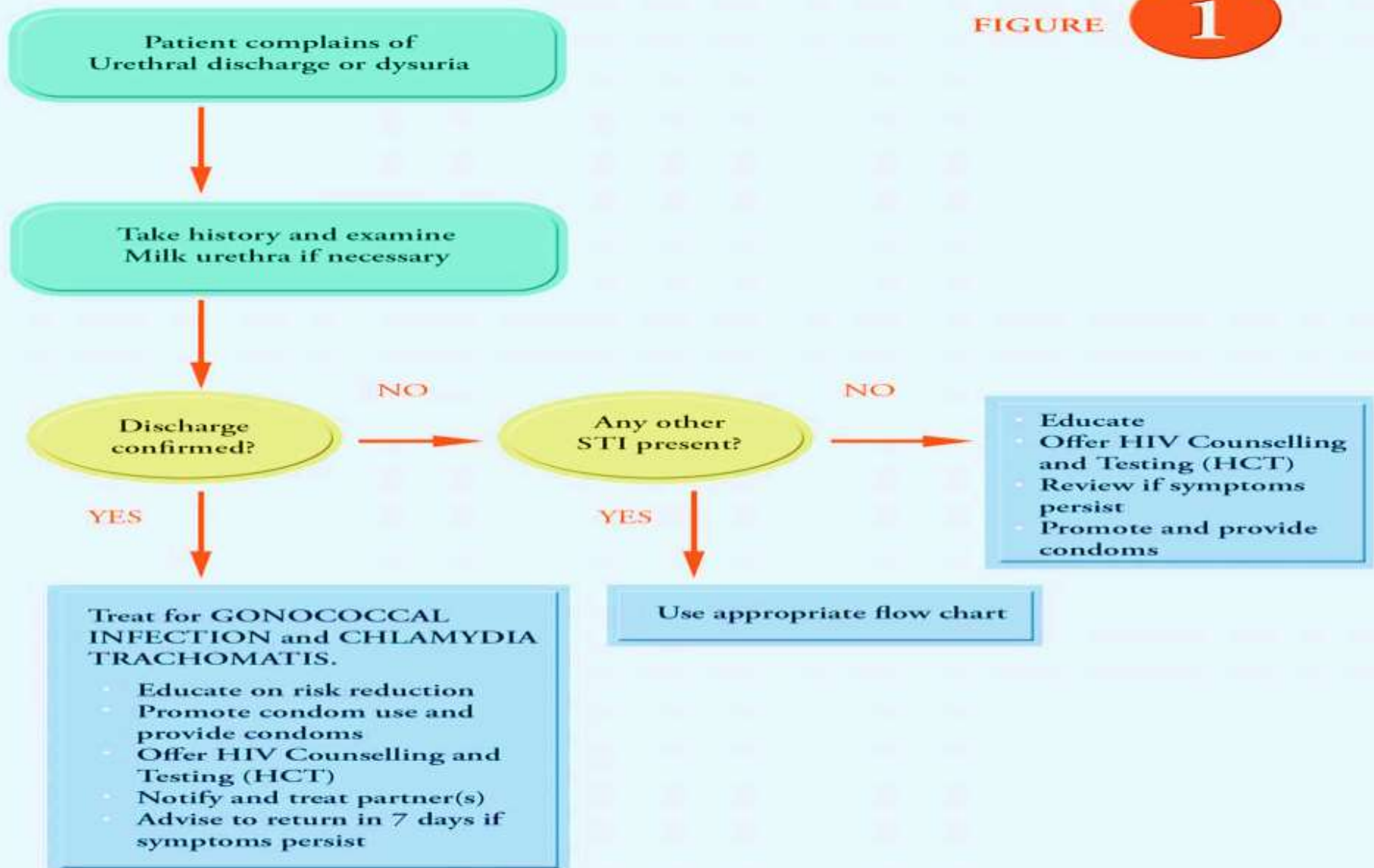
- Burning sensations (dysuria) during micturition
- Increased frequency and urgency of urination
- Itching sensation of urethra
- **Urethral discharge**
- **Purulent or mucoid, clear, white, or yellowish-green**
- Meatus excoriation

Urethral discharge



FIGURE

1



Recommended Treatment for Urethral Discharge and Burning on Urination

Ceftriaxone 250mg IM stat/Spectinomycin 2 gm IM stat
Plus

Azithromycin 1gm po stat/Doxycycline 100 mg po bid for 7 days/Tetracycline 500 mg po qid for 7 days/Erythromycin 500 mg po qid for 7 days in cases of contraindications for Tetracycline (children and pregnancy)

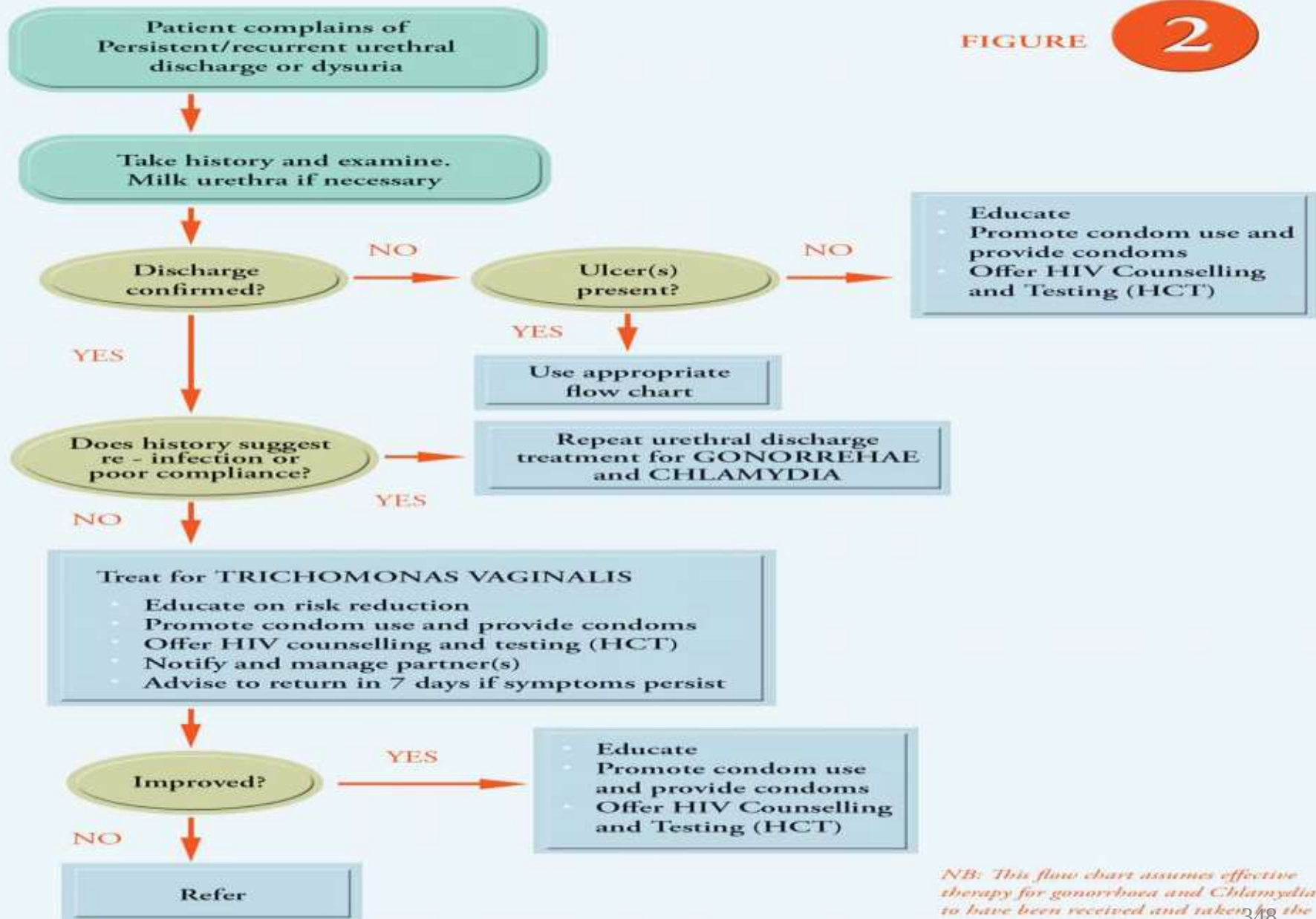
Note: The preferred regimen is Ceftriaxone 250mg IM stat plus *Azithromycin 1gm po stat*

2. Persistent/Recurrent Urethral Discharge

- Complain of persistent or recurrent burning sensation on urination, with or without discharge, due to various reasons:
 - Inadequate treatment or poor compliance
 - Re-infection (**partner/s is not managed**)
 - T. vaginalis is also known to cause urethritis in men
 - Infection by drug-resistant organisms (N.gonorrhea)
- **Metronidazole** 2 gm p.o. stat or **Tinidazole** 1gm po once for 3 days (Avoid Alcohol)

FIGURE

2



NB: This flow chart assumes effective therapy for gonorrhoea and Chlamydia to have been received and taken by the patient prior to this consultation.

3. Vaginal Discharge

- Abnormal Vaginal discharge in terms of quantity, color or odor could be most commonly as a result of vaginal infections
- The most common causes of vaginal discharge are
 - **Neisseria gonorrhoeae,**
 - **Chlamydia trachomati,**
 - **T. vaginalis, Gardnerella vaginalis and**
 - **C. albicans**
- One or more of the following are risk factors for STIs related cervicitis in Ethiopia:
 - **Multiple sexual partners in the last 3 months**
 - **New sexual partner in the last 3 months**
 - **Ever traded sex**
 - **Age below 25 year**

Clinical Manifestations

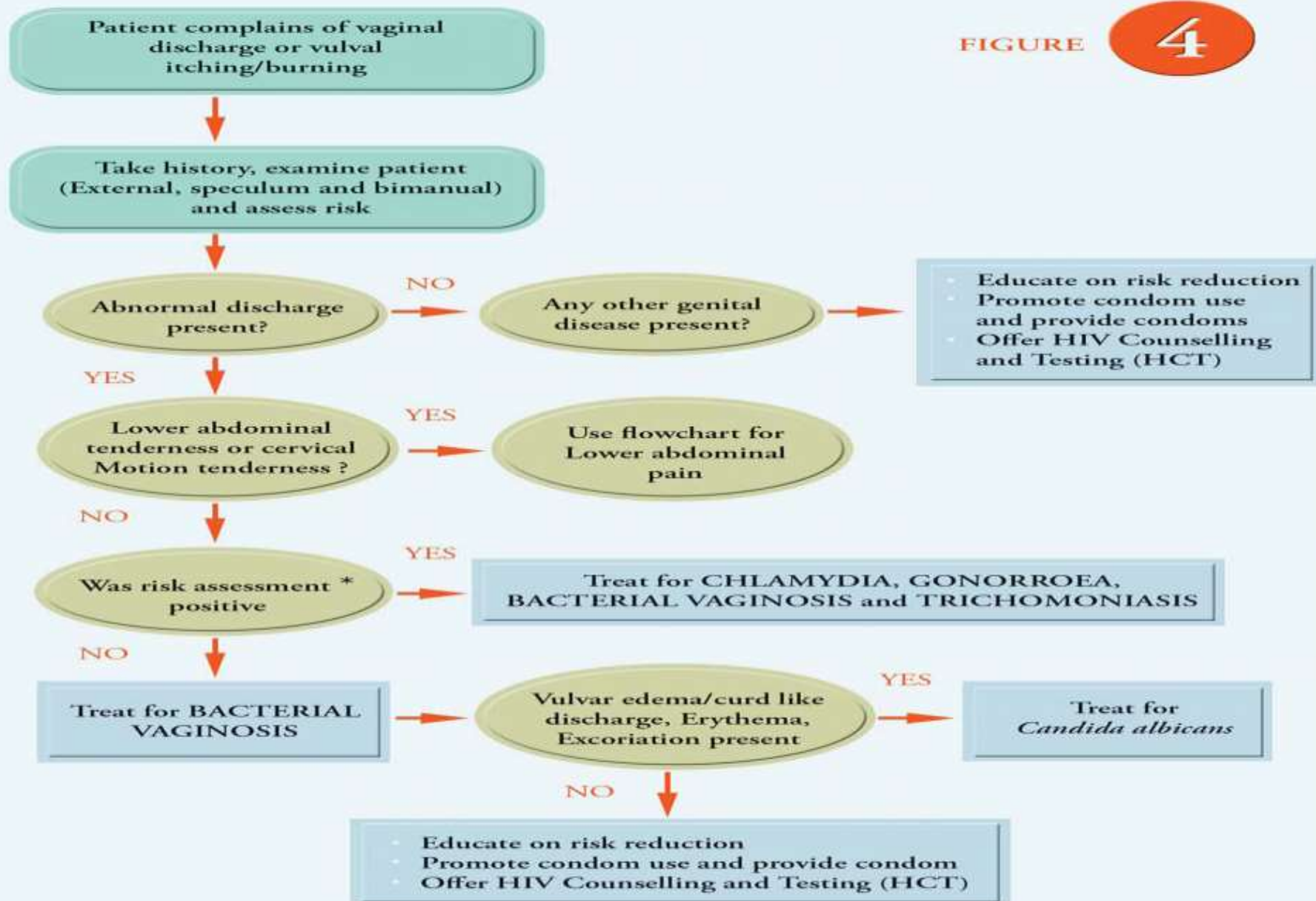
- The classical manifestation of vaginal discharge is **discharge from the vagina.**
- The discharge can be
 - Thin, homogenous whitish discharge with fishy odor
 - Thick, profuse, malodorous, yellow-green, frothy itchy
 - Purulent exudate from the cervical Os
 - White, thick and curd like discharge coating the walls of the vagina
- The other manifestations are **vulvo-vaginal pruritus**, irritation of vulva, **dyspareunia**, dysuria, and frequency of urination

Vaginal Discharge



FIGURE

4



* Risk factors include age < 25 years, trading sex, multiple or new partner in the last 3 months.

Recommended Treatment for Vaginal Discharge

Risk Assessment Positive

Ceftriaxone 250mg IM stat/Spectinomycin 2 gm IM stat

plus

Azithromycin 1gm po stat/Doxycycline 100mg po bid for 7 days

Plus

Metronidazole 500mg bid for 7 days

If discharge is white or curd-like add Clotrimazole vaginal pessary 200 mg at bed time for 3 days

Note: The preferred regimen is Ceftriaxone 250mg IM stat plus Azithromycin 1gm po stat plus Metronidazole 500 mg bid for 7 days

Risk assessment Negative

Metronidazole 500mg bid for 7 days

If discharge is white or curd-like add Clotrimazole vaginal pessary 200 mg at bed time for 3 days

4. Lower Abdominal Pain Due to PID (Pelvic Inflammatory Disease)

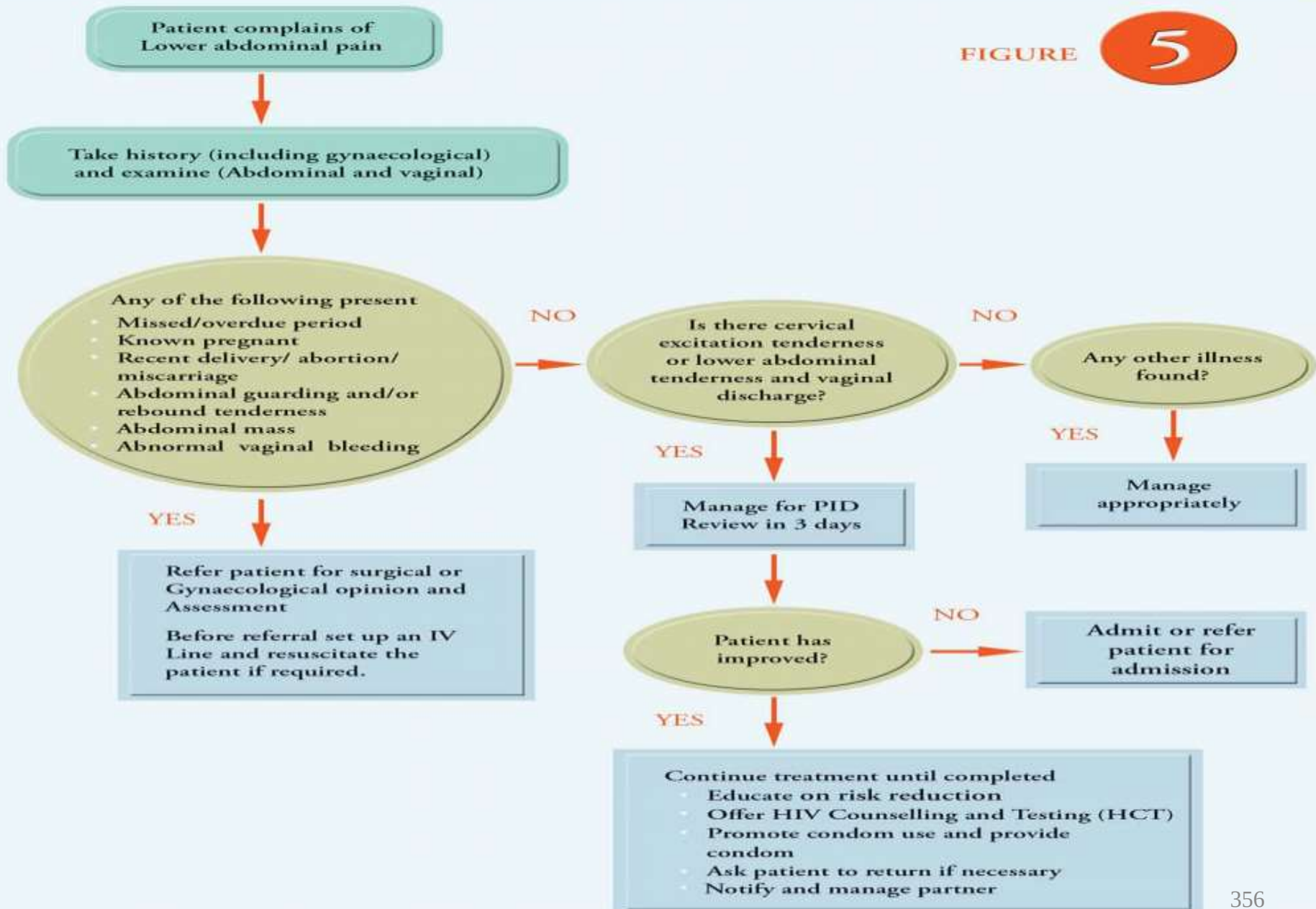
- PID is ascending infection of the upper genital tract (uterus, tubes, etc) from the cervix and/or vagina
- Common etiologies:
 - Sexually transmitted:
 - *Neisseria gonorrhea*,
 - *Chlamydia trachomatis*,
 - *Mycoplasma hominis*
 - Others (non-STI): streptococci, *E. coli*, etc
- Vaginal discharge is often present

Clinical Manifestation

- The commonest manifestations of PID:
 - **Lower abdominal pain**
 - **Abnormal vaginal discharge**
 - **Inter-menstrual or post coital bleeding**
 - Dysuria
 - Backache
 - Fever, nausea and vomiting
 - Cervical excitation tenderness
 - Adnexal mass, tenderness and
 - Rebound tenderness

FIGURE

5



Recommended treatment for PID

Recommended treatment for PID

Out patient

Ceftriaxone 250mg IM stat/Spectinomycin 2 gm IM stat plus

Azithromycin 1gm po stat/Doxycycline 100mg po bid for 14 days plus

Metronidazole 500mg bid for 7 days

Admit if there is no improvement within 72 hours

Note : The preferred regimen is Ceftriaxone 250mg IM stat plus Azithromycin 1gm po stat plus Metronidazole 500 mg bid for 7 days

Inpatient

Ceftriaxone 250mg IV/IM daily /Spectinomycin 2gm IM bid plus

Azithromycin 1gm po daily / Doxycycline tablet 100 mg bid for 14 days

Plus

Metronidazole 500mg bid for 14 days

Indication For Admission

- ❖ The diagnosis is uncertain
- ❖ Surgical emergencies such as appendicitis and ectopic pregnancy cannot be exclude
- ❖ Pelvic abscess is suspected
- ❖ Severe illness precludes management on an outpatient basis
- ❖ The patient is pregnant
- ❖ The patient is unable to follow or tolerate an outpatient regimen
- ❖ Patient has failed to respond to outpatient therapy
- ❖ PID in HIV patients

5. Scrotal Swelling

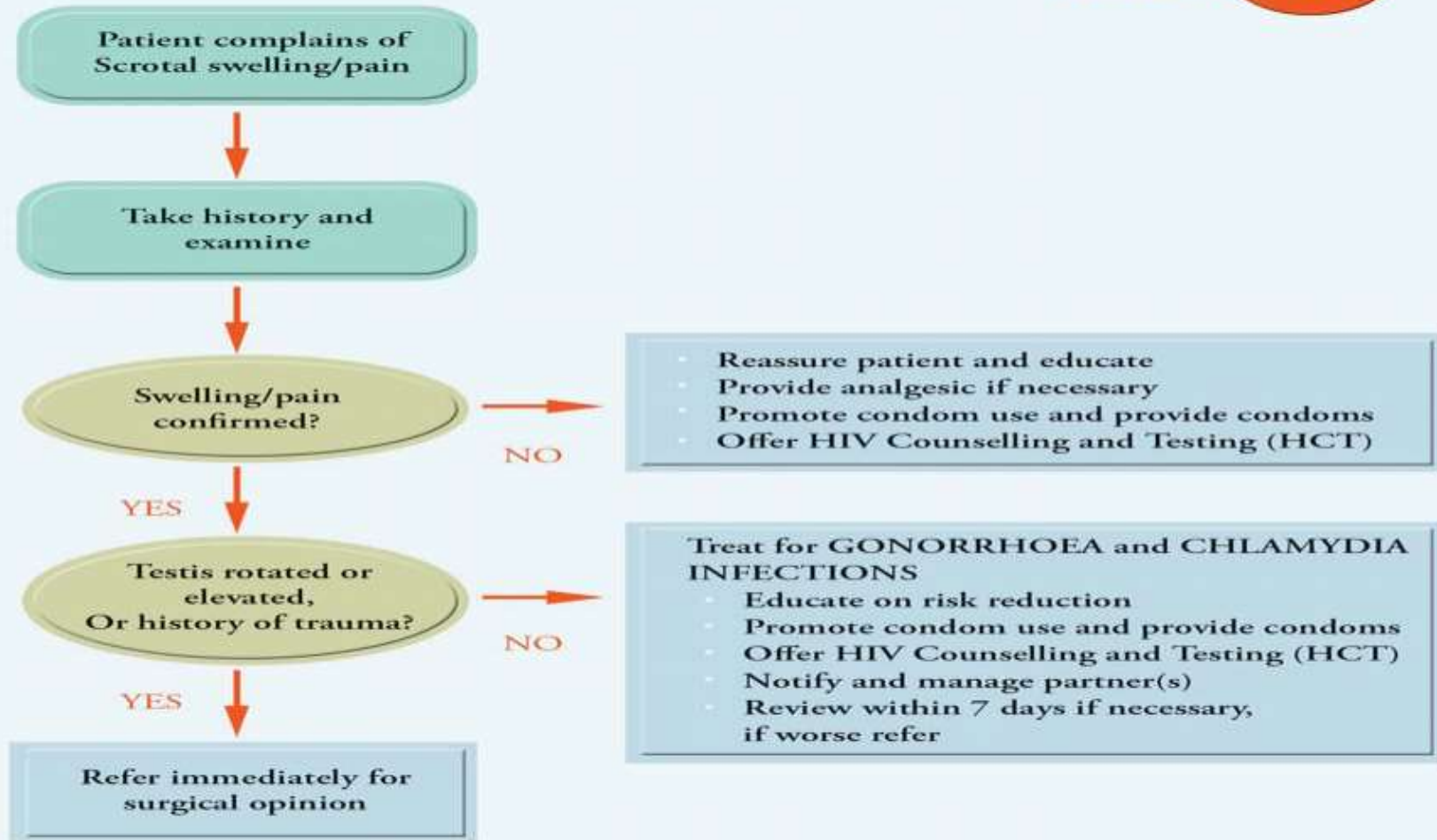
- Mostly the inflammation of the epididymis is caused by sexually transmitted diseases
- Common STI causes of scrotal swelling are similar to those of urethral discharge
 - Neisseria gonorrhea
 - Chlamydia trachomatis
- Exclude non-STI causes of scrotal swelling:
 - TB, Pseudomonas aeruginosa , Filarial diseases
 - Inguinal hernia

Clinical manifestations

- **Some of the signs and symptoms are:**
 - Pain and swelling of the scrotum
 - Tender and hot scrotum on palpation
 - Edema and erythema of the scrotum
 - Dysuria
 - Sometimes frequency and urethral discharge can be there

FIGURE

6



Scrotal Swelling Recommended Therapy

- **Non-Pharmacologic:** scrotal support
- **Pharmacologic**

Recommend treatment for Scrotal Swelling

Ceftriaxone 250mg IM stat/ Spectinomycin 2gm IM stat

plus

Azithromycin 1gm po stat/Doxycycline 100mg bid PO for 7 days

Note :The preferred regimen is Ceftriaxone 250mg IM stat plus

Azithromycin 1gm po stat

6. Genital Ulcer Diseases (GUD)

- Clinical manifestation and patterns of GUD may vary with presence of HIV infection.
 - The causes of genital ulcer are
 - *Treponema Pallidum* (**syphilis**),
 - **HSV2** infection,
 - *Haemophilus ducreyi* (**chancroid**),
 - *C. trachomatis* serovar L1, L2 & L3 (**LGV**)
- **Syphilis**
 - Clinically has three stages (primary, secondary, tertiary)
 - The ulcer starts during the primary stage of the disease as papules & rapidly ulcerative
 - The ulcer is typically **painless, clean base and raised border.**

Genital Ulcer Disease



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■ Genital Herpes

- Herpes simplex virus is the most common causes of genital ulcer world wide
- Produces life long infection after the primary infection (latency)
- **Lesions are painful, erythematous macules** which progressively form vesicles, pustules, ulcer and crusts

■ Chancroid

- is also the common cause of genital ulcer in developing countries.
- The lesion **started as painful papules** and pustules which **ulcerate with dirty base and soft edge**
- Inguinal fluctuant adenopathy (buboes) may occur following ulcer

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❖ Lymphogranulomaveneurum (LGV)

- The disease **starts as painless papules** that develops an ulcer
- After a few days painful regional lymphadenopathy develop and associated systemic symptoms may occur

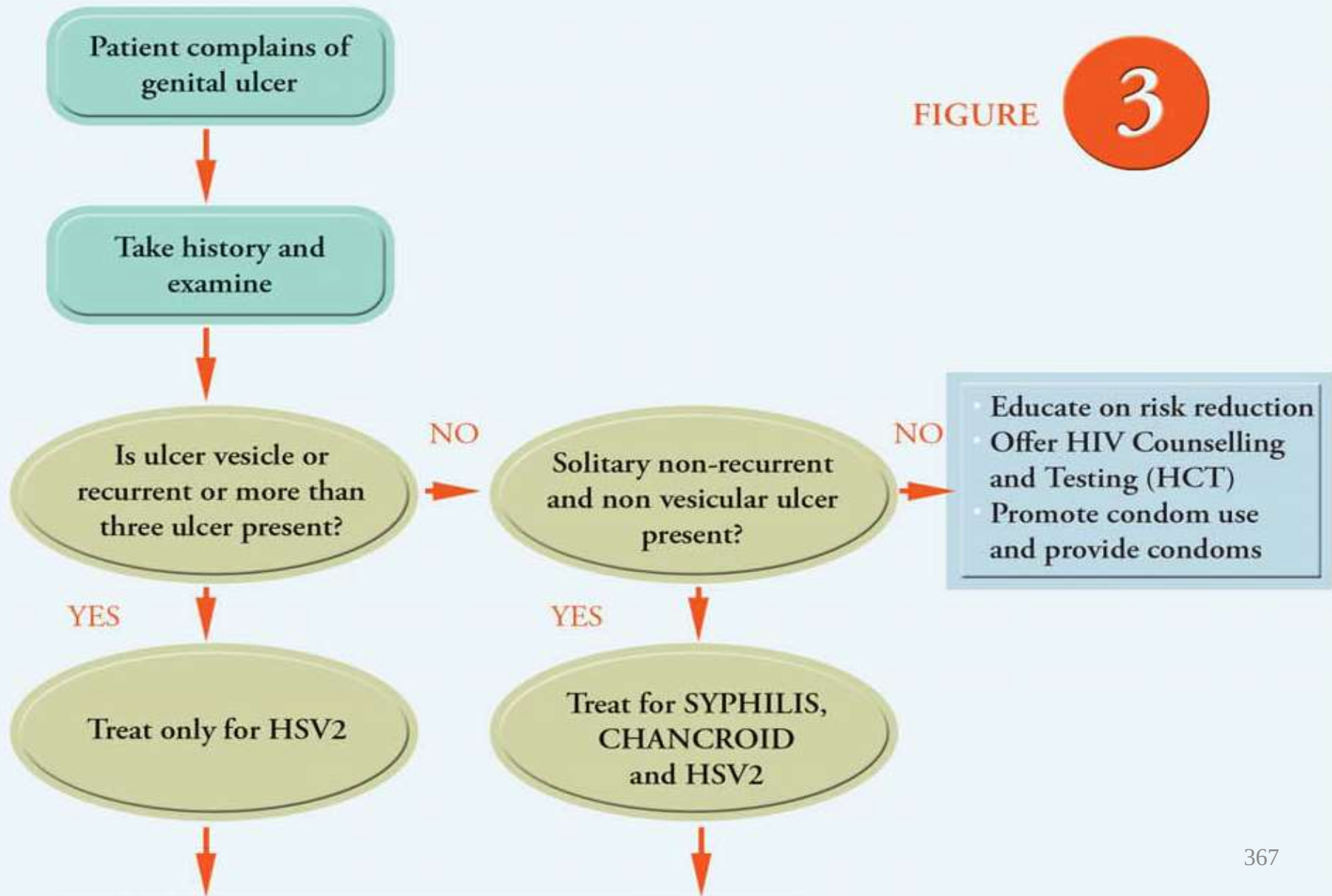
❖ Granuloma inguinale (Donovanosis)

- is chronically progressive ulcerative disease without systemic symptoms, **presents with non-suppurative painless** genital ulcer and beefy-red appearance

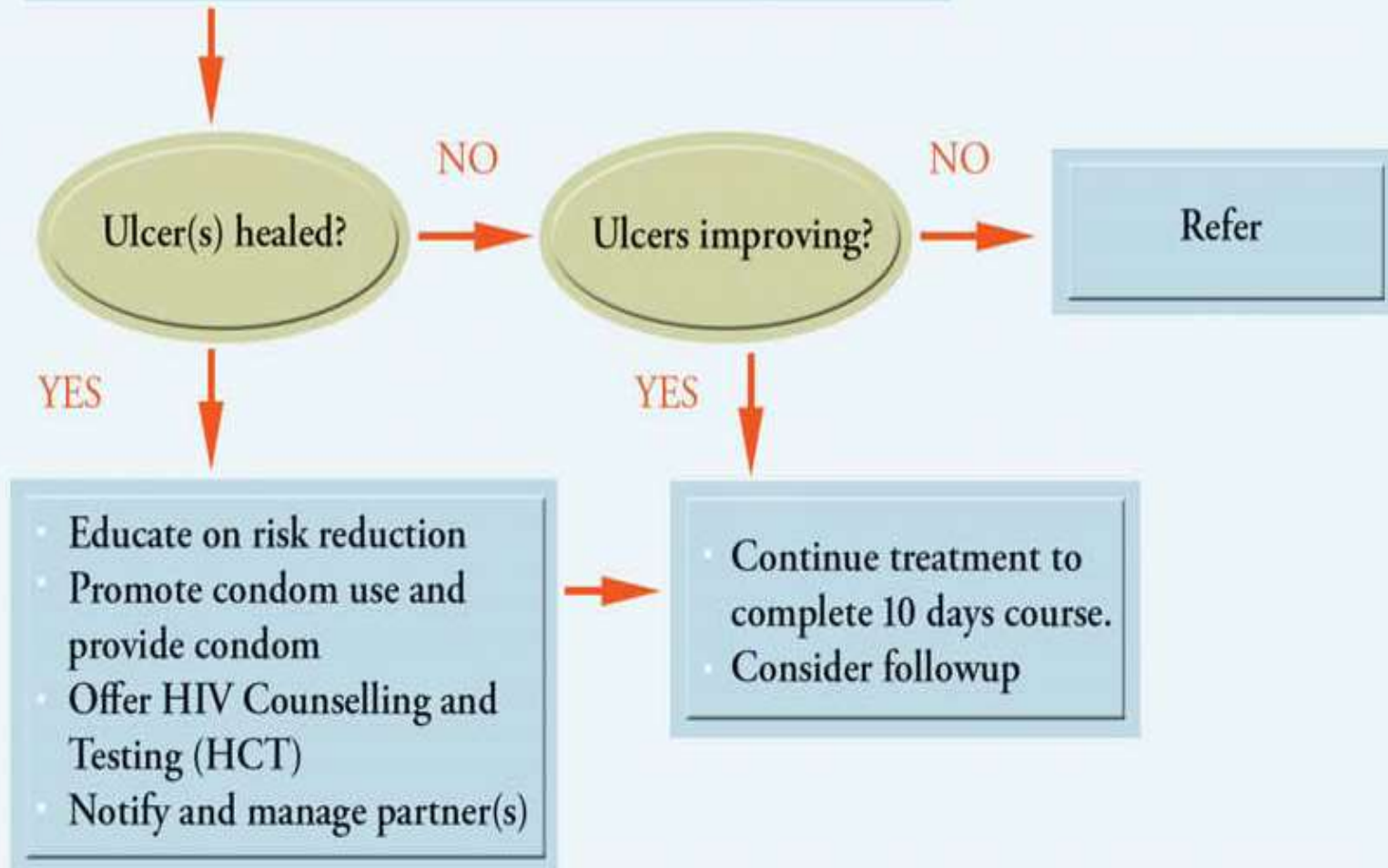
GENITAL ULCERS SYNDROME

FIGURE

3



- Educate on risk reduction
- Promote condom use and provide condoms
- Offer HIV Counselling and Testing (HCT)
- Ask the patient to return in 7 days
- Notify and manage partner(s)



Genital Ulcer Disease Treatment

Recommended treatment for non-vesicular genital ulcer

- Benzathin penicillin 2.4 million units IM stat /Doxycycline(in penicillin allergy) 100mg bid for 14 days plus
- Ciprofloxacin 500mg bid orally for 3 days /Erythromycin 500mg tab qid for 7 days

Plus

- Acyclovir 400mg tid orally for 10 days (or 200mg five times per day of 10 day)

Recommended treatment for vesicular multiple first episode genital ulcer

- Acyclovir 400mg tid for 10 days
- or
- Acyclovir 200 mg 5 times per day for 10 day

Severe Chronic Herpes Simplex Ulcers



HSV



HSV circumferential ulcer



Courtesy of CDC/ Dr. M. F. Rein; Susan Lindsley

HSV Treatment

- Primary infection
 - Acyclovir 200 mg PO 5x/day for 7-14 days
or
 - Acyclovir 400mg PO tid for 7-14 days
or
 - Famciclovir 500 mg PO bid for 7-14 days
or
 - Valacyclovir 1 gm PO bid 7-14 days

...

- **Episodic treatment of recurrent episodes**

- Treatment should be initiated during prodrome or immediately after onset of symptoms.
- **Non pharmacologic:** Local care: Keep affected area clean and dry
- **Pharmacologic: Acyclovir** 400 mg P.O. TID for 5 days,
- **Suppressive treatment:** recommended for patients with 6 recurrences or more per year
 - **Acyclovir, 400mg P.O. BID for 1 year**
 - N.B. The need for continued suppressive therapy should be reassessed

7. Inguinal Bubo

- This is a **painful, fluctuant, swelling of the lymph nodes in the inguinal region (groin)**
- Swelling of inguinal lymph nodes as a result of STIs (or other causes)
- Common causes:
 - Chlamydia trachomatis serovar L1,L2, L3 (LGV)
 - Hemophilus ducreyi (chancroid)
 - Calymatobacterium granulomatis (granuloma inguinale)
 - Treponema pallidum (syphilis)

Clinical Manifestations

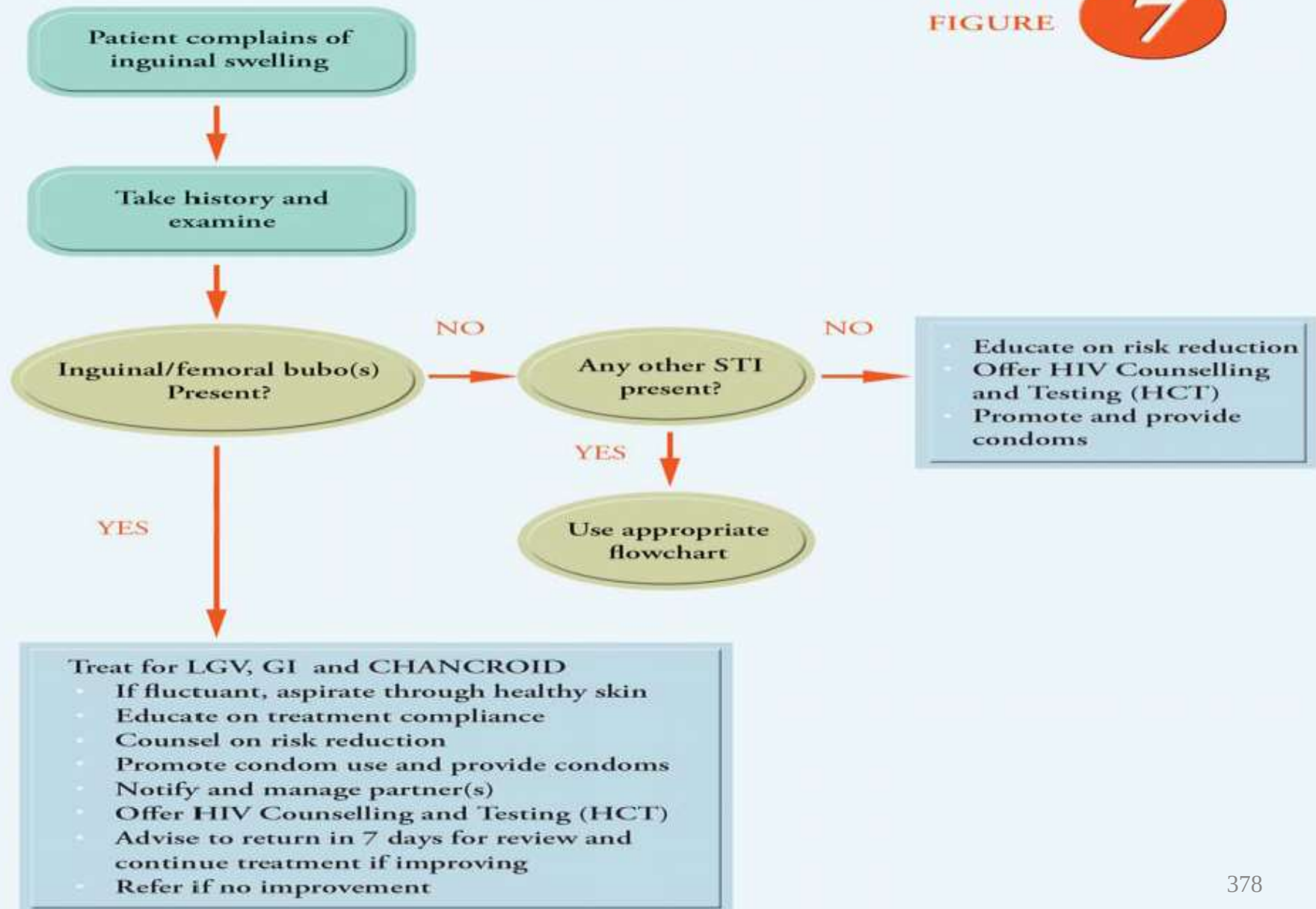
- Constitutional symptoms of fever, headache and pain
- Tender unilateral or bilateral lymphadenopathy forms a classical “groove sign” in the inguinal area
- Fluctuant abscess formation which form a coalesce mass (bubo)
- Some time concomitantly occur with genital ulcer

Inguinal Bubo



FIGURE

7



Inguinal Bubo

Recommended treatment for Inguinal bubo

Ciprofloxacin 500mg bid for 3 days

plus

Doxycycline 100mg bid orally for 7 days/

Erythromycin 500mg qid orally for 14 days

NB. * surgical incisions are contraindicated, aspirate pus with hypodermic needle through the healthy skin

Neonatal Conjunctivitis

- Infection of the eyes of the neonate as a result of genital infection of the mother, transmitted during birth
- Causes:
 - *Neisseria gonorrhoea*
 - *Chlamydia trachomatis*

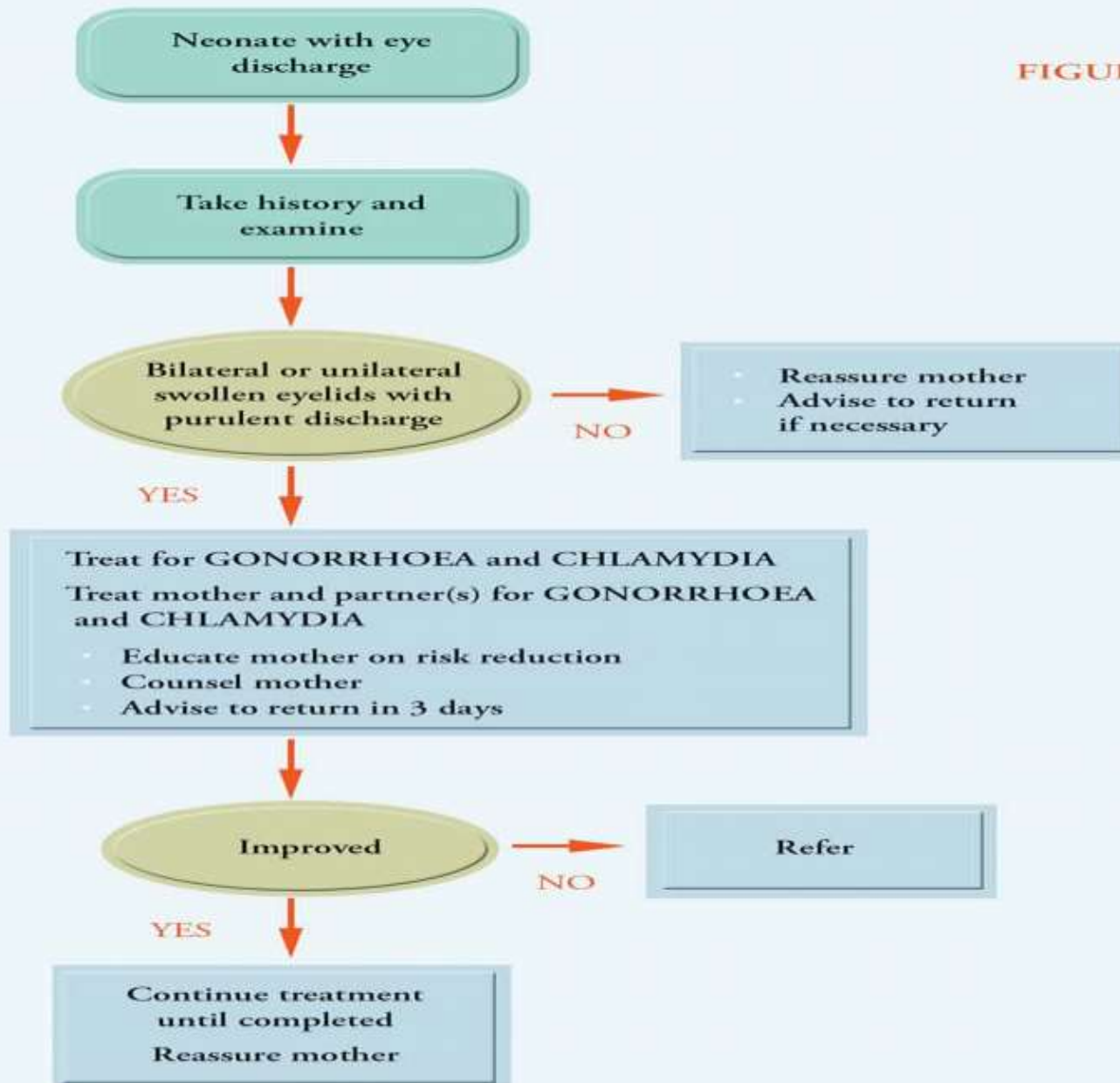


Clinical Manifestations

- The common clinical presentations of neonatal conjunctivitis (ophthalmia neonatorum) are:
 - Red and edematous conjunctiva
 - Edematous eye lead
 - Discharge which may be purulent
 - Orbital cellulitis in more serious cases

FIGURE

8



Neonatal Conjunctivitis: Treatment

Recommended Treatment for Neonatal Conjunctivitis

- **Ceftriaxone 125mg IM stat maximum dose / Spectinomycin 25 mg/kg IM stat maximum dose 75mg**
plus
- **Erythromycin 50mg/kg orally in four divided doses for 14 days**
- **Note: TTC is used as prophylaxis for neonatal conjunctivitis but not for treatment**

Chapter 8

ZOONOTIC DISEASES

Introduction

- A zoonosis is an animal disease that is transmissible from vertebrate animals to humans
- Humans are usually **an accidental host**
- acquires disease **through close contact** with an infected animal
- In most of these diseases man is the dead end of the transmission cycle
 - Man will not infect other human beings

1. Taeniasis

- 2 main species of *Taenia* for which humans are the **obligatory final/ definite** hosts
 - ***Taenia saginata***: the **beef tapeworm**; **cattle** are the intermediate hosts
 - ***Taenia solium***: the pork tapeworm; **pigs are its intermediate host**

Mode of transmission

- **Eggs** of *Taenia saginata* passed in the stool of an infected person are infectious only to cattle
- In the **flesh of cattle**, the parasites develop into “**cysticercus bovis**” larva stage of *Taenia saginata*
- Humans are infected by eating **undercooked meat** containing **cysticerci**
- In the intestinal wall **cysticerci develops in to proglottids**
- These proglottids become gravid and hatch
- **Incubation period: 8-14 weeks**
 - eggs appear in stool in both species

Clinical symptoms

- Most humans with adult tapeworm are **asymptomatic**
- Patients may intermittently pass **proglottids** either with their **stool (T. solium)** or **spontaneously (T. saginata)**
- Patients notice the segment in the toilet bowl or may feel the **spontaneous movement of proglottids through the anus**
- nonspecific GI symptoms including nausea, anorexia, or epigastric pain

Diagnosis

- Based upon the patient's description of **passing mobile proglottid segments**
- **Microscopy** — The diagnosis of *T. solium* or *T. saginata* is usually confirmed by finding eggs or proglottids in the feces

Treatment

- **Praziquantel**-The first line treatment for all tapeworms
- 600mg P.O or 10mg/Kg, single dose

Alternative

- Niclosamide, 2gm P.O in a single dose

Prevention and control

1. Educate the public to: Prevent fecal contamination of soil, water, human & animal foods
2. Cook beef and pork thoroughly. ~~f~~Use latrines
3. Identification and immediate treatment of cases.
4. Deny swine access to latrines and human feces

2. Rabies

- Rabies is a preventable zoonotic disease
- The etiologic agents are neurotropic RNA viruses lyssaviruses
- **lyssaviruses are deposited in peripheral wounds**
- **Retrograde passage** occurs from the periphery to the dorsal root ganglia and then to the brain
- **Viral replication occurs in the central nervous system**
- Lyssaviruses have a predilection to neural tissue
- **Transmission** : Almost all cases of rabies are transmitted from rabid animals through a bite or contact
- **Reservoir** : mammals; **primarily dogs**
- **Incubation period**: **7 days to >1 year** (average 1- 2mths)

Clinical Manifestations

- After initial signs have appeared, progressively worsening **encephalopathy and death**

Has 4 phases

1. Prodromal

- Nonspecific (Headache, fever, nausea)
- tingling sensation and paresthesia

2. Acute encephalitic phase

- Confusion, hallucination, bizarre behavior,
- Muscle spasm response to touch, noise
- Meningismus, opisthotonus posturing, seizure, phobia.....

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3. Paralytic (dumb) rabies

- Profound dysfunction of brain
- Cranial nerve involvement
- **Foaming in the mouth due to swallowing muscle paralysis**
- quadriparesis with sphincter involvement
- Airphobia, hydrophobia
- Coma

4. Death and in rare cases recovery

- **It is almost always fatal**

Diagnosis

- Fresh brain biopsy studies of the offending animal.
- Post mortum brain biopsy
 - **Negri bodies**, intracytoplasmic inclusion bodies

Treatment

- No specific treatment
- cornerstone of rabies prevention is **wound care**
 - **Immediately scrub with alkaline soap and water**, and **flush with water for 15 minutes**
 - Irrigate with Povidone-iodine
- TAT and antibiotic given
- Only supportive care

Post exposure prophylaxis

- If Known domestic animal (dogs) can be observed for 10 days, and if healthy no need for PEP
- If animal is rabid, or suspected, or rabid during 10 days observation, PEP should be given for the victim
- **All wild animals**, and status unknown ones **should be regarded as rabid** and PEP given for the victim

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- Decide on post-exposure vaccination and immunoglobulin use depending on the type of contact with the rabid animal

1. Category I:

- **Touching or feeding** animals,
- Licks on the intact skin

2. Category II:

- nibbling of uncovered skin,
- **minor scratches or abrasions without bleeding**

3. Category III:

- Single or multiple transdermal bites or scratches,
- Licks on broken skin,
- Contamination of mucous membrane with saliva from licks; exposure to bat bites or scratches

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Treat according to category of contact:

- Category I: No treatment is required
- Category II: **Immediate vaccination**
- Category III: **Immediate vaccination** and administration of **rabies immunoglobulin**
- **Immunoglobulin:**
 - **Human rabies immune globulin 20 IU/kg OR**
 - **Equine rabies immunoglobulin 40 IU/kg**
- NB: It is mostly injected at the site of the bite.
 - If any is leftover, inject IM at a distant site.
 - This can be given up to 7 days post-exposure if not available immediately

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Vaccination

- Modern cell culture vaccine on days **1, 3, 7, 14 and 28 days**
- All intramuscular injections must be given into the deltoid region
- The vaccine should never be administered in the gluteal region

Prevention and control

- 1) Immunize all dogs
- 2) Detain and clinically observe for 10 days any healthy appearing dog or cat known to have bitten a person.
- 3) Post exposure prophylaxis
- 4) Keep dogs and cats at home
- 5) Destroy stray animals where rabies is endemic

3. Toxoplasmosis

- Toxoplasmosis is a systemic protozoal disease that can be either acute or chronic type with intracellular parasite.
- Toxoplasma gondii is responsible for the development of **lymphadenopathy, myocarditis and encephalitis**.

Infectious agent

- Toxoplasma gondii

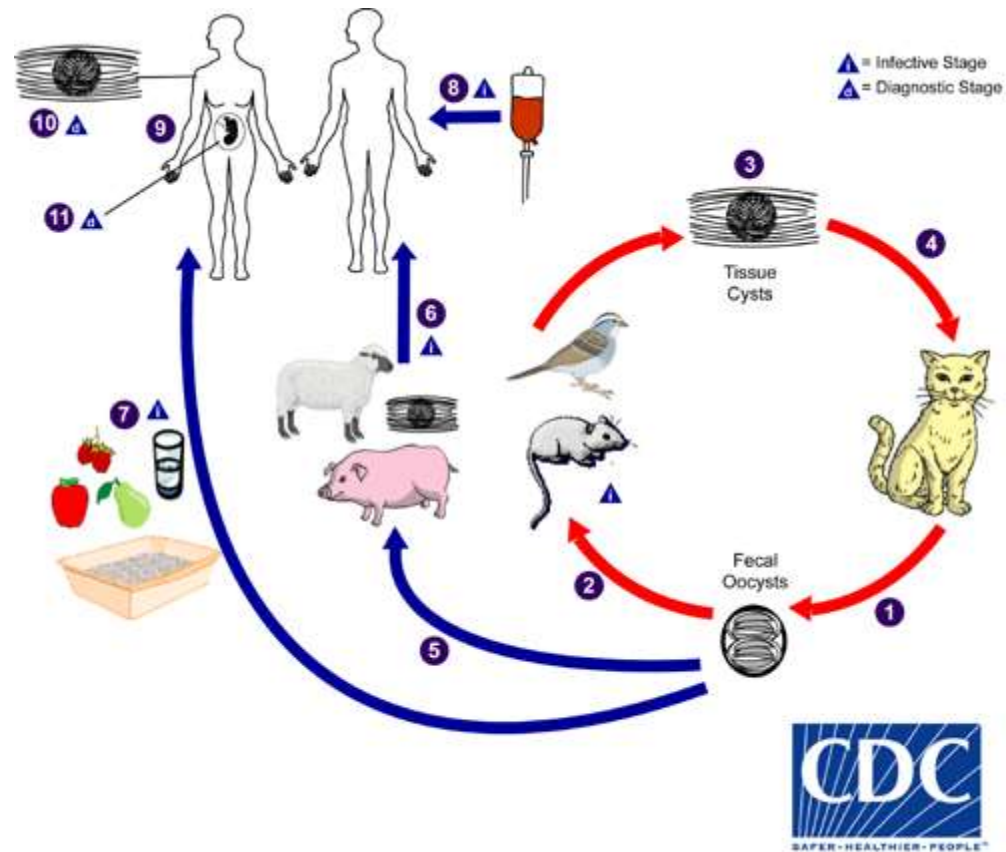
Epidemiology:

- Worldwide in mammals and birds. Infection in man is common.

Reservoir

- The definitive hosts are **cats and other felines**, which acquire the infection mainly from eating infected mammals (especially rodents) or birds and rarely from feces of infected cats.
- The intermediate hosts of T. gondii include **sheep, goats, rodents, cattle, chicken and birds and man**.

Life cycle



Mode of Transmission

- **Ingestion of cysts in raw or under-cooked meat**
- Ingestion of oocysts in food, drink or from hands contaminated with feces of an infected cat.
- Transplacental/congenital
- Blood transfusion
- Organ transplantation

Incubation period:

- From 10-23 days. One common source outbreak from ingestion of under-cooked meat is possible.

Period of communicability

- Not directly transmitted from person to person, except in utero. Oocysts shed by cats sporulate and become infective 1-5 days later and may remain infective in water or moist soil for about a year.
- **Cysts in the flesh** of an infected animal remain infective as long as the meat is edible and uncooked.

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Susceptibility and resistance

- Susceptibility to infection is general, but immunity is readily acquired and most infections are asymptomatic.
- Duration and degree of immunity are unknown, but are assumed to be long-lasting or permanent.
- Antibodies persist for a year, and probably for life.
- Patient undergoing cytotoxic or immuno-suppressive therapy or **patients with AIDS are at risk of developing the disease.**

Clinical manifestation

- **General symptoms:** are mild and mimic those seen in cases of infectious mononucleosis.
- **Acute form:** fatigue, lymphadenitis, chills, fever, headache and myalgia.
- **Chronic disease:** maculopapular rash, **encephalomyelitis** and **hepatitis**; retinochoroiditis with subsequent **blindness** has been known to occur on rare occasions.
- **Congenital Toxoplasmosis:** **hydrocephaly, microcephaly, choreoretinitis**, convulsion and psychomotor disturbance.
- Most of these infections ultimately result in mental retardation, severe visual impairment or blindness.

Diagnosis

- Clinical sign and symptom
- Serological test
- Demonstration of the agent in body fluid or tissue biopsy or
- Cell culture

Treatment

1. Treatment is not routinely indicated for a healthy immunocompetent host, except in an initial infection during pregnancy or the presence of active choreoretinitis and myocarditis or other organ involvement.
2. Severe symptomatic disease is: **Pyrimethamine** combined with **sulfadiazine** and folinic acid for four weeks.
3. For pregnant women, Spirmycin is commonly used to prevent placental infection. If ultrasound or other studies indicate that fetal infection has occurred, Pyrimethamine and sulfadiazine should be considered.

Treatment for infants

- Pyrimethamine, Sulfadiazine, Folinic acid

4. Anthrax

- Anthrax is caused by *B. anthracis*
- a gram-positive, nonmotile
- **spore-forming rod** that is found in soil
- Predominantly causes disease in herbivores such as **cattle, goats, and sheep**
- Anthrax spores can remain viable for decades
- human infection is generally the result of contact with anthrax-infected animals or animal products
- **Goat hair in textile mills** or **animal skins used in making drums**
- The three major clinical forms of anthrax
- **gastrointestinal, cutaneous, and inhalational**

Mode of transmission

- *Gastrointestinal anthrax* typically results from the **ingestion of contaminated meat**
- *Cutaneous anthrax* typically begins as a papule following the **introduction of spores through an opening in the skin**
- **Inhalational anthrax**: inhalation of spores that become deposited in the alveolar spaces
- **Incubation period**-A few hours to seven days; most cases occur **within 48 hours of exposure**

Clinical manifestation

- **Cutaneous**
 - **95% cases of human anthrax**
 - Localized skin lesion with **central blackened necrotic** eschar and surrounding marked **non pitting edema** on exposed areas of the skin
 - In severe cases extensive edema associated with shock occurs
- **Spontaneous healing** → 80 to 90% of untreated cases



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Inhalational Anthrax

- Wool sorter's disease
- *B. anthracis* spores in airborne particles → deposited directly into the alveoli or alveolar ducts
- After 1-3 days of exposure → acute phase with increasing fever, dyspnea, stridor, hypoxia & hypotension → **death within 24 hours**

Gastrointestinal Anthrax

- Results from **ingestion of inadequately cooked meat** from infected animals
- Symptoms (variable): fever, nausea, vomiting, abdominal pain, bloody diarrhea,

Laboratory Diagnosis

- Needs careful specimen handling
- B.anthraxis : Gram's staining, or culture
- Antibody to B. anthracis serology test

Treatment

- Anthrax can be successfully treated if the disease is promptly recognized and initiated early
- **Penicillin, ciprofloxacin, and doxycycline**

Prevention

- Toxoid vaccine for high risk individuals

5. Brucellosis

- A systemic bacterial disease with acute or insidious onset transmitted to humans from infected animals.
- **Infectious agent**
Brucella melitensis (most common worldwide): acquired primarily from **goats, sheep and camels**.
B. abortus from cattle
B. suis from pigs
B. canis from dogs
- These are small aerobic gram-negative bacilli, intracellular parasites.

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Epidemiology

- Worldwide.
- Predominantly an **occupational disease** of those working with infected animals or their tissues especially **farm workers, veterinarians and abattoir workers**, which is more frequent among males.
- Outbreaks can occur among **consumers of raw milk and milk products**, especially unpasteurized soft cheese from cows, sheep and goats.
- **Reservoir:** cattle, swine, goats and sheep, pet dogs.

Mode of transmission

- by **contact with tissues**, blood, urine, vaginal discharges, aborted fetuses and especially placentas (through breaks in the skin).
- Most commonly through **ingestion of raw milk and dairy products from infected animals** (raw meat or bone marrow).
- Airborne infection occurs in humans in laboratories and abattoirs.
- **Incubation period**- may last about 1-3 weeks but may be as long as several months.
- **Period of communicability**: no evidence of communicability from person to person.
- **Susceptibility and resistance**: severity and duration of clinical illness are subject to wide variation. Duration of acquired immunity is uncertain.

Clinical manifestation

- Abrupt onset of symptoms
- Most common symptoms are: Fever, chills, diaphoresis, headache, myalgia, fatigue, anorexia, joint and low back pain, weight loss, constipation, sore throat, and dry cough.
- Physical examination reveals
 - Often no abnormalities and patient looks well
- Some are acutely ill, with pallor, lymphadenopathy, arthritis, hepatosplenomegally,, **spinal tenderness, epididymoorchitis, skin rash, meningitis, cardiac murmurs, or pneumonia**
- Reactive asymmetric polyartaritis (knees, hips, shoulders, sacroiliac and sternoclavicular joints)

Diagnosis

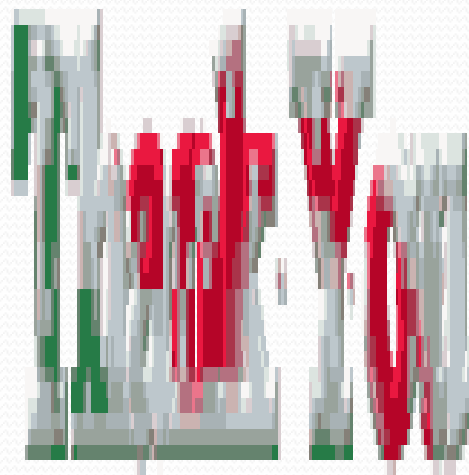
- Exposure and consistent clinical features
- Serology- raised levels of B. agglutinin
- Blood or bone marrow culture

Treatments

- Doxycycline + aminoglycoside for 2 weeks followed by
- Doxycycline + Rifampicin for 4-8 weeks is the most effective regimen.
- In pregnancy and in children less than 7 years, Bacterium and Rifampicin for 8-12 weeks
- **N:B** 4-14 days after the initiation of therapy, patients become afebrile and constitutional symptoms disappear but enlarged liver and spleen return to normal size within 2-4 weeks.

Prevention and Control

1. Control depends on elimination of the disease among domestic animals.
- 2. Educate people not to drink untreated milk or eat products made from untreated milk.**
3. Educate farmers and slaughterhouse workers and those in meat processing plants and butcher shops as to the nature of the disease and the risk in the handling of carcasses and products of potentially infected animals.
- 4. Educate hunters to use barrier precaution (gloves and clothing)**
5. Eliminate infected animals.
- 6. Pasteurize milk; cook meat and bone well.**
7. Proper disposal of placenta, discharges or fetus from an aborted animal. Disinfect contaminated areas.



Chapter 9

Other infectious diseases

1. Tetanus

- **Gram positive, spore forming, anaerobic rod**
- **Reservoir:** soil
- **Transmission:** puncture wounds, trauma, human bites

Pathogenesis

- Spores germinate in tissues and produce toxins
 - Tetanospasmin – **neurotoxin**
 - Tetanolysin – **haemolytic, tissue lysis**
- Release of tetanospasmin from vegetative *C. Tetani* →
- **the toxin binds to peripheral motor neuron terminals** →
- Retrograde intra-axonal transport of toxin to the spinal cord and brain stem →
- **Blockade of inhibitory neurotransmitters (glycine and GABA)** release in presynaptic terminals →
- ↑resting firing rate of motor neurons → **rigidity** → Simultaneous recruitment of agonist and antagonistic muscles 2⁰ to limited glycinergic activity → **Spasm**
- ↑Preganglionic sympathetic activity → **Sympathetic hyperactivity and ↑ circulating catecholamine level**
- **Blockade of neurotransmitter release at the NMJ** → paralysis

Forms of Tetanus

- ✓ Generalized,
- ✓ localized,
- ✓ cephalic,
- ✓ neonatal

- Definition of terms:

- ❖ IP: time gap b/n injury and **first symptom** (**trismus**)
- ❖ Onset time: time between trismus and first spasm (time from first symptom to first generalized spasm)
- ❖ **Trismus** (**lockjaw**) 2⁰ to masseter muscle hypertonicity
- ❖ **Opisthotonus: backward arched posture**
- ❖ **Risus sardonicus**: recession of the lips backward and a grinning like grimace

opisthotonus





Risus sardonius



Risus sardonicus



Generalized tetanus

- Most common presentation of tetanus
- Trismus (lockjaw) 2^o to masseter muscle hypertonicity
- Neck, shoulder and back muscle stiffness and pain
- Rigid abdomen and stiff proximal limb muscles
- Risus sardonicus,
- Opisthotonus
- Paroxysmal generalized muscle spasm → **apnea /Cyanosis** /laryngospasm (spontaneous or provoked)
- Hyperpyrexia with clear mentation

Severity of generalized tetanus

1. Mild tetanus	2. Moderate tetanus	3. Severe tetanus
- IP > 14 days - Onset time > 7 days - Mild trismus - Localized spasm and rigidity - No autonomic dysfn	- IP = 7-14 days - Period of onset > 3-6 days - Marked trismus - Mild dysphagia - Rigidity and spasm - No autonomic dysfn	- IP < 7 days - Period of onset < 3d - frequent explosive spasms - spontaneous spasm - asphyxia, dysphagia - Autonomic dysfunction

Neonatal tetanus

- Generalized form of tetanus
- Develops in neonates born in unimmunized mothers after unsterile treatment of the umbilical cord stump
- Occurs **within 2 weeks of neonatal life**
- Manifests with *poor feeding, rigidity and spasm*
- High rate of mortality

Localized tetanus

- Uncommon form of tetanus
- Manifests with **localized muscle contraction near the wound**
- Good prognosis

Cephalic tetanus

- Rare form of local tetanus
- Follows **head injury or ear infection**
- Manifests with **trismus and CN palsy**
- High mortality

Diagnosis of tetanus

- Entirely on clinical findings
- Spatula test – gag stimulation causes masseter muscle spasm

Complications

- **Laryngospasm**: involuntary/uncontrolled muscular contraction of the vocal cords
- **Bone fracture-spinal**
- HAI
- Blockage of main artery of the **lung(pulmonary embolism)**
- **Aspiration Pneumonia**
- Breathing difficulty
- **Death**

Treatment

- **Principle**
 - Cardio-respiratory support
 - Eliminate the source of toxin
 - Neutralize unbound toxin
 - Prevent muscle spasms

General measures

- **Admit in a quiet ICU** with a possibility for continuous monitoring but minimal disturbance
- Hydrations
- Nutritional support (IV/PO)
- Physiotherapy to prevent contracture
- Psychological support, analgesia
- Treatment of superimposed infection eg aspiration pneumonia
- Treatment of source infection

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- Antibiotic therapy to eradicate vegetative C.Tetani; choice are:
 - **Metronidazole,**
 - **Penicillin,**
 - **Clindamycin,**
 - **Erythromycin**
- Lower dose of penicillin should be used as high dose has a GABA inhibitor effect
- Wound debridement

1. Antitoxin

- To neutralize circulating toxin and unbound toxin in the wound
- Human TIG, Pooled IV Ig, Equine TAT

2. Control of muscle spasm

- Benzodiazepines (**Diazepam**, midazolam), Barbiturates, chlorpromazine, propofon, dantrolene, baclofen

3. Respiratory care

- Intubation / **Tracheostomy + mechanical ventilation** for adequate oxygenation

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4. Autonomic dysfunction

- for sympathetic over activity
 - **Labetalol** – both α and β blockers are preferred
 - **Magnesium sulphate** can control the autonomic dysfunction and spasm
- **Vaccine** - Tetanus toxoid
 - Age <7 years – DPT vaccine
 - Age > 7 years – DT Vaccine

Thank
Very
Much